DF1725IED
User's Manual

Dongfang Electronics Co., Ltd.

Introduction

Yantai Dongfang Electronics Co., Ltd. is to provide smart solutions to optimize transmission and distribution operations for the electric power industry, using a wide range of automation products and systems to help utilities automate, monitor, control and maintain transmission and distribution properties safely and efficiently with innovative substation systems and specialized services and support. Dongfang's products have been used in substations, switching substations at different voltage levels, and power plants.

With more than 20 years of experience and efforts, Dongfang has earned the recognition and respect of many loyal customers, developed a series of reliable automation products, and cultivated a professional R&D team. We have confidence in our ability to meet the changing demands in the electrical field and dedicated to meeting the needs and serving the interest of our customers.

Dongfang adopts the state-of-the-art technology in the design and manufacture of automation products. Dongfang is one of the largest suppliers and service providers in China and is branching out to the international market.The vast pools of experienced technical professionals of Dongfang are continuously progressing in the design, development and manufacture of automation products and systems to meet customers' needs.

DF1725IED has advanced functions, flexible configuration, reliability and easy maintenance.

This manual mainly provides detailed information on the installation, operation and maintenance of the DF1725IED for engineers and technicians. It is designed to help you understand how the DF1725IED works, as well as to help in troubleshooting DF1725IED.

There are 2 sections in this User's Manual.

Section 1: Technical Description of DF1725IED

Section 2: Installation Guide of DF1725 IED

In Section 1, the composing mode, features and functions of DF1725IED system are described, including each part modules within DF1725IED.

In Section 2, Installation Guide gives the step-by-step procedures for mounting, installation and activation of DF1725IED operation.

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1. Technical Features and Buildup

1.1 Technical Features of DF1725IED

1.1.1Centralized and distributed structure design

Within the subrack of DF1725IED, different function boards are provided to be selected and configured such as Power supply board (POWER-220/110V, POWER-24V and POWER-48V), Main Control Unit board (MCU-1,MCU-2, etc.), serial COM port extension board (COM-6), Digital input board (DI-16-220V , DI-16-110V ,DI-16-24V and DI-16-48V for 16 digital inputs), Control Output board (CO-8 for 8 control outputs), Analog Input board (DCAI-8 for 8 DC analog inputs and ACAI-3U3I/6I for 6 AC analog inputs), Analog Output board (AO-2 for two analog ouputs), etc. There are totally 11 slots for the configuration of function modules in one 1725IEDSubrack 19" (6 slots for 2/3 19"Subrack), so commonly for a low I/O requirement distribution substation, one 1725Subrack is enough. If one subrack cannot cover the I/O points in one substation, other 1725IED Subracks(One MCU board and one power supply board are needed for one subrack) with their I/O boards can be added. All the 1725IEDSubracks can be connected through Ethernet ports. So the structure of DF1725IED can be the type of mixed of centralized mode and distributed mode, the configuration for DF1725IED is very flexible and the application range of DF1725IED is wide. DF1725IED can collect more than 10240 points as the Data Concentrator.

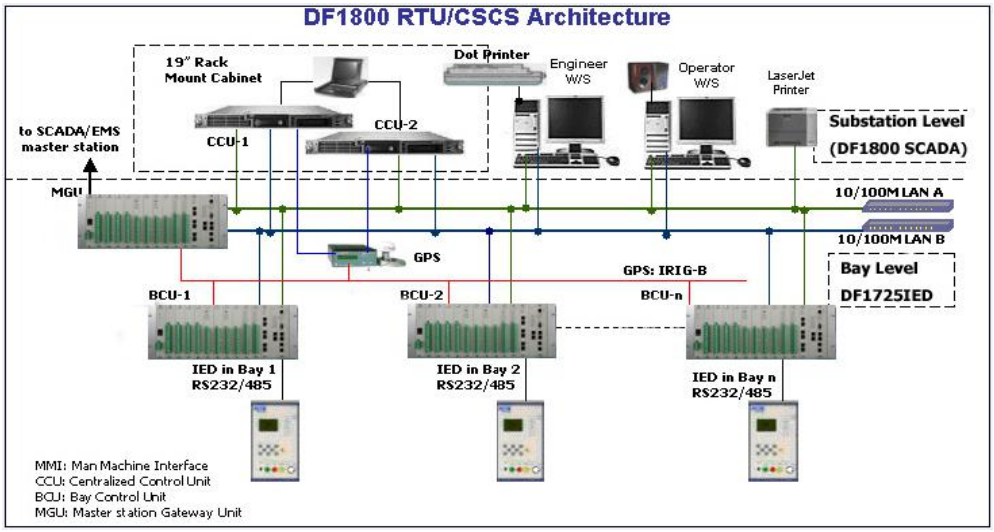


Fig1.1 Distributed application as MGU and BCU

1.1.2 Compact and flexible design

1725IEDSubrack size in 19'' width and 4U height (or 2/3 19" width and 4U height), it can be flash mounted in cubicle or be wall mounted. In each 1725IEDSubrack, there are 11 slots for different types of I/O boards, and maximum 8 same type I/O boards can be managed, that is, maximum 128DI or 64DO or 64DCAI or 48ACAI

or 16AO can be configured in one subrack. Any function board can be replaced without reprogramming, redefinition of configuration parameters, or rewiring.

1.1.3 Advanced network communication technique

If more 1725IED Subracks are configured, in each subrack, one MCU and one power supply board are needed. They are connected with each other by Ethernet ports. DF1725IED can prevent unauthorized access and support webserver.

1.1.4 Convenient remote and local maintenance function

The 1725IED can be maintained by DF1000v maintenance software (Refer to *DF1000v user's manual*) on portable computer at site or even through the public telephone or Internet. All the things can be done in remote as in the substation, such as monitoring the real time data, checking the sending and receiving buffers, changing the setting values, diagnosing the equipment, upgrading the software of the whole system, etc. It is not necessary to go to the site and many things can be solved in your office. No erase or write to the EPROM is required. With the remote maintenance function, our technical supporter can also help you to maintain the DF1725IED running in correct mode.

1.1.5 Support mixed kinds of analog sampling

The DCAI-8 board is designed for the DC analog sampling with a very high accuracy, Voltage and current input can be injected into one board at the same time. ACAI-3U3I/6I AC analog input board is designed for flexible sampling of the AC signal without transducers with a high accuracy and stability, It can be set flexibly with different voltage inputs for the calculation of active power, reactive power, power factor, frequency, etc. If ACAI-3U3I/6I is selected as the sampling proposal in substation/plant, a lot of transducers and corresponding interface cabinets and connecting wire can be omitted.

1.1.6 High capacity CPU

1) The CPU of the MCU main control unit board adopts international standard ColdFire 5275(32bits,500MHz). It owns strong processing ability and easy to upgrade. Ethernet card is equipped in the main communication processing board.

2) Multiple serial ports (RS232/RS422/RS485) can work simultaneously in different protocols. There is an abundant protocol library available for selection, which includes IEC61850, IEC870-5-101, IEC870-5-102, IEC870-5-103, IEC870-5-104, DNP3.0 (security, serial, TCP/IP), WISP+, Modbus, etc. It also contains protocols of different kinds of protection relays and intelligent energy meters. The MCU can communicate with multiple Master Stations, or the station layer computer simultaneously. It also can be configured as the sub-master for the other slave RTU/IEDs or communicate

with intelligent equipment in the substation to compose the substation automation system.

3) Ethernet ports are available for the communication with dispatch centers, or with IEDs, or any places required. It supports TCP/IP and address can be configured locally or remotely.

1.1.7 Support soft PLC functions

DF1725IED supports soft PLC functions which complying with the IEC1131-3 standard.

1.1.8 Sub-Master capability

1) DF1725IED can act as a Sub-Master. It can poll a number of slave RTU/IEDs through its different communication ports in point to point or multi-drop configuration. The database of the slave RTU/IED can be mapped in the Sub-master RTU/IED for transmission to Master Station and the Master Station can retrieve the information from Sub-Master RTU/IED at any time.

2) DF1725IED can communicate with its slave RTU/IEDs by two communication channels in redundant mode. In case there is no response from Slave RTU/IED, the Sub-Master RTU/IED will send message to Master Station. The message will carry virtual digital input point to identify the address of the slave RTU/IED.

3) By using DF1725IED maintenance software, the data of Slave RTU/IED can be monitored from Sub-Master RTU/IED and control commands can be executed from Sub-Master RTU/IED as well.

1.1.9 Interface to intelligent electronic devices (IEDs)

1) DF1725IED can communicate with IEDs via optical-electrical isolated interface configured with IEC 60870-5-103 protocol or other protocols. The configuration can be ‘point to point’ or ‘multi-drop’, depending on pointed protocol, with a transmission rate from 300bps to 19200bps.

2) DF1725IED can communicate with maximum 32 IEDs on its single channel under RS485 mode.

3) DF1725IED can receive data and disturbance records from IEDs for retrieval by Master Station or a PC. It can transmit commands to IEDs in response to command from Master Station as well.

1.1.10 Self-contained alarm message

1) When MCU detects the malfunction of any circuit board within the DF1725 IED, host communication fail and lack of response from any Slave RTU or Serial I/O Device, it will activate a contact from MCU. On clearing the fault,

the alarm is automatically deactivated.

2) The DF1725IED provides access to alarm status information via DF1000v maintenance software.

3) Support watch-dog for device running

1.1.11 VQC

DF1725IED Can be used as VQC device. It can be used to realize the VQC function for transformer or used as the distributed VQC module for control unit.

1.1.12 Object-Oriented Communication Unit

There are 5 serial ports on each DF1725IED module, When DF1725IED used as BCU, all other IEDs in one bay, such as protect relay, can be connected to DF1725IED.

1.1.13 Feeder switch Monitor& Control

DF1725IED can be installed on pole-top to monitor & control the outdoor switch and realize the functions of FTU.DF1725IED can also apply to realize the monitor & control function for ring main unit (RMU).

The conformal coating of DF1725IED comply the IEC 60068-2-38 and IEC 60721 standard.

1.1.14 Distribution Power Fault Detect and Automatic Isolation

Through the parameters configure, DF1725IED can detect the shorted fault and earth fault, the faulted part can be isolated automatically by DF1725IED through DMS

The following fault type can be detected:

Overcurrent (ANSI 50/51, 50G/51G)

Sensitive earth fault (ANSI 50SEF)

Directional relay (ANSI 67)

Broken conductor (ANSI 46BC or ANSI 47BC)

Negative sequence voltage (ANSI 47)

DF1725IED can record at least 512 fault Events, associated events with time stamp and wave recording in COMTRADE format.

1.2 Main Component of the DF1725IED

DF1725IED has a full series of components to cover all the functions required in power automation system. The main components and its brief description are listed below.

Rack and power supply units		
1	1725IED-rack	For mounting plug-in I/O Boards
2	Power	Power Supply Boardt
	Power-D	Power Supply Board, Dual mode
3	LCD	LCD for local display
Communication processing units and their accessories		
1	MCU-1	Main Communication Processing Unit, 0.5/ 2ppm
2	MCU-4	Main Communication Processing Unit, RTU slave subrack mode and FRTU mode
3	MCU-5	Main Communication Processing Unit, FRTU+GPRS mode
4	MCU-6	Main Communication Processing Unit, 0.5/ 2ppm
5	MCU-G	Main Communication Processing Unit, Gateway mode
6	COM-6	Expansion Com port board, 6 serial ports included
Input and Output Cards		
1	DI-16	Digital input board (16 digital channels),DC 24/48 V input
2	DI-32	Digital input board (32 digital channels), DC 24/48V input
3	CO	Digital output board (2/8/16 contacts)
4	DCAI-8	DC analog input board (8 current/voltage DC inputs)
5	ACAI-3U3I/6I	AC analog input board (3 voltage & 3 current or 6 current AC inputs)
6	ACAI-4U4I/8I	AC analog input board (4 voltage & 4 current or 8 current AC inputs)
7	GPS	Satellite synchronization signal receiver
8	AO-2	Analog output board (2 current/voltage outputs)
9	LPT	Communication channel lightning protection terminal
10	PLPT	Power supply lightning protection terminal block
Maintenance Tools		
1	DFET	1725IED maintenance software

Table 1.1 Components list of 1725IED

1.3 DF1725IED System Composition and Configuration Mode

DF1725IED adopts the 19’’or 2/3 19” rack-mounted structures. All the functional boards can be plugged in the 1725IEDSubrack. The dimension of 1725IEDSubrack is 19’’×4U or 2/319’’×4U. All the IO functional boards get secondary power supply and fulfill the data communication through the internal bus board in the 1725IEDSubrack. All the signals from the site can be directly connected with the boards, No other terminal is needed for DF1725IED.

1725PSB

Technical features

- Diode protection against reverse DC polarity
- Input Transient over voltage protection by TVS and Varistor.
- Overload protection
- Reverse polarity protection and isolation facility to provide power supply output to all function modules.
- Since all function modules can be plugged in and out with the power on, it is not necessary to provide isolating facilities at each function module.
- The power supply unit can keep a stable output for all I/O modules even when the input varies in a wide range. Overvoltage and under voltage protection components like TVS, Auto-Resetable-Fuses, and varistors are designed in all modules. The power supply monitor circuit is also provided in each function module to protect the RTU internal logic.
- Provide power fail alarm contact output

Specification

POWER-220/110V
Input Type and Range: 220VAC,220VDC or 110VDC (-15% ,+ 20%)
Output Voltage and Current: 5 V DC and 3.3V
Power Consumption: 25 W
Insulation: 2000 V AC
Impulse Withstand: 5KV AC



Fig1.2 Power board

POWER-48V
Input Type and Range: 48 V DC (36V DC ~ 72V DC)
Output Voltage and Current: 5 V DC and 3.3V
Power Consumption: 25 W
Insulation: 2000 V AC
Impulse Withstand: 5KV AC

POWER-24V
Input Type and Range: 24 V DC (18V DC ~ 36V DC)
Output Voltage and Current: 5 V DC and 3.3V
Power Consumption: 25 W
Insulation: 2000 V AC
Impulse Withstand: 5KV AC

MCU series

The MCU series Main Communication Processing Unit is suitable for acquiring, processing and transmitting communications information within the substation. The communications may adopt the serials or the Ethernet network. The unit can

be flexibly installed in either centralized configuration in the DF1725IED cabinet or de-centralized configuration within the substation.

The MCU module is the CPU of the DF1725IED system. Based on a 32-bit ColdFire 5275 micro-controller chip that provides high computational throughput and communications speed, it allows the unit's sophisticated software to process information in real time.

Technical features

- Based on an advanced industry compatible chip, electrical isolation and electromagnetic shielding both conform to the corresponding international standards.
- The PCB board design is simplified by using EPLD and FLEX (Electrically-erasable Complex Programmable Logic Device).
- The main chips and boards adopt SMT and four-layer PCB
- Equipped with a real time clock with full calendar, inclusive of leap year support.
- Supported by a dedicated battery backup, time and date can be maintained for at least a year when there is no power supply.
- The internal clock accuracy is 1.2 PPM and will not drift more than 5ms per hour without GPS synchronization via IRIG-B code.
- The clock resolution is 1ms. Software algorithms are employed to counter inaccuracies and drift resulting from crystal ageing or other factor like temperature, etc.
- It has stainless steel membrane keypads to access all measured parameters and status information, as well as for programming functions that provide a unique and flexible user interface.
- Configurable size of the rolling Main SOE buffer with time stamps for digital input modules to store at least ten times the number of installed digital input points, together with overflow alarm indication.
- One MCU can communicate with the Master Station using two communication channels in redundant mode.
- Alarm status information accessible via the maintenance port.



Fig1.3 MCU-1

Specifications

Microprocessors:

- ColdFire 5275
- 32 bit, 500.0 MHz

Memory

- Flash 1G bytes
- Nand flash 2G bytes
- RAM 512M bytes

Com1 Port (RS232)

- 100 to 115200 baud for Maintenance/Configuration
- TXD, RXD, GND
- RS232 DB9 connector

Com2 Port (RS232)

- 100 to 115200 baud
- TXD, RXD, GND
- RS232 RJ45 connector

Com3 Port (RS232)

- 100 to 115200 baud
- DSR, RTS, GND, TXD, RXD, DCD, CTS, and DTR
- RS232 RJ45 connector

Com4 Port

- 100 to 115200 baud
- TXD, RXD, GND; Ta, Tb
- RS232/485 RJ45 connector

Com5 Port

- 100 to 115200 baud
- TXD, RXD, GND; Ta, Tb
- RS232/485 RJ45 connector

Three Ethernet ports:

- 10/100 M bps
- RJ45 connector

One IRIG-B time input port: RJ45 connector

Delay point: one status alarm

LED Indicators

- POWER: To indicate power supply OK
- RUN indication of the normal or error state
- ERR indication of the error state
- IRIG-B indication of the IRIG-B signal state
- ETH1 Send & Receive LED, flashing when the ETH1 send & receive data
- ETH2 Send & Receive LED, flashing when the ETH2 send & receive data
- COM1: Send & receive LED, flashing when the com1 sends and receives data

- COM2: Send & receive LED, flashing when the com2 sends and receives data
- COM3: Send & receive LED, flashing when the com3 sends and receives data
- COM4: Send & receive LED, flashing when the com4 sends and receives data
- COM5: Send & receive LED, flashing when the com5 sends and receives data

One ALARM port: RJ45 connector.

DCAI-8

The DCAI-8 DC analog input board is the analog input interface module, which is capable of processing voltage and current inputs continuously, and has the following features:

Technical features

- 8 differential mode input.
- One jumper configurable for the differential mode input or common mode input.
- Programmable Input Ranges: ±10 mA, ±20 mA, 4 - 12 - 20 mA, 4 - 20 mA, ±5V and ±10V.
- A jumper setting is provided to select whether to include Precision Resistor for current inputs or to exclude Precision Resistor for voltage inputs.
- One Precision Resistor (250ohm) for current up to 20mA or two Precision Resistors (total resistance 500ohm) for current up to 10mA jumper configurable for current inputs. The range of current inputs can be further configured between the 10mA range and 20mA range via the maintenance software.
- When the current input is more than 10mA (or 20mA) or voltage input is more than 5V, the exceeded voltage or current value will be reported accordingly up to 20mA (or 40mA) in case of current and 10V in case of voltage.
- Common mode noise rejection: at least 90dB from 0 t0 50Hz.
- Normal (differential) mode rejection: at least 60 dB at 50Hz.
- Integrity check and auto-calibration.
- Current input impedance is such that the voltage across the input terminals does not exceed 5 volts (with full-scale input current) and Current input over range: 100% over current (full scale) continuous. For voltage inputs, the input impedance is infinite.
- Computed values: minimum and maximum values, average values and rates of value changes of analogue inputs stored in the flash memory of MCU, such data accessible to Master Station or a printer connected to the MCU.
- Analog deadbands configurable 0 to 100% of full scale.

- Force To Zero parameter configurable from 0 to 100% of full scale to eliminate noise fluctuations around the zero value.
- Smoothing factors configurable for analog inputs to ensure that fast transient interferences do not cause large fluctuations in actual reported values, avoiding incorrect reporting of limit transgressions.
- Low Alarm and High Alarm limits configurable from 0 to 200% of full scale with limit transgression alarms.
- Maintenance software disabled analogue input reports.

Specifications

Sampling Capacity

- 8 Voltage/Current (differential) mode inputs.
- Voltage: ± 5V, or ±10V.
- Current: ±1 mA,±10 mA, ±20 mA, 4 - 12 - 20 mA, 4 - 20 mA, etc.

A/D References

- 5V and 10V precise reference voltage, long-term stability better than 0.02%; software auto calibration, and no manual adjustments required
- Sampling Rate 10ms per point and 80 ms for 8 Points

Accuracy

- 0.1% current accuracy references
- 0.1% voltage accuracy references

Overload Rating

- Current: 200% continuous
- voltage: 200% continuous (@ ±5 V)

Standard Analog Calculations

- Capability for 8 analog voltage/current values
- Equivalent to 16 bits resolution including sign bit, Overall range is –10000 to +10000 by ADCs

Site Addressing

- 3-bit via jumper blocks
- 7 discrete addresses



Fig1.4 DCAI-8

Address Decoding: Address decoding is performed by an EPLD.

ACAI-3U3I/6I

Technical features

Sampling each AC input at high speed and analyzing its harmonic.

Under 3 watt-hour meters, each ACAI-3U3I board can get 6 direct value, that are Ia, Ib, Ic, Ua, Ub, Uc; and other important values can be calculate such as Uab, Ubc, Uca, total P, total Q, power factor, kWh, and kVarh.

Under 2 watt-hour meters, each ACAI-3U3I board can get 4 direct value, that are

Ia, Ic, Uab, Ucb; and other important values can be calculate such as total P, total Q, power factor, kWh, and kVarh.

Specifications

Sampling Capacity

- 6 AC inputs
- Input current: current transformer (CT) or line-post sensor 1A or 5A
- Voltage inputs: voltage transformer (PT) or line-post sensor 100V or 220V



Fig1.5 ACAI-3U3I

Maximum Load

- 0.25 VA for current transformer
- 0.25 VA for voltage transformer
- A/D References: 16-bit (resolution)

Sampling Rate

- 32/64(50/60Hz) times per AC cycle
- 128 times per cycle for high speed sampling

Accuracy

- 0.2% current accuracy references
- 0.2% voltage accuracy references
- 0.5% Active Power, Reactive Power
- Frequency 0.01Hz

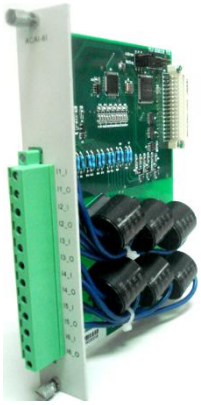


Fig1.6 ACAI-6I

Overload Rating

- Current: 2000% continuous
- Voltage: 120% continuous

Site Addressing

- 3-bit via jumper blocks
- 7 discrete addresses

DI-16-220/110/48/24V

The DI-16-220/110/48/24V Digital Input Peripheral obtains and processes digital input states from field devices..

The 16 digital inputs are organized in two groups of eight electrically isolated inputs, each group with one common return. The digital inputs can use 220V,110V,24V or 48V as wetting voltage. Using the DF1000V Maintenance Software, each input can be configured from a downloaded database to be processed as a single-bit input, double-bit input, Binary Coded Decimal (BCD) input or counter input. Each double-bit and single-bit input can be configured as an SOE. Counter input can be configured as Form A, Form B or Form C inputs. BCD can be configured up to 8-digit decimal number without sign and decimal point.

Technical features

- Diode protected inputs
- 16 photo coupled digital inputs freely configured as one of the single-bit points, double-bit points, counter inputs and BCD inputs
- Auto-Resetable-Fuse and varistor to protect each digital input from earth faults and incorrect wiring which introduces external voltages into the DF1725IED.
- Digital inputs scanned once per 1 ms. Each input is possible to invert its digital signal on a per point basis so that a normally closed contact might be reported as normally open contact.
- Software Debouncing: configurable, on a per point basis, to any value between 0 and 1,000 ms, in 1 ms increments. Any pulse longer than 3ms in width can be detected by DI-16-220/110/48/24V.
- Chatter Filtering: A digital input is disabled if the number of state changes encountered during a defined time interval is excessively high.
- Report Limiting: A double bit processing time parameter configurable to any value between 0 and 255 seconds before a report is sent.
- Any DI point configurable as an SOE point. Single-bit or double-bit points are SOE time tagged.
- Change of state identified with a point identification number, type of event and time of occurrence. The Event Time Tagging Error is not more than 1 ms.
- Time resolution: 1 ms.
- Automatic roll-over 32-bit counters
- Counter inputs configurable via DF1000 maintenance software system as: Form A, Form B, and Form C inputs incrementing on each pulse cycle (count on 0 to 1 transitions only) or on each pulse half cycle (count on 0 to 1 and 1 to 0 transitions)
- FREEZE & RESET commands. The time period is user configurable in minutes on a channel-by-channel, minimum freezed interval is 5 minutes.

Specifications

Sampling Capacity

- 16 inputs in 2 groups of 8
- Max capability for 16 pulse counters.
- Status Point Input
- 5.0 mA typical input current
- Common ground for each group of 8 channel and each ground is independence of each other
- 220V,110V,24V or 48V contact wetting voltage can be applied
- Threshold Detection
- <12V=off, >18V=on for 24V nominal input



- Overload Withstand
- Reverse polarity protection
- 2000 V AC insulation

Fig1.7 DI-16

- When error occurs on DI-16, alarm message will be sent .LED Indicators

16 Status Indicators

CO-8

Technical features

Among its security features, the CO-8 offers:

- 4 Trip/Close outputs
- Select-Before-Operate (SBO) functionality to provide highly secure operation which is especially essential in critical situations, such as Trip/Close operations of power circuit breakers; only one point can be operated at a time, and each command involves a series of steps to confirm and validate controls received from the master.
- Direct Operate Operation: Only one control point in a DF1725IED can be selected or executed at any given time. In the event DF1725IED receives a “select” (or direct operate) command from the second host station after it receives “select” command from the first host station, the DF1725IED will send a signal to the second the host station to reject the command.
- Support “Long Pulse” and “Short Pulse” commands as specified in the IEC101 protocol.
- Each control output provides a form A contact output up to 8A at 30V DC
- Control outputs are provided with LEDs to confirm the status of each control relays, protected against transients, surge withstand, and immunity to impulse transients
- Control outputs can be freely configured to one of the two-operation modes - Select Before Operate (SBO) operation or Direct Operate operation

Specifications

Control Output Points

- Select Before Operate (SBO) output
- Direct Operate output.
- 8 relays output, 1 NO contact per relay
- Having disconnect switch allowing disabling output relay
- Maximum switching voltage 110Vdc
- Maximum switching power: 240W, 2000VA (30Vdc)
- Maximum maintaining current: 8A (230Vdc)



Fig1.8 CO-8

Overload Withstand

- 150% nominal continuous
- Reverse polarity protection
- 2000V AC insulation

Site Addressing

- 3-bit via jumper blocks
- 7 discrete addresses

LED Indicators: 8 Status Indicators

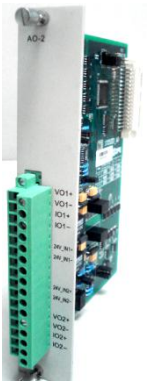


Fig1.9 AO-2

AO-2

Technical features

- Two analog outputs per AO-2, expandable
- AO-2 stabilizing within 100 ms.
- Analog output would not be changed when 1725IED power off.
- All analog outputs are equipped with optical isolation to protect against ground loops and power surges.

Specifications

Output Channels

- 2 channels in one AO-2, additional sub-module can be added to expand to Output types, its maximum overload is 750 ohm for current output
- 0 - +5V, 0 - +10V voltage
- 0-1mA,0 - 10mA, 0 - 20mA, 4 - 20mA current(Selected by jumpers)

D/A resolution for output & A/D resolution for feedback input

- 14 bits including sign bit

Threshold Detection

Accuracy Input:

- Output: 0.1%

Site Addressing

- 3-bit via jumper blocks
- 7discrete addresses

COM-6 port expansion board:



Fig1.10 COM-6

- Six RJ45 communication ports can be selected. (COM1 COM2, COM3, COM4, COM5, COM6)
- COM1 RS232, FULL signal
- COM2, COM3,COM4 RS232
- COM5, COM6 RS232/RS485
- Baud rate: 300~38400 bps

2. Detailed board description

2.1 MCU

2.1.1 Applications

The MCU Main Communication Processing Unit is suitable for acquiring, processing and transmitting communications information within the substation. The communications may adopt the Serial ports or the Ethernet network. The unit can be flexibly installed in either centralized configuration in the DF1725IED cabinet or distributed configuration in the substation.

2.1.2 Introduction

The MCU module is the CPU of the DF1725IED system. Based on a 32-bit ColdFire 5275 chip that provides very high computational throughput and communications speed, it allows the unit's sophisticated software to process information in real time.

Interfaces

The unit is fixed by two M3 screws. The front panel and rear interfaces are shown in the appended figure.

The rear panel has the following:

- Three ethernet ports: E1, Ethernet network settings configurable and support TCP/IP.
- One IRIG-B inputs.
- One COM1 asynchronous port, it is only RS232, DB9 connector.
- One COM2 asynchronous port, it is RS232, RJ45 connector.
- One COM3 asynchronous port, it is RS232, RJ45 connector.
- One COM4 asynchronous port, it is RS232/RS485, RJ45 connector.
- One COM5 asynchronous port, it is RS232/RS485, RJ45 connector.
- One ALARM port, RJ45 connector.

To extension serial COM port, COM-6 boards can be added.

2.1.3 Time-Synchronization with or without Redundancy

The following can be realized along with GPS:

- When GPS is working normally, MCU will synchronize time with GPS directly
- When there is no GPS, MCU will synchronize time with selected Master Station

MCU will synchronize time with GPS when GPS is available. When GPS is not available, the MCU will synchronize time with the selected Master Station (configurable by maintenance software). In the event MCU is not able to synchronize time from both GPS and Master Station within a specific time period

(configurable by maintenance software), the MCU will use its own internal clock as system time and label all messages (to indicate it is acquired during out of synch. Period) after this period. Once the time-synchronization is resumed, the message sent out will not be labeled anymore.

When synchronized with GPS, the Time Synchronization Error is less than 1 ms within the clock of MCU and GPS.

2.1.4 External Programmable Algorithms Function

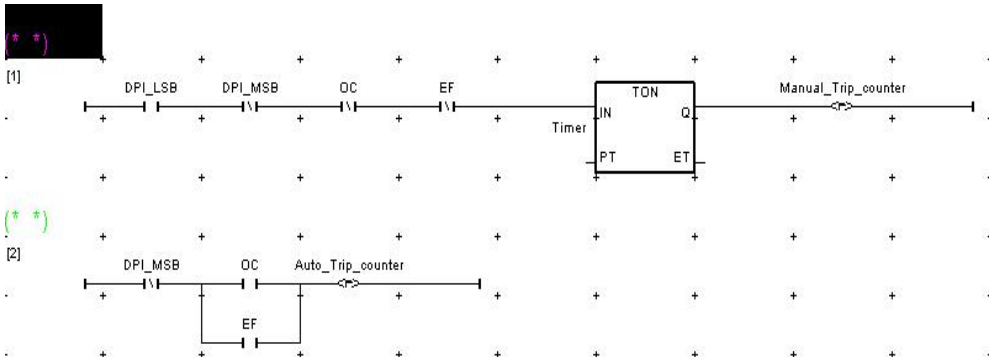


Fig2.1 IEC61131-3: Ladder Diagram(LD)

MCU uses ISAGRAF software to perform PLC function.

Thus MCU is capable of being programmed to perform logical, mathematical, and trigonometric operations within the program on accumulator, as well as to input values and store results in memory. The memory locations are accessible to all programmed algorithms.

A Windows-based graphical programming tool, that is compliant with the Open Standard PLC programming languages to IEC 1131-3 (soft PLC) standard, is used to develop logic programs. It supports IEC 1131-3 programming model that includes Configuration, Resources, Task, Programs and Functions. It also supports the Functional Block Diagram, Ladder Diagram and Structured Text. New function blocks and libraries can be developed on top of the standard function block.

All I/O points within the DF1725IED, including virtual I/O data and I/O data from satellite RTUs or IEDs, are made accessible to the programming tool to be used in the programs.

The MCU is capable of storing and executing multiple algorithms, which might be event-driven or periodic at selectable periods and priorities.

Algorithms residing in the DF1725IED have access to the internal RTU clock for controlling time sensitive operations.

The memory area in which the programmable logic algorithm executes is protected. This feature protects the memory area of the DF1725IED, preventing any malfunction or system deadlocks and synchronization problems.

The memory area executing the PLC function is protected. This feature will protect the memory area of the RTU, preventing any malfunction or system deadlocks and synchronization problems.

The programming tool has the facility to diagnose, debug and simulate programming logic programs.

2.1.5 Other Features and Functions

2.1.5.1 Technical features

The unit integrates the state-of-the-art technology in monitoring and control systems to high voltage and extra-high voltage substations. It includes the following features:

Based on an advanced industry compatible chip, electrical isolation and electromagnetic shielding both conform to the corresponding international standards. Its hardware is extremely stable. The PCB board design is simplified by using EPLD and FLEX (Electrically-erasable Complex Programmable Logic Device).

The main chips and boards adopt SMT and four-layer PCB

Equipped with a real time clock with full calendar, inclusive of leap year support. Supported by a dedicated battery backup, time and date can be maintained for at least a year when there is no power supply. The internal time accuracy is 1.2 PPM and will not drift more than 5 ms per hour without GPS synchronization via IRIG-B code.

The clock resolution is 1ms. Software algorithms are employed to counter inaccuracies and drift resulting from crystal ageing or other factor like temperature, etc.

Featuring a large, high-visibility, LCD display for text and graphic with long-life span. It has stainless steel membrane keypads to access all measured parameters and status information, as well as for programming functions that provide a unique and flexible user interface.

DF1725IED can be managed through maintenance software locally or remotely including downloading and uploading parameters, etc.

2.1.5.2 Application features

There is a Main SOE buffer in MCU. This Main SOE buffer collects events from all the digital input modules and subsystem within the MCU. The DF1725IED uses this Main SOE buffer to report SOE data to the Master station. Event is stored in the chronological order in which they occur. When the cyclic buffer becomes full of entries (that have not been reported to the Host), the event archiving ceases and the DF1725IED notes, “it is now ‘losing’ events”. An alarm indication will be activated whenever the DF1725IED SOE data buffer has overflowed. Events that have been reported to the Host can be overwritten by new events. The size of the Main SOE buffer is configurable and has enough memory to support archiving of at least ten times the number of installed digital input points.

The MCU can be provided with dual hot backup automatic switching. There may be two MCU in one DF1725IED Subrack, with one configured as main machine while the other as standby. The main and standby machines monitor each other. When there is a failure on the main machine, The main MCU switches to the

standby machine automatically. The original main machine will now assume the position of the standby machine. Both MCU share the ports of COM-6 boards.

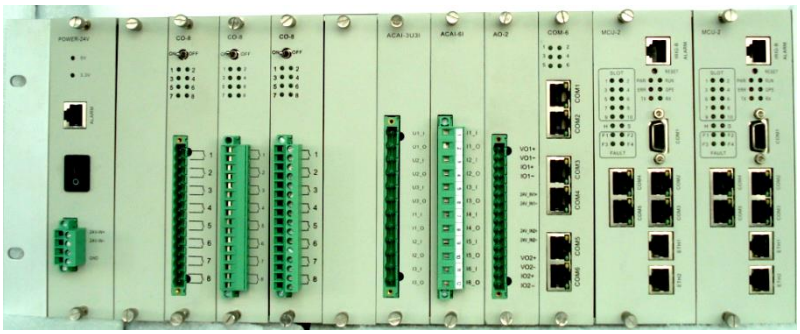


Fig2.2 Dual MCU for one SubRack

In the event that only one MCU is used, it can communicate with the Master Station through two communication channels in redundant mode.
MCU can provide access to alarm status information via maintenance port.

2.2 DCAI-8

2.2.1 Circuit Description

There are eight ADCs for 8 channels. One ADC is used for reference power. There are some isolators between the digital signal of each ADC and CPLD .

At any one time, there will be one channel signal sent to MCU from ADC.

Voltage Sampling
Voltage signal is inputted to the ADC through connector, TVS, magnet and capacitor. The signal is divided via resistance. Then the voltage signal is converted to digital signal by ADC CS5460A and this digital signal would be sent to MCU from via CPLD.

Current Sampling
Via the pin 1 and 2 of connector, current signal is sent to the ADC and through sampling resistance, TVS, magnet and capacitor, the signal voltage is divided via resistance. Then the signal is converted to digital signal by ADC CS5460A and this digital signal would be sent to MCU via CPLD.

2.2.2 Other Features and Functions

The DCAI can be configured to accept analogue inputs in the range of ±10 mA, ±20 mA, 4 - 12 - 20 mA, 4 - 20 mA, ±5V and ±10V.

A jumper setting is provided to select, whether to include Precision Resistor for current input or to exclude Precision Resistor for voltage input.

In case of current input, the same jumper setting can be used to include only one

Precision Resistor (410 ohm total resistance) for current up to 20mA By using the maintenance software, the range of current can be further modified between the 10mA range and 20mA range.

When the current input is more than 10mA (or 20mA) or voltage input is more than 5V, the exceeded voltage or current value will be reported accordingly up to 20mA in case of current and 10V in case of voltage.

The DCAI module is equipped with an integrity check and auto-calibration feature.

For current inputs, the input impedance is such that the voltage across the input terminals does not exceed 200mV (with full-scale input current) and no damage occurring for sustained 100% over current. For voltage inputs, the input impedance is infinite.

The minimum and maximum value, average value and rate of change value can all be got. All these data will be stored in flash memory of MCU. All these data can be sent to Master Station as well.

The dead band for each analog input point can be configured in DCAI-8. The dead band is configured in the range of 0 to 100% of full scale. This dead band can be configured via maintenance software or downloadable from Master Station.

The Force To Zero value can be configured for each analog input point in a way such that the DCAI-8 reports a zero value for any readings within zero and the Force to Zero value. The Force To Zero parameter is configurable from 0 to 100% of full scale. This parameter can be configured via maintenance software or downloadable from Master Station.

There are a Low Alarm parameter and High Alarm parameter for each channel. Transgression of the alarm limits will generate an alarm/event that can be reported to the Master Station. These parameters can be configured via maintenance software or downloaded from Master Station with range from 0 to 200% of full scale.

Each analogue input point in DCAI-8 can manually disable by using maintenance software.

2.3 ACAI-3U3I/6I

2.3.1 Circuit Description

Sampling Part

The AC analog input section of the 1725IED uses one monolithic ADC converters (ADS8364). The ADS8364 includes six 16-bit, 250KHz ADCs (Analog-to-Digital

converters) with 6 fully differential input channels grouped into two pairs for high-speed simultaneous signal acquisition.

The control of the ADC (ADS8364) is provided by a MCU which calculates the parameters for each input, once every cycle.

Adopting precision components to ensure the stability of long-time running.

Sampling each AC input at high speed and analyzing its harmonic.

The small AC signals come from the mini-transformers on the ACAI boards. They are isolated and magnified by the input process circuit. The discrete sample digital codes are obtained after passing through the ADC converter. The MCU calculates these data in some algorithms like Fast Fourier Transformer to get the various AC circuit parameters like current, voltage, active power, reactive power, power factor, frequency, etc.

Each ACAI can fulfill the sampling and calculation of six AC circuits Ia, Ib, Ic, Ua, Ub, Uc, Uab, Ucb, total P, total Q, power factor, KWH, KvarH.

There are two kinds of ACAI cards: ACAI-3U3I and ACAI-6I.

2.3.2 ACAI-3U3I

Three mini VTs (voltage transformer) on ACAI-3U3I are used for the insulation and conversion of the input AC 0~150V voltage signals to secondary 2.5V small signals. The secondary voltage signal flows into the ADC.

Three mini CTs (current transformer) on ACAI-3U3I are used for the insulation and conversion of the input 5A/1A current signals to secondary 2.5V small signals. The secondary signal flows into the ADC.

2.3.3 ACAI-6I

Six mini CTs (current transformer) on ACAI-6I are used for the insulation and conversion of the input 5A/1A current signals to secondary 2.5V small signals. The secondary signal flows into the ADC

2.4 DI-16-220/110/48/24V

2.4.1 Application

The DI-16 has 16 digital inputs in two groups of eight. Each group has one separate negative common terminal. The digital inputs can use 220V, 110V, 24V, or 48V as wetting voltage. Using the Maintenance Software, each input can be configured as single-bit input, double-bit input, Binary Coded Decimal (BCD) input or counter input. All double-bit and single-bit input can be assigned as SOE. Counter input can be configured as Form A, Form B or Form C inputs. BCD can be configured from 1 decade to 8 decade (0000 0000 - 9999 9999) without sign

and decimal point.

2.4.2 Circuit Description of the Digital Inputs

2.4.3 Input Protection

For the first channel on DI-16 board, the capacitor minimizes high-frequency noise by providing a low impedance by-pass to the ground. Capacitor has a high di-electric rating which allows it to provide surge withstand capability (SWC) for the digital input. Resistor limits current passing through it while inductor reduces noise from coupling to the input line.

The common terminal for digital input channel 1 to 8 is at pins 17 and the common terminals for digital inputs 9 to 16 is at pins 18. Each 8 digital inputs share one common terminal.

2.4.4 Input Recognition

Digital input one is connected to the diode of opto-coupler. When the voltage applied on digital input one exceeds the required threshold at the diode of the auto-coupler, the output transistor of E1 turns on and the collector switches to 0 V DC.

When the voltage applied on digital input one fails to reach the threshold to turn on the output transistor of E1, resistor network pulls the collector to +5 V DC. The collector of E1 then connects to the first input of the tri-state buffer of the 74LCX245

2.4.5 Reading the Digital Inputs

When the enables of 74LCX245 becomes active (0 VDC), the states of the 8 digital inputs of group 1 can be read by the MCU over the data bus. Note that the state read by the MCU is the complement of the digital input point, high for OFF and low for ON. When the enables of 74LCX245 are inactive, the outputs of 74LCX245 are in a high impedance state. The state of the digital inputs is read every 1 ms by MCU.

2.4.6 Other Features and Functions

All digital input circuits are individually optically isolated. DI-16 uses varistor to protect each digital input from earth faults and incorrect wiring which introduce external voltages into the DF1725IED. In this way, supply to other modules will not be affected and damage to any of the DF1725 IED modules can be avoided.

All digital inputs are scanned by the MCU once every 1 millisecond. Each input is possible to invert its digital signal, on a per point basis, so that a normally closed contact might be reported as normally open contact.

The DI-16 provides debounce filter that can be configured by the user. The setting range is from 0 to 1000 ms in width for the pulses, and in a step of 1ms.

Any pulse longer than 3ms in width can be detected by MCU. However the default for debounce filter setting is 5 ms.

The DI-16 provides chatter filter that can be configured by the user. It is equipped with anti-chatter algorithm to reduce nuisance reporting of noisy or chattering plant contacts for chatter of a relay point. The DI-16 provides three configurable parameters for the relay chatter filter. The number of changes, Nc and the chatter filter period, Tc, are used to define its filtering function. These parameters are configured by the maintenance software system. If a point changes state more than Nc times within a period of Tc, the system will mark the point as chattering and disable it so that no further changes of state or events will be recorded until the point recovers from chattering. After chattering occurs, a lockout period that is Nl time of Tc starts. When the lockout period ends, the relay chatter filter can automatically restore monitoring of the point when it detects that it is no longer in a chattering state.

A double bit processing time parameter is provided. In this interval the detecting system waits when a status change occurs for a double bit point to ensure that double bit transition, or intermediate states that occur during the interval, is not reported. The transition states are “00” and “11”. Valid states are ‘01’ and ‘10’. When a double-bit point changes state, the detecting system waits for a time interval equal to the double-bit processing time. If the point changes state to a transition state and then to a final valid state during that interval, the digital input subsystem will not report the transition state. If the point fails to reach a valid state at the end of the interval, the subsystem will report the transition state as the final state. The double-bit processing time is a configurable parameter ranging from 0 to 255 seconds.

All single-bit or double-bit points are equipped to provide SOE time tagging in addition to the normal function of the DI points. It is possible to assign any DI data points in the DF1725IED as an SOE point.

When a change of state occurs, the event is identified with a point identification number, the type of event and the time of occurrence in hours, minutes, second, and milliseconds.

The Event Time Tagging Error is not more than 1 ms.

The separating capability of the DF1725IED is 1 ms for events occurring within a substation.

The time resolution of the DI-16 is 1 ms.

For the DI-16’s double-bit points configured as SOE points, the detecting system time-tag the event recorded at the time of the event in the following manner:

If the point changes state to a transition state and then to a final valid state during that interval, the detecting system will time-tag the event at the time that the point changes state to a valid stat.

If the point fails to reach a valid state and remains in the transition state until the end of the time interval, the detecting system will then time-tag the event at the time when the point initially changes state to the transition state.

32-bit counters are provided which will automatically reset and count from 0 again when they are full (Automatic roll-over).

Each counter is able to accumulate any input occurring while the counter is being read so that no input counts will be lost.

Each channel can be configured as single-bit input (or double-bit input if we use two channels) and counter input at the same time so that the required number of plant contacts and D/I channel can be reduced.

The DI-16 is capable of responding to FREEZE commands generated internally within the 1725IED at a regular time (through maintenance software system) intervals. The FREEZE command will copy the content of the counter to a buffer and send this value to Master station at the same time. When the next FREEZE command is received, the content of the counter will be copied to the same buffer and override the previous content of the buffer, sending the new content to Master Station at the same time.

The DI-16 is capable of responding to 'RESET' commands generated internally within the 1725IED at a regular time. The RESET command effectively zeroes the counter.

And the DI-16 is capable of responding to FREEZE & RESET commands generated internally within the DF1725IED at a regular interval. In this case the counter will be reset once it has been frozen and reported. The time stamp at the time of "freeze" is recorded.

The time period is user configurable in minutes on a channel-by-channel basis in the range from 1-255 minutes (default 30 minutes).

2.5 CO-8

2.5.1 Control Output Process

The control output section operates the control outputs of the DF1725 IED. The operation of each control point follows an operational sequence:

- Select a point.
- Check the points selected.
- Execute the point and Check the operation.

Select a Point

When the MCU receives a "select" order from the host station, the MCU will write a byte of data onto the data bus with the selected address for the point to be selected. The MCU then strobes enable signal to a low state and subsequently brings them back to a high state. On the rising edge, the latching devices

74AC273 latch the data on the data bus. Then CPLD receives data from the micro-controller to choose these enable signals. Thus one of the 8 object relays will be selected via these driver monolithics and activate.

Check the Point Selected

These 74LCX245 read the states of the selected relay contacts. For a point that is selected, the corresponding output of open collector driver signals is low. The MCU can read the states of the relay contacts at any time by setting these enable signals low. When these signals are low, the buffers of these driver monolithics are enabled, placing the relay point signal onto the data bus to be read by the MCU. The micro-controller will check these signals and return a correct feedback signal to the host station. Otherwise MCU will cancel this control operation and MCU will return a false feedback signal to the host station.

Executing an Selected Point

After the feedback information is sent to the host station and the MCU received an executing order from the host station in defined time (the time can be set from 0 to 255 seconds), the CPLD writes a group of pulses to the negative voltage circuit. This will cause a negative volt, turning on the diode of photo-coupler. The MC1413 will supply +5volt to the relay, turning it on. The micro-controller writes a byte of data onto the data bus via these 74AC273 and the driver monolithics. These relays are then turned on.

These eight relays make up a series of executing relays. The micro-controller checks the operation and sends the operational information to the host station.

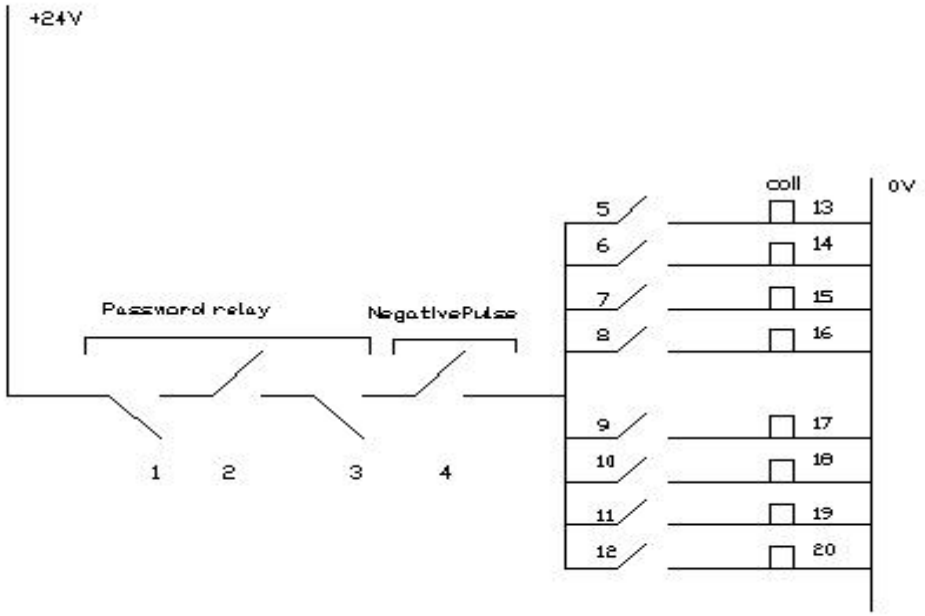


Figure9.1 Digital output relay combination

2.5.2 Direct Operate

The MCU can directly operate a point on CO-8. After the MCU receives the direct operate order from the host station, it writes a byte of data onto the data bus. Via these driver monolthics this will turn on the object relay and executing relay. Subsequently the MCU checks the operation and send the operational information to the host station.

2.5.3 Control Security

To summarize the above and further explain the security measure provided by MCU, a process starting from the receiving of command from the host station till output execution is described as below.

Select Before Operate (SBO) operation:
Upon receiving “select” command from the host station, the corresponding object relay will activate (object relay are in series and they are selected based on per channel basis). Subsequently the MCU will check the closing state of this and other relays. A signal is feedback to the host station if everything is ok. If the “execute” signal is not received from the host station within a time period (configurable from 1-255s), the process will be reset and the relays selected will be de-activated. If the “execute” signal is received within that time period, 1 normally open and 2 normally close executing relays, that are connected in series with the object relay, will be activated. The use of 1 NO and 2 NC executing relays is to ensure the circuit will remain open in case of any disturbance that can cause unintentional activation of the relays. There is another NO executing relay connected in series with the above-mentioned 4 relays. This relay will get its negative supply through the generation of pulses by MCU upon receiving the ‘execute’ command.

Only when all the above-mentioned 5 relays are activated can the output be sent out and hence ensuring that no false output will result in any case.

That is,There are 6 relays related with the control output:3 password relays,1 negtive voltage relay,1 object relay and 1 final relay which can drive the breaker.

Direct Operate Operation:
Once the MCU receives a direct operate command from the host station, the object relay and all executing relays will close at the same time to provide output.

Only one control point in a CO-8 can be selected or executed at any given time. In the event MCU receives a “select”(or direct operate) command from the second host station after it receives “select” command from the first host station, the MCU will send a signal to the second the host station to reject the command.

Both SBO and Direct operations
The DF1725IED can support “Long Pulse” and “Short Pulse” commands as stipulated in IEC101 protocol.

2.5.4 Contacts of the Control Outputs

Each Channel provides a form A contact output. Each contact switches up to 8A at 30V DC or 0.8A at 250V AC.

Each of the eight channels is protected against transients in a similar manner. The protection of point one is described here.

To provide surge withstand capability protection for GND terminal, the capacitor provides a low impedance path to the ground at high frequencies. To provide immunity to impulse transients, metal oxide varistors clamp high voltage transients to the ground. Inductor prevents transient volts from being coupled to the input line, lowering the conducted emissions.

2.5.5 Other Features and Functions

- Ensure single point operation at any given time.
- Ensure no mal-operation or false output occurrence in any case.
- Provide both Select Before Operate (SBO) operation and Direct Operate operation. All channels in the same CO-8 module can be freely configured to one of the two operating modes.
- All control outputs (Secure and Direct-Operate) are equipped with individual LEDs to confirm the status of each control relays.
- The control output can be disabled in specific condition.

2.6 AO-2

2.6.1 Circuit Description for Analog Output

On AO-2 board, each analog output channel consists of two D/A converters (AD5551), one A/D converter (AD7894) and a multiplexer (AD1334). These two monolithic D/A converters output voltages to the Multiplexer. One of the D/A converters gives output-to-output circuit through the Multiplexer based on the last command received. Upon receiving new “select” command from Master Station, the encoded value will be sent to second the D/A converter, causing it to send output to the Multiplexer based on the values specified in the command. The Multiplexer will switch this signal to the A/D converter and eventually send to the MCU as a feedback signal. After the outputs of the converters stabilized (less than 10us) and MCU verified that the value of this feedback signal is the same as the value in the command, a “correct” signal will be sent back to the host station. Upon receiving “operate” command from the host station, the Multiplexer will switch the output of the second D/A converter to outside equipment. The output of first D/A converter will then be connected to the feedback circuit.

For operation that does not require select before operate procedure, once the encoded value is transferred to the D/A converter that is connected to the feedback circuit, the Multiplexer will switch to its output-to-output circuit.

Because the D/A's settling time is less than 10ms, the output circuit can stabilize within 10ms and hence the total settling time is much less than 100 ms.

DC 24V supply must be connected to each channel of AO-2 through its terminal board.

The Figure below shows the above-described operation.

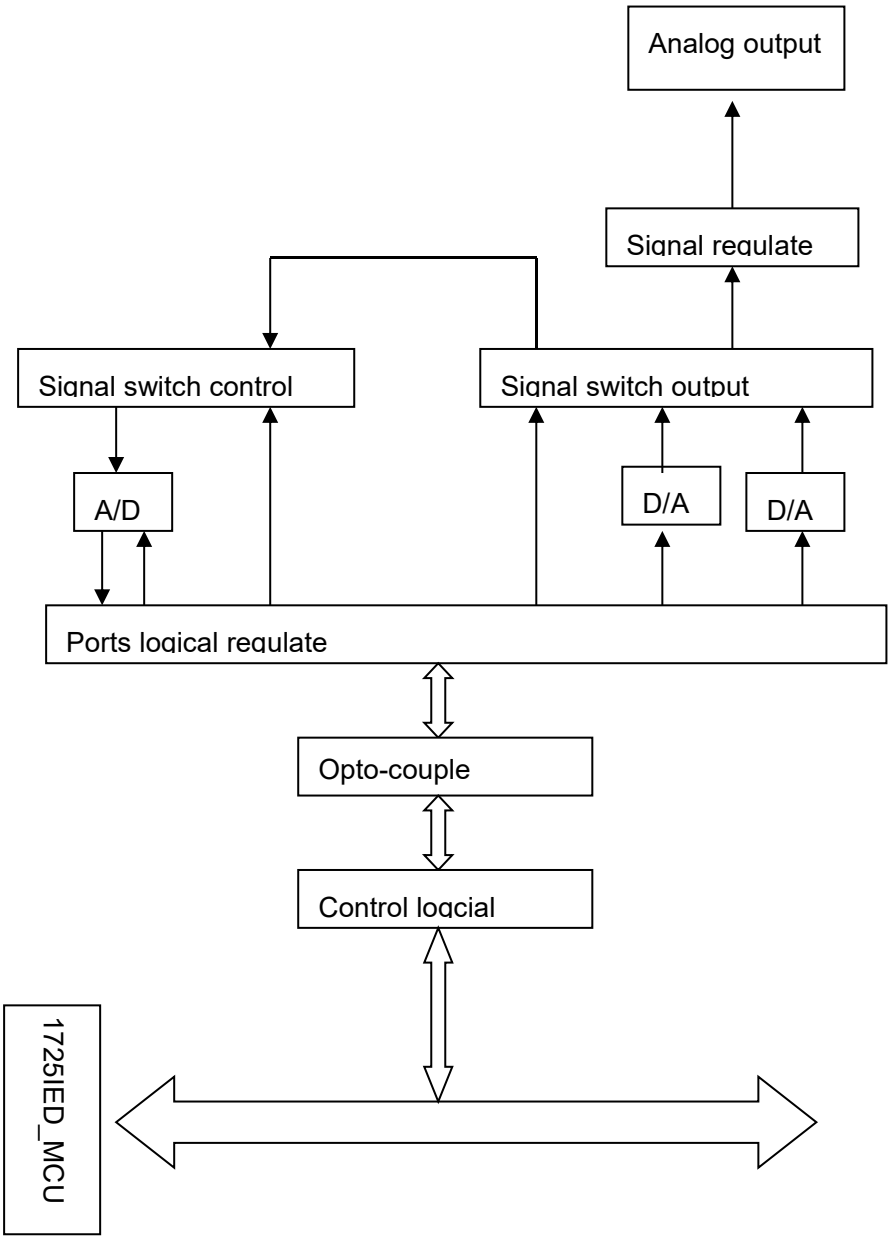


Fig 7.1 Block Diagram of AO-2

2.6.2 Other Features and Functions

Each analog output can be configured as voltage output (0-5V, 0-10V) and current output (0-1mA,0-10mA, 0-20mA, 4 -20mA) using jumper setting and maintenance software.

All analog outputs are equipped with optical isolation to protect against ground loops and power surges.

2.7 GPS

The GPS card handles time & date. Satellite signal will be received via antenna. GPS card can synchronize the time of DF1725IED device and send time synchronization message to other devices.



Specifications

4(four) IRIG-B output channels

1(one) PPS output

1(one) RJ45 port for SNTP

LED indicator for running status and satellite signal

BNC interface for antenna

Alarm potential free contact for losing satellite signal.

3. Definition of Interface Ports

3.1 MCU

To install the MCU, please do some basic wiring on each board and connect them to your master station and other devices.

Note: The wiring method you use must comply with all international and local laws and regulations.

To connect the MCU, please:

- Connect the serial ports
- Connect the Ethernet ports (optional)

Pins Definition of Interface Ports

(1) COM1

Terminal Type	RS232 DB9 connector (female)					
Pin No.	2	3	5	7	8	Others
Pin definition	RXD	TXD	GND	RTS	CTS	No use

(2) COM2

Terminal Type	RJ45			
Pin No.	3	4	5	others
Pin definition	GND	TXD	RXD	No use

(3) COM3

Terminal Type	RJ45			
Pin No.	3	4	5	others
Pin definition	GND	TXD	RXD	No use

(3) COM4(RS232/RS485)

Terminal Type	RJ45					
Pin No.	1	2	3	4	5	Others
Pin definition	485-A	485-B	GND	TXD	RXD	No use

(4) COM5(RS232/Rs485)

Terminal Type	RJ45					
Pin No.	1	2	3	4	5	Others
Pin definition	485-A	485-B	GND	TXD	RXD	No use

(5) IRIG_B(Alarm)

Terminal Type	RJ45				
Pin No.	1	2	7	8	Others
RS232	IRIG_B(+)	IRIG_B(-)	ALARM(+)	ALARM(-)	No use

(6) ETH1

Terminal Type	RJ45				
Pin No.	1	2	3	6	Others
Pin definition	TX(+)	TX(-)	RX(+)	RX(-)	No use

(7) ETH2

Terminal Type	RJ45				
Pin No.	1	2	3	6	Others
Pin definition	TX(+)	TX(-)	RX(+)	RX(-)	No use

(8) ETH3

Terminal Type	RJ45				
Pin No.	1	2	3	6	Others
Pin definition	TX(+)	TX(-)	RX(+)	RX(-)	No use

Serial COM1

- The COM1 is asynchronous RS232 serial ports using a DB9/F female connector.
- Baud rate: 100~115200 bps for the maintenance equipment

Serial COM2

- The serial port COM2 is asynchronous RS232 port using a RJ45 connector.
- Baud rate: 100~115200 bps

Serial COM3

- The serial port COM3 is asynchronous RS232 port using a RJ45 connector.
- Baud rate: 100~115200 bps

Serial COM4

- The serial port COM4 is asynchronous RS232/RS485 port using a RJ45 connector.
- Baud rate: 100~115200 bps

Serial COM5

- The serial port COM5 is asynchronous RS232/485 port using a RJ45 connector.
- Baud rate: 100~115200 bps

Ethernet Ports

- The MCU has three 10BASE-T Ethernet ports featuring an RJ45-8 connector.
- Pins layout: 1, TX+; 2, TX-; 3, RX+; 6, RX-
- Baud rate: 10M bps

MCU-1 Jumper Description:

X1: set IRIG-B's termination resistor mode

XS2: set COM4's termination resistor mode

XS3: set COM5's termination resistor mode

3.2 ACAI-3U3I

Pins Definition of Interfaces

Definition for terminal of 1725IED_ACAI-3U3I.

1	2	3	4	5	6	7	8	9	10	11	12
Ua	Un	Ub	Un	Uc	Un	I1_i	I1_o	I2_i	I2_o	I3_i	I3_o

Jumpers Setting

Setting the board address of ACAI-3U3I:

Set the ACAI-3U3I address by placing jumpers on the three jumper blocks. The

jumper blocks represent a 3-bit binary address.

Set the address by placing the jumpers provided over the appropriate pins on jumper blocks. See Table below for the binary weighting of each jumper position.

No jumper represents binary 0.
Jumper on represents binary 1.
Jumpers setting:

Jumper Position	Binary Address
X1	1
X2	2
X3	4

X4-X6:ON means AC220V input
OFF means AC100V input
X7-X12:Factory use

3.3 ACAI-6I/6U

Pins Definition of Interfaces

Definition for terminal of ACAI-6I.

1	2	3	4	5	6	7	8	9	10	11	12
I1_i	I1_o	I2_i	I2_o	I3_i	I3_o	I4_i	I4_o	I5_i	I5_o	I6_i	I6_o

Jumpers Setting

Setting the board address of ACAI-6I:

Set the ACAI-6I address by placing jumpers on the three jumper blocks. The jumper blocks represent a 3-bit binary address.

Set the address by placing the jumpers provided over the appropriate pins on jumper blocks. See Table below for the binary weighting of each jumper position.

No jumper represents binary 0.
Jumper on represents binary 1.
Jumpers setting:

Jumper Position	Binary Address
X1	1
X2	2
X3	4

3.4 DCAI-8

Pins Definition of Interfaces

(1) Definition for terminal of DCAI-8

1	2	3	4	5	6	7	8
AI1+	AI1-	AI2+	AI2-	AI3+	AI3-	AI4+	AI4-
9	10	11	12	13	14	15	16
AI5+	AI5-	AI6+	AI6-	AI7+	AI7-	AI8+	AI 8-

Jumpers Setting

Setting the board address DCAI-8

Set the DCAI-8 address by placing jumpers on the three jumper blocks. The jumper blocks represent a 3-bit binary address.

Set the address by placing the jumpers provided over the appropriate pins on jumper blocks. See Table below for the binary weighting of each jumper position.

No jumper represents binary 0.

Jumper on represents binary 1.

Jumpers setting:

Jumper Position	Binary Address
X1	1
X2	2
X3	4

Voltage/Current select:

Jumpers	U/I Select
X5-X8 ,X10 X11 X13 X15 X16 X`19 X20 X22 X23 X25 X26 X28-X30,X32 X33 X35 X37 X38 X40 X41 X43 X44 X46-X50	Current:1,2 short Voltage: 2,3 short
X4 X9 X14 X21 X27 X31 X36 X42	Current:2,3 short Voltage:1,2 short

3.5 DI-16-220/110/48/24V

Pins Definition

The DI-16-220/110/48/24V has 16 digital inputs in a group of sixteen (two groups of eight). These inputs are connected directly to DI board.

Note: All digital inputs are negative (-) common. Each group of eight inputs has a separate common.

Apply the appropriate voltage (220VDC,110VDC,24VDC or 48VDC) to the digital inputs.

1) Wire the positive (+) side of each digital input to the terminal according to table as shown in figures as below:

2) Wire the common (-) of digital inputs 1 to 8 to terminal.

3) Wire the common (-) of digital inputs 9 to 16 to terminal.

4) Jumper X4 and X5 is between the common of 1 to 8 and the common of 9 to 16.

X4&X5: to connect or disconnect both common grounds of the first 8 and last 8 digital inputs.

- Position ON: to connect both common grounds of the first 8 and last 8 digital inputs.
- Position OFF: to disconnect both common grounds of the first 8 and last 8 digital inputs.

Defintion for terminal of DI-16-220/110/48/24V:

Digital input	Terminal
Digital 1	1
Digital 2	2
Digital 3	3
Digital 4	4
Digital 5	5
Digital 6	6
Digital 7	7
Digital 8	8
Digital 9	9
Digital 10	10
Digital 11	11
Digital 12	12
Digital 13	13
Digital 14	14
Digital 15	15
Digital 16	16
COM	17
COM	18

Jumpers Setting

Setting the board address DI-16-220/110/48/24V

Set the DI-16-220/110/48/24V address by placing jumpers on the three jumper blocks. The jumper blocks represent a 3-bit binary address.

Set the address by placing the jumpers provided over the appropriate pins on

jumper blocks. See Table below for the binary weighting of each jumper position.
No jumper represents binary 0.
Jumper on represents binary 1.
Jumpers setting:

Jumper Position	Binary Address
1	1
2	2
3	4

X4&X5: to connect or disconnect both common grounds of the first 8 and last 8 digital inputs.

- Position ON: to connect both common grounds of the first 8 and last 8 digital inputs.
- Position OFF: to disconnect both common grounds of the first 8 and last 8 digital inputs.

3.6 CO-8

Pins Definition of Interfaces

The CO-8 has four relay control outputs with a Form A contact for each relay.
Wire the control outputs to your control device using a Form A.
Note: The contact rating is 8 amps at 30V DC.

Definition doe terminal of CO-8

1	2	3	4	5	6	7	8
CO1		CO2		CO3		CO4	
9	10	**11**	12	13	14	15	16
CO5		CO6		CO7		CO8	

Jumpers Setting

Setting the board address of CO-8:

Set the CO-8 address by placing jumpers on the three jumper blocks. The jumper blocks represent a 3-bit binary address.

Set the address by placing the jumpers provided over the appropriate pins on jumper blocks. See Table below for the binary weighting of each jumper position.

No jumper represents binary 0.
Jumper on represents binary 1.
Jumpers setting:

Jumper Position	Binary Address
1	1
2	2
3	4

3.7 AO-2

The AO-2 has two analog outputs. This output connects to terminal XS1.

Pins Definition

1	2	3	4	5	6	7	8
VO1+	VO1-	IO1+	IO1-	NC	24V1+	24V1-	NC
9	10	11	12	13	14	15	16
NC-	24V2+	24V2-	NC	VO+	VO-	IO+	IO-

Jumpers Setting

Setting the board address of 1725IED_AO-2:

Set the AO-2 address by placing jumpers on the three jumper blocks. The jumper blocks represent a 3-bit binary address.

Set the address by placing the jumpers provided over the appropriate pins on jumper blocks. See Table below for the binary weighting of each jumper position.

No jumper represents binary 0.

Jumper on represents binary 1.

Jumpers setting:

Jumper Position	Binary Address
1	1
2	2
3	4

Setting the voltage/current and the current range:

Output selection for AO-2:

Jumper	Output select
X4, X7	0-20mA,0-20mA,0-10mA,0-5mA:2,3 short 0-10V,0-5V:1，2 short 0-1mA:2,3 short
X6, X8	0-20mA,0-20mA,0-10mA,0-5mA： 1， 2 short 0-10V,0-5V:2,3 short 0-1mA:1,2 short

X5,X10	0-20mA,0-20mA,0-10mA,0-5mA:1,2 short 0-10V,0-5V:1,2 short 0-1mA:2,3 short

3.8 COM-6 COM port expansion board

- COM1 RS232,FULL signal
- COM2,COM3,COM4 RS232
- COM5, COM6 RS232/RS485
- Baud rate: 300~38400 bps

Pins Definition

COM1:

Type	RJ45							
Pin No.	1	2	3	4	5	6	7	8
Name	DSR	RTS	GND	TXD	RXD	DCD	CTS	DTR

COM2:

Type	RJ45							
Pin No.	1	2	3	4	5	6	7	8
Name	NC	RTS	GND	TXD	RXD	NC	CTS	NC

COM3-COM4:

Type	RJ45							
Pin No.	1	2	3	4	5	6	7	8
Name	NC	NC	GND	TXD	RXD	NC	NC	NC

COM5-COM6:

Type	RJ45							
Pin No.	1	2	3	4	5	6	7	8
Name	RS485-A	RS485-B	GND	TXD	RXD	NC	NC	NC

Jumpers Setting

X1, X2, X3 Setting the board address COM-6:

Set the COM-6 address by placing jumpers on the three jumper blocks. The jumper blocks represent a 3-bit binary address.

Set the address by placing the jumpers provided over the appropriate pins on jumper blocks. See Table below for the binary weighting of each jumper position.

No jumper represents binary 0.

Jumper on represents binary 1.

Jumpers setting:

Jumper Position	**Binary Address**

X1	1
X2	2
X3	4

COM-6 Jumper Description:
X4: set COM5's termination resistor mode
X5: set COM6's termination resistor mode

4. Maintenance software instruction

4.1 General

4.1.1 Product introduction

4.1.1.1 Function introduction

The engineer tool software may help you use the RTU/IED system of DONGFANG conveniently. You may maintain the main processor and all I/O modules by serial port, Ethernet, etc.

Station maintenance

●Add and delete station

●Select the station to be maintained

Processor maintenance

●Maintain database parameters

●Edit and check files in the main communicating processor

●Check real time data

●Check operation status

Module maintenance

●Modify the parameters of module database

●Check real time data

●Do control output, analog output and BCD code output operation

Channels include the following types: Ethernet and serial port, which can be used to connect all input and output function modules. Besides, there are virtual channels, which can't be used in communication and only provide entrance for some special software

arithmetic.

There are three kinds of modules, which are physical module, user module and logic module and can be used to deal with various kinds of tasks agilely.

Physical module refers to the input and output module existed actually, such as the input and output module of Dongfang RTU DF1725IED series.

User module is used to store the real time data created by logic and extends the real time base created by physical module. User module usually stores the RTU or IED data of slave stations connected with RTU main processor, historical data, arithmetic data, etc.

Physical module and user module are all real modules.

Logic module is used to select the data of real module and reflect these data to the logic module database. Logic module is usually used to reflect these data to the protocol database communicated with the main station or to reflect from the real base of some special protocols under virtual port.

4.1.1.2 Term and abbreviation

The following terms are adopted in this manual:

DF1725IED RTU

This product series include DF1725IED main communicating module, etc.

Main communicating processor

Refers to RTU/IED of Dongfang main communicating module, such as 1725IED_MCU, etc. and hereinafter refers to main processor.

Download

Transmit parameter files from upper machine to processor or I/O module.

Upload

Transmit parameter files from processor or I/O module to upper machine.

Physical module (or I/O module)

Physical module refers to the modules connected with the processor by internal bus. These modules collect real time data or carry out output operation, such as 1725IED_6I，1725IED_AC， 1725IED_AC_2， 1725IED_AC_3， 1725IED_AO， 1725IED_CO，1725IED_DA，1725IED_DC，1725IED_DI etc.

User module

User module is used to store the real time data created by logic and extends the real time base created by physical module. User module usually stores the RTU or IED data of

slave stations connected with RTU processor, historical data, arithmetic data, etc.

Real module

Physical module and user module are all real modules.

Logical module

Logic module is used to select the data from real module and reflect these data to the logic module database. Logic module is usually used to reflect these data to the protocol database communicated with the main station or to reflect from the real base of some special protocols under virtual port.

4.1.1.3 Version

The version of DFE Engineer Tool software is 1.01.001

4.1.2 Installation and operation

4.1.2.1 Operation environment

The computer configuration required for the software is as follows:

● Intel X86 PC compatible computer (above Pentium III)

● 256 MB system memory

● 1 G HD

●LCD or SVGA display, 800×600×256

●Ethernet card (if communication with the main processor by Ethernet is required)

●One RS232 serial port (a converter from USB to RS232 is required if only has USB port)

●Internal or external mouse

●Microsoft Windows 2000/xp/win7

4.1.2.2 Installation

For the first installation of this software, the default installation directory is D:\DFETS. This directory can be modified if required.

The installation is very easy. Just double click DFETSSetup.exe and then do as the indication on screen.

Installation files

Installation procedure automatically copies the following files to the installment directory. (See List 1-1)

List 1-1: Installment files

Path	File Name	Description
..\bin	tableinfo.xml	List box setting

	combodic.xml	Bay level dictionary
	prtclset.xml	Protocol module setting
	model.xml	model for 61850
	cli_logcfg.xml	config file of 61850 client
	cli_osicfg.xml	config file of 61850 client
	logcfg.xml	config file of 61850
	osicfg.xml	config file of 61850
..\dll	qpmgr.exe	Executable program
	DataSetFilter.cfg	Config file of 61850 dataset
	*.dll	Dynamic link library
..\templet\DF1725_IED	1725IED_CO.MLT	DF1725 Series CO module
	1725IED_CO_16.MLT	DF1725 Series CO module
	1725IED_6I.MLT	DF1725 Series ACAI module
	1725IED_AC_2.MLT	DF1725 Series ACAI module
	1725IED_AC_3.MLT	DF1725 Series ACAI module
	1725IED_AC_61850.MLT	DF1725 Series ACAI module
	1725IED_AO.MLT	DF1725 Series AO module
	1725IED_BSCO.MLT	DF1725 Series CO module
	1725IED_DA.MLT	DF1725 Series DA module
	1725IED_DC.MLT	DF1725 Series DCAI module
	1725IED_DI.MLT	DF1725 Series DI module
	1725IED_DI_32.MLT	DF1725 Series DI module
..\templet\Other_Device	61850Spare.MLT	1725IED 61850 module
	SelfCheck.MLT	1725IED selfcheck module
	DF7120_S.MLT	DF7120S module
	E0320J.MLT	E0320J module
..\uif\i18n	language	Language choose
	qpmgr_en.qm	Language support

	qpmgr_zh.qm	Language support
..\uif\ini	qpmgr_index	Index config
	qpmgr_table_set	Name table config file
	qpmgr_table_style	Style config file
..\baypara		Store all substation parameter files

4.1.2.3 Uninstall software

- Open the Windows control panel
- Run add/delete procedure
- Select Parameter Manager Tool
- Click add/delete button to uninstall the procedure

4.1.3 Run Parameter Manager Tool

After installing Parameter Manager Tool, the icon will be displayed in the desktop. Run the software, the main window is as follows:

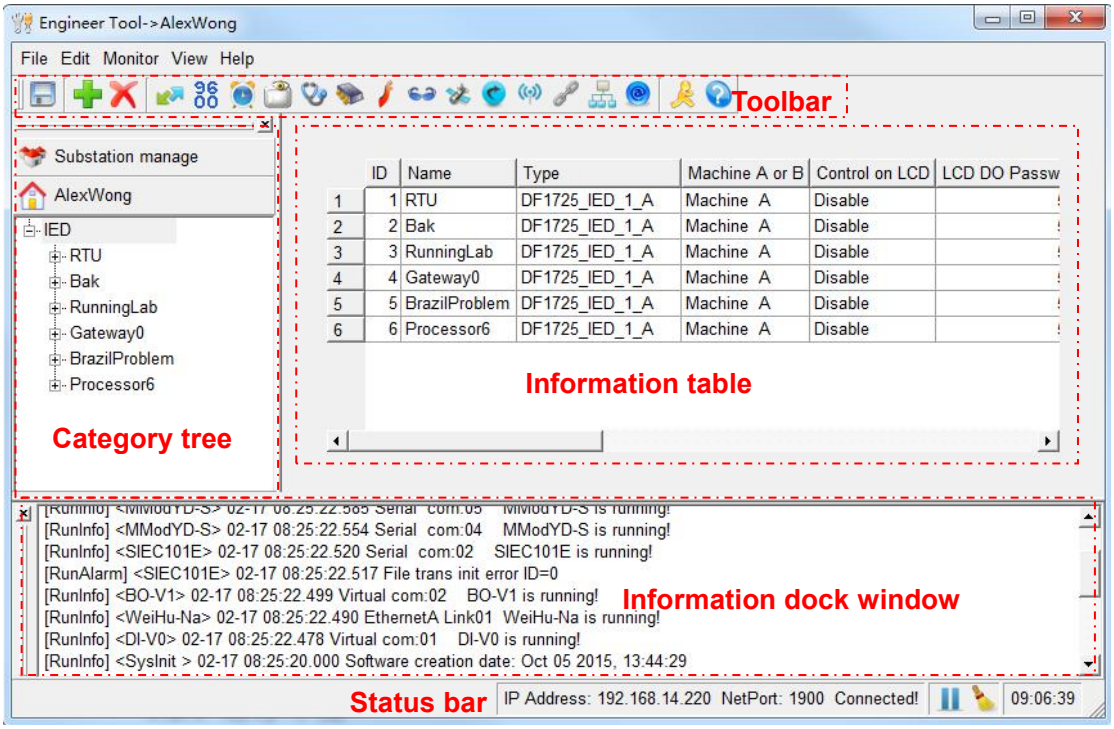


Fig.1-1 Parameter Manager Tool main window

Tool bar & Status bar

Tool bar is in the top of the window and includes main function options of the maintenance software. Besides, click the list item by the right button and the corresponding slave menu will pop up. Then carry out specific function.

For convenient usage, there are some buttons in toolbar to realize some functions easily. They are:

List 1-2: Buttons

Save	Add a record	Delete a record
Communication test	Baud rate change	Module clock
Real time data	Diagnose information	Device memory
Flash program	Watch RT frames	GPS setting
Reset	Search device	Link
FTP client	Exit	About
Pause	Continue	Clear

4.1.4 Maintenance steps

The following steps are usually required for configuring and testing system:

- ***Run*** DFE Engineer Tool Software
- ***Add a station*** and edit its name
- ***Add a processo***r and set its property
- ***Add channels*** under the processor and set their properties
- ***Add physical modules*** in the related channels and set their databases
- A***dd user modules*** in the related channels and set their databases
- ***Add logic modules*** in the related channel and set their databases
- ***Download*** all parameter files to the main processor module
- ***Rese***t main processor module
- ***Check*** the real time data of the module
- ***Test*** the output control of the module

Notice: The detailed configuration steps will be introduced in each function introduction.

4.2 Software Introduction

4.2.1 Substation management

Click the “substation management” in the right side of main interface to open the management list of substation.

	Code	Name	Running mode	Maintain this substation
1	a0	AlexWong	Bay level	YES
2	a1	India	Bay level	NO

Fig.2-1 Substation list

4.2.1.1 Add a substation

Click the icon  in the tool bar to add a substation.

4.2.1.2 Delete a substation

Click the icon  in the tool bar to delete the selected substation.

4.2.1.3 Modify the substation property

1) "Station code" is the identification ID of the appointed station and is the main key of station.
2) "Station name" is the substation name displayed in the main interface of appointed software.
3) "Running mode" is not used for DF1725RTU.
4) "Maintain this station" means whether this station is the substation to be maintain currently. If it is, this station will be displayed in the left side of the main interface, or it won't be displayed in the main interface.

4.2.2 Processor management

Select a station in the left list of the main interface and assume to be "AlexWong". Click "IED" and the system will load the database of this station and display all the processors and their contents of this station. The right list gives the detailed property of each processor.

	ID	Name	Type	Machine A or B	Control on LCD	LCD DO Password	Lcd admin password	Time delay(s)
1	1	RTU	DF1725_IED_1_A	Machine A	Disable	5555	5555	20
2	2	Bak	DF1725_IED_1_A	Machine A	Disable	5555	5555	20

Fig.2-2 Processor list

4.2.2.1 Add a processor

Click the icon  in the tool bar to add a processor.

4.2.2.2 Delete a processor

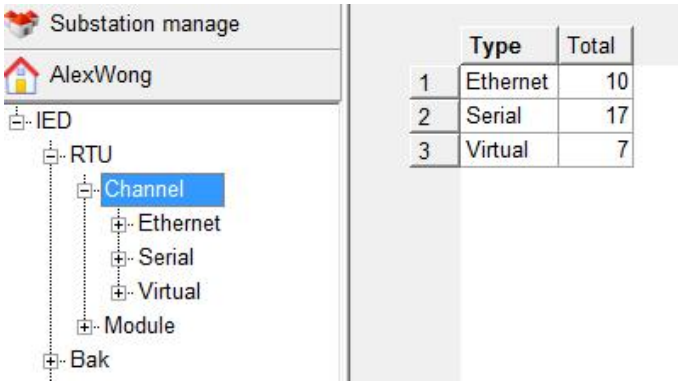
Click the iron  in the tool bar to delete the selected processor.

4.2.2.3 Modify the processor property

1) ***ID：*** Processor ID is the main key created by the system automatically and can't be modified.
2) ***Name：*** Processor name is the appointed subdirectory name of this processor in the directory of this substation and can be modified without repetition.
3) ***Type：*** Processor type is the type of Dongfang RTU main processor to be maintained by appointed maintenance tool. Selectable: Dongfang RTU processor type.

4.2.3 Channel management

Click “channel” and the types of all channels in the processor will be displayed and the types and amount of all channels will be displayed in the right side. Include Ethernet, serial port and virtual channel.



	Type	Total
1	Ethernet	10
2	Serial	17
3	Virtual	7

Fig.2-3 Channel list

4.2.3.1 Ethernet

Click “Ethernet” and all Ethernet cards and their properties will be displayed in the right list.

	Name	RTU IP Addr	Subnet mask	Scan interval(ms)	Use SNTP	SNTP mode	Transmit mode
1	EthernetA	10.240.0.4	255.255.255.224	30	Use SNT	SNTP Client	Auto transmit
2	EthernetB	10.240.0.226	255.255.255.224	30	Use SNT	SNTP Client	Auto transmit

Transmit mode	SNTP sync interval(s)	SNTP port addr	SNTP server IP
Auto transmit	64	123	0.0.0.0
Auto transmit	64	123	0.0.0.0

Fig.2-4 Property of Ethernet card

1) ***Name:*** Main processor supports 8 Ethernet cards.
2) ***RTU IP Addr:*** The IP address of RTU’s Ethernet card.
3) ***Subnet Mask:*** The Subnet Mask of RTU’s Ethernet card.
4) ***Scan interval (ms):*** Time interval of channel scanning. Input range: [0~65535]. Unit: [ms].
5) ***Use SNTP: S***elect whether using SNTP or not.
6) ***SNTP mode:*** SNTP Client or SNTP Server.
7) ***Transmit mode:*** Not used for SNTP Client. If select Auto transmit, That is to say, SNTP Server will send time automatically with time interval defined in “SNTP sync interval”. If Auto transmit has not been selected, That is to say, SNTP Server would not send time automatically, in this case, SNTP Server will send time only after receiving time synchronization from SNTP Client.

8) ***SNTP sync interval(s):*** *I*nput range [0~65535], unit [s]. If Auto transmit on SNTP Server be selected, SNTP Server will send time automatically with this time interval.

9) ***SNTP port Addr:*** The address of SNTP port.

10) ***SNTP server IP:*** Used only by SNTP client, It is no need to be set on SNTP Server.

If it is set on SNTP client, That is to say, SNTP client will request time synchronization from SNTP Server with time interval defined in “SNTP sync interval”. If it is not be set on SNTP client, That is to say, SNTP will not request time synchronization

Setting for Ethernet card

Take Ethernet card B for example, clicks “Ethernet A” and all connections of this Ethernet card and their properties will be displayed in the right list.

	Name	Protocol select	Description	Standby channel	Mirrored channel	Standby mode
1	EthernetA, Link1	19.Master_DNP	2K21P	None	None	No spare, single channel
2	EthernetA, Link2	19.Master_DNP	1T87P	None	None	No spare, single channel

mode	R&T Under Main Machine	R&T Under Standby Machine	Main channel?	Breaking time(ms)	Target IP Addr
, single channel	Normal	Receive only	YES	3000	10.240.0.6
, single channel	Normal	Receive only	YES	3000	10.240.0.7

Subnet mask	Port addr	Gateway IP	Use gateway	Work mode	Is leader?	Leader name	Is emergency channel?
255.255.255.224	20000	0.0.0.0	NO	TCP Client	YES	None	NO
255.255.255.224	20000	0.0.0.0	NO	TCP Client	YES	None	NO

Fig.2-5 Property of Ethernet Link

1) ***Name:*** Link 1～100 of this Ethernet card can be selected.
2) ***Protocol select:*** Select protocol for this link.
3) ***Description:*** Write required description and 59 characters can be input to the max (excluding the termination character of character string).
4) ***Standby channel:*** The name of the standby channel of this channel. Choices: Ethernet, serial port, virtual channel, etc. Default: “None”.
5) ***Mirrored channel:*** The name of mirrored channel of this channel. Choices: Ethernet, serial port, virtual channel, etc. Default: “None”.
6) ***Standby mode:*** Set Standby mode adopted for this channel.

   “No spare, single channel” means no standby channel for this channel.

   “Both have task” means there are two channels standby with each other.
7) ***R&T Under Main Machine:*** The status of receive and transmit under main machine. Default: “Normal”. It means both receive and transmit.
8) ***R&T Under Standby Machine:*** The status of receive and transmit under

standby machine. Default: “Receive only”.

9) ***Main channel:*** Whether this channel is the main channel or not.
10) ***Break time (ms):*** The link will break overtime, if there’s no message on this link.
11) ***Target IP Addr:*** IP address of the opposite side Ethernet card which connected with this Ethernet card.
12) ***Subnet Mask:*** The Subnet Mask of opposite side IP.
13) ***Port addr:*** Setting according to requirement.
14) ***Gateway IP:*** This should be set if “Use gateway” be selected.
15) ***Use gateway:*** Choices: Yes or No. Default: “No”.
16) ***Work mode:*** “TCP server”, “TCP client”, “UDP, Unicast”, “UDP broadcast” and “IP Multicast “can be selected.
17) ***Is leader?:*** whether this Link is main link or not. If not, set the “Leader link name” item below and select the main Link of this connection. Choices: “Yes” or “No.” Default: “Yes”.
18) ***Leader name:*** leader link name shall be selected if this link isn’t leader link.
19) ***Is emergency channel?:*** whether this Link is emergency link or not. Choices: “Yes” or “No.” Default: “No”.

4.2.3.2 Serial port

Click “Serial” and all included serial ports and their properties will be displayed in the right list.

	Name	Protocol select	Description	Scan interval(ms)	Standby channel	Mirrored channel
1	Serial1	1.Slave_Maintain	43SC	200	None	None
2	Serial2	239.Slave_IEC101Eh	IHM	100	None	None

Standby mode	R&T Under Main Machine	R&T Under Standby Machine	Main channel?	Breaking time(ms)
No spare, single channel	Normal	Receive only	YES	3000
No spare, single channel	Normal	Receive only	YES	3000

Serial type	Baudrate	RTS control mode	RTS advanced time	RTS delayed time	Communicate mode	Data bits	Parity bits
RS232	9600	High	10	10	Asyn	8	NULL
RS232	9600	High	10	10	Asyn	8	EVEN

Parity bits	Stop bits	Via MODEM	Flow control	Syn chars Num	Syn char1	Syn char2
NULL	1	NO	NULL	2	0xEB	0x90
EVEN	1	NO	NULL	2	0xEB	0x90

Fig.2-6 Property of Serial port

1) ***Name:*** Serial port 1～50 can be selected.
2) ***Protocol select:*** Select protocol for this serial port.
3) ***Description:*** Write required description and 59 characters can be input to the max (excluding the termination character of character string).
4) ***Scan interval (ms):*** The interval time to scan the serial port receive buffer.

5) ***Standby channel:*** default, “None” means no Standby channel. FDKBUS, CANBUS, Ethernet, virtual channel, LPT port, etc. can be selected as standby channel.

6) ***Mirrored channel:*** The name of mirrored channel of this channel. Choices: Ethernet, serial port, virtual channel, etc. Default: “None”.

7) ***Standby mode:*** set Standby mode adopted for this channel.

   a) “No spare, single channel” means no standby channel for this channel.

   b) “Both have task” means there are two channels standby with each other.

8) ***R&T Under Main Machine:*** The status of receive and transmit under main machine. Default: “Normal”. It means both receive and transmit.

9) ***R&T Under Standby Machine:*** The status of receive and transmit under standby machine. Default: “Receive only”.

10) ***Main channel:*** Whether this channel is the main channel or not.

11) ***Breaking time (ms):*** The link will break overtime, if there’s no message on this link.

12) ***Serial type:*** Choices, RS232, RS485 and RS422.

13) ***Band rate:*** Range, 110～115200. Default, 9600.

14) ***RTS control mode: Choices:*** “High”, “Low”, “High level valid” and “Low level valid”. Default, “High”.

15) ***RTS advanced time:*** Not need to be changed.

16) ***RTS delayed time:*** Not need to be changed.

17) ***Communicate mode:*** syn(synchronization), asyn(asynchronism) and HDLC mode. Default: “syn”.

18) ***Data bits:*** Choices: 5, 6, 7 and 8. Default: 8.

19) ***Parity bits:*** Choices: null, odd parity, even parity, Mark and Space. Default: “null”.

20) ***Stop bits:*** Choices: 1, 1.5 and 2. Default: “1”.

21) ***Via MODEM***: Whether need to via the modem. Default: “No”.

22) ***Flow control:*** choices: null, hardware and Xon/Xoff. Default: “null”.

23) ***Frame interval:*** interval between sending data frame. Input range: [0～65535]. Unit: [ms].

24) ***Other Items:*** Not used for DF1725RTU and DF1725IED.

4.2.3.3 Virtual channel

Click “Virtual”, then the inclusive virtual channels and their properties will be showed on right side.

	Name	Protocol select	Description	Standby channel	Mirrored channel
1	Virtual1	101.Arith_DI/Counter Gathering	ENTRADAS DIGITAIS	None	None
2	Virtual2	104.Arith_DO/CO	SAIDAS DIGITAIS	None	None

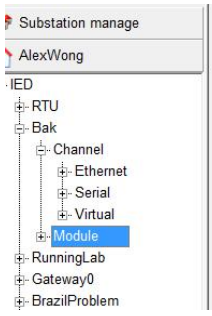
Standby mode	R&T Under Main Machine	R&T Under Standby Machine	Main channel?	Breaking time(ms)
No spare, single channel	Normal	Receive only	YES	3000
No spare, single channel	Normal	Receive only	YES	3000

Fig. 2-7 Property of virtual channel

1) ***Name:*** Virtual channel 1 ~ 100 is available for selection.
2) ***Protocol select:*** Select protocol for this virtual port.
3) ***Description:*** Write necessary description here. Max. Of 59 characters (not including ending character of character string) are allowed to input.
4) ***Standby channel:*** default, "None" means no Standby channel. FDKBUS, CANBUS, Ethernet, virtual channel, LPT port, etc. can be selected as standby channel.
5) ***Mirrored channel:*** The name of mirrored channel of this channel. Choices: Ethernet, serial port, virtual channel, etc. Default: "None".
6) ***Standby mode:*** set Standby mode adopted for this channel.
    c) "No spare, single channel" means no standby channel for this channel.
    d) "Both have task" means there are two channels standby with each other.
7) ***R&T Under Main Machine:*** The status of receive and transmit under main machine. Default: "Normal". It means both receive and transmit.
8) ***R&T Under Standby Machine:*** The status of receive and transmit under standby machine. Default: "Receive only".
9) ***Main channel:*** Whether this channel is the main channel or not.
10) ***Breaking time (ms):*** The link will break overtime, if there's no message on this link.

4.2.4 Module management

Click Module list, then all modules in current processor and their property will be displayed in right side.



	ID	Type	Name	Main	Standby channel	Addr1	Addr2	Sync time mode	Selfcheck reported?
1	14	User Module	User014	EthernetB, Link1	None	1	0	Disable	NO
2	3	Logic Module	Logic003	Serial2	None	41	0	Disable	NO
3	8	User Module	User008	Serial4	None	10	0	Disable	NO
4	10	User Module	User010	Serial4	None	11	0	Disable	NO
5	12	User Module	User012	Serial4	None	12	0	Disable	NO
6	4	User Module	User004	Serial5	None	20	0	Disable	NO
7	5	User Module	User005	Serial5	None	21	0	Disable	NO
8	6	User Module	User006	Serial5	None	22	0	Disable	NO
9	1	1725IED_DI	IEDDI001	Virtual1	None	0	0	Disable	NO
10	7	1725IED_DI	IEDDI007	Virtual1	None	1	0	Disable	NO
11	2	1725IED_CO	IEDCO002	Virtual2	None	0	0	Disable	NO
12	9	1725IED_CO	IEDCO009	Virtual2	None	1	0	Disable	NO

Fig. 2-8 Module list

1) ***ID:*** Serial number of this module in the processor, System default setting. Editing is forbidden.
2) ***Type:*** i.e. the module type when adding the module. System default setting.

Editing is forbidden.

3) ***Main:*** Select the channel which will be the main channel of this module.
4) ***Standby channel:*** Not used for RTU, Editing is forbidden.
5) ***Addr1:*** Source address, setting is necessary according to protocol. Input range [an integer which consists of 4 bytes]
6) ***Addr2:*** Destination address, setting is necessary according to protocol. Input range [an integer which consists of 4 bytes]
7) ***Sync time mode***: RTU's Clock can be modified or not.
8) ***Self check reported***: There would be some information appeared when fault meet in this module, this item decide to write those information into alarm module or not.

4.2.4.1 Add a module

Method 1: Select Module list and click the icon to add a module. Select its initial main channel from the channels in the processor.

Method 2: Select a channel from the channel list and click the icon to add a module. Its initial main channel is the current channel.

Method 3： Select Module List, Right click on the right window, Select a wanted module from the pull–down menu.

4. 2.4.2 Delete a module

Method 1: Select Module list and click the icon to delete the selected module.

Method 2: Select a module of a channel in the channel list, and then click the icon to delete the selected module.

4.2.4.3 Module's property

4.2.4.3.1 Setting for the property of physical module

Add a Physical module; its basic setting is as follows.

	AI	BCD	Physical DI	SPI	DPI	Counter	DO	Syn control	BCO
1	99	0	0	191	0	0	10	0	0

AO	Event report	AI for relay	Protection setting	Element enable	Assistant setting
0	0	0	0	0	0

Fig. 2-9 Property of Physical module

1) ***AI:*** The number of analog input. Only valid for Dongfang RTU. . Input range [0～65535]
2) ***BCD:*** The number of step position information, Input range [0～65535]
3) ***Physical DI:*** The number of digital input. Only valid for Dongfang RTU. Editing is forbidden.
4) ***SPI:*** The number of Single digital input, Input range [0～65535]
5) ***DPI:*** The number of Double digital input, Input range [0～65535]

6) ***Counter:*** Counter input number, Input range [0～65535]
7) ***DO:*** Digital output number, Input range [0～65535]
8) ***Syn control:*** Synchronous control number. Input range [0～65535]
9) ***BCO:*** Binary output number: Input range [0～65535]
10) ***AO:*** Analog output number: Input range [0～65535]
11) ***Event report:*** The number of sequence events report. Input range [0～65535]
12) ***AI for relay:*** The number of protection analog input. Input range [0～65535]
13) Protection setting: The number of protection setting. Input range [0～65535]
14) Element enable: The number of soft board. Input range [0～65535]
15) Assistant setting: The number of protection assistant setting. Input range [0～65535]

4.2.4.3.2 Logic module property

Open one logic module from the module list. Various data quantities that allowed add to this module show. The right side list shows module property.

	AI	BCD	SPI	DPI	Counter	DO	BCO	AO	Event report	Protection AI
1	58	0	460	0	0	213	0	0	0	0

Fig. 2-10 Property of logic module

Note: The number of logic module can't be manual edit, but it can introduce from physical module

4.2.5 Data Setting of Physical Module

4.2.5.1 Analog input (DCAI)

4.2.5.1.1Steps for adding DCAI module

Step one: Add a virtual channel under channel list.

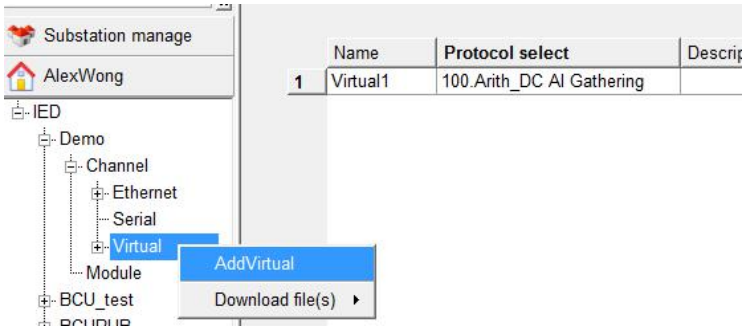


Fig. 2-11 Add a virtual channel

Step two: Select "100.Arith_DC AI Gathering" for this virtual channel.

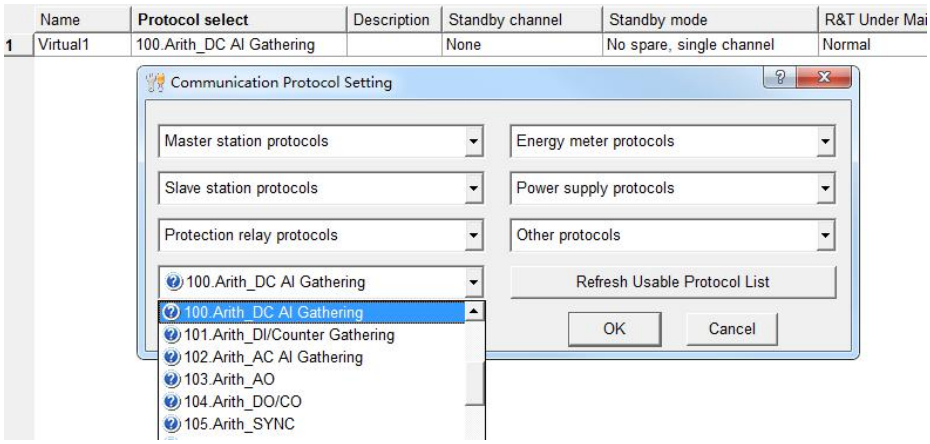


Fig. 2-12 Select protocol for Virtual channel

Step three: Add a DCAI module from module template.

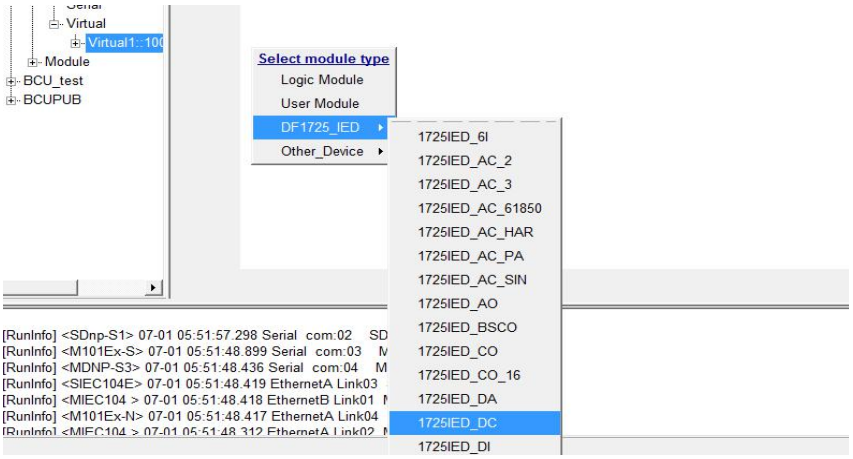


Fig. 2-13 Select module for a Virtual channel

4.2.5.1.2 Setting for public parameters

	AI multiply coe	AI event buffer size
1	NO	8

Fig. 2-14 AI pubic parameters setting

- ✓ ***AI multiply coe:*** YES or No.
- ✓ ***AI event buffer size:*** Default: 8 bytes, range [0~65535].

4.2.5.1.3 Setting parameters point by point

	AI name	Short name	Obj addr	AI type	High alarm	Low alarm	2th high alarm	2th low alarm	3th hig
1	Ia	0 AI	0x00000000	NULL	NO	NO	NO	NO	NO
2	Ib	1 AI	0x00000001	NULL	NO	NO	NO	NO	NO

3th high alarm	3th low alarm	IO force to 0	Main force to 0	High limit	High limit recover	Low limit	Low limit recover	2th
NO	NO	NO	NO	0	0	0	0	
NO	NO	NO	NO	0	0	0	0	

2th high limit	2th high limit recover	2th Low limit	2th low limit recover	3th high limit	3th high recover	3th low limit	3th
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	

3th low recover	Force to 0	Dead band	Smooth F	Coe.a	Coe.b
0	0	0	100	1	1
0	0	0	100	1	1

Fig. 2-15 Point by point parameter of analog input

1) ***AI name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.
3) ***Obi adder:*** Setting is necessary according to protocol. Input range [an integer which consists of 4 bytes], Not used for physical module.
4) ***AI type****:* You may select NULL,－5～＋5V, －10～＋10V,－10～＋10mA, 0～20mA, 4～20mA, or 4～12～20mA, 0~1mA. Default: NULL.
5) ***High alarm:*** Used to select whether alarm or not for this measured value is greater than normal operational guidelines
6) ***Low Alarm:*** Used to select whether alarm or not for this measured value is less than normal operational guidelines
7) ***2th high alarm:*** Used to select whether alarm or not for this measured value is greater than the "2th high" design
8) ***2th low alarm:*** Used to select whether alarm or not for this measured value is less than the "2th low" design
9) ***3th high alarm:*** Used to select whether alarm or not for this measured value is greater than the "3th high" design
10) ***3th low alarm:*** Used to select whether alarm or not for this measured value is less than "3th high" design
11) ***IO force to 0:*** Whether the value defined in "Force to 0" used by IO or not.
12) ***Main force to 0:*** Whether the value defined in "Force to 0" used by Main processor or not.
13) ***High limit:*** High Operational Limits, Readings greater than these limits indicate a deviation from normal operational guidelines, if "High alarm" has been set as "Yes", then this value should be set. Range [Long integer]
14) ***High limit recover:*** If "High alarm" has been set as "Yes", then this value should be set. Range [Long integer]
15) ***Low limit:*** Low Operational Limits, Readings less than these limits indicate a deviation from normal operational guidelines, if "Low alarm" has been set as "Yes", then this value should be set. Range [Long integer]
16) ***Low limit recover:*** If "Low alarm" has been set as "Yes", then this value should be set. Range [Long integer]
17) ***2th high limit:*** High Emergency Limits, Readings greater than these limits indicate that the equipment is operating outside its design, if "2th high alarm" has been set as "Yes", and then this value should be set. Range [Long

integer]

18) ***2th high limit recover:*** If “2th high alarm” has been set as “Yes”, then this value should be set. Range [Long integer]
19) ***2th low limit:*** Low Emergency Limits, Readings less than these limits indicate that the equipment is operating outside its design, If “2th low alarm” has been set as “Yes”, Then this value should be set. Range [Long integer]
20) ***2th low limit recover:*** If “2th low alarm” has been set as “Yes”, then this value should be set. Range [Long integer]
21) ***3th high limit:*** High Emergency Limits, Readings greater than these limits indicate that the equipment is operating outside its design, if “3th high alarm” has been set as “Yes”, and then this value should be set. Range [Long integer]
22) ***3th high limit recover:*** If “3th high alarm” has been set as “Yes”, then this value should be set. Range [Long integer]
23) ***3th low limit:*** Low Emergency Limits, Readings less than these limits indicate that the equipment is operating outside its design, If “3th low alarm” has been set as “Yes”, Then this value should be set. Range [Long integer]
24) ***3th low limit recover:*** If “3th low alarm” has been set as “Yes”, then this value should be set. Range [Long integer]
25) ***Force to 0:*** Measured value less than the set value will be reported as 0. Range [0~32767].
26) ***Dead band:*** It’s used for transmission process of protocol. The value will be transmitted only when the difference between this value and last transmitted value greater than or equal to “dead band”. Range [0～65535].
27) ***Smooth F:*** Smoothing factor. It shall be applied as,

$$Value_{SmoothNew} = F * Vaule_{New} + (1 - F) * Value_{SmoothOld}$$

28) ***Coe.a:*** Precision adjustment coefficient of molecules.
29) ***Coe.b:*** Precision adjustment coefficient of the denominator
30) ***Recorder length:*** Input range [0～65535].

NOTE:

a) “Low limit recover” should be greater than “Low limit”
b) “High limit recover” should be less than “High limit”
c) “2th low limit recover” should be greater than “2th low limit”
d) “2th high limit recover” should be less than “2th high limit”
e) “2th low limit” should be less than “Low limit”
f) “2th high limit” should be greater than “High limit”
g) “3th low limit recover” should be greater than “3th low limit”
h) “3th high limit recover” should be less than “3th high limit”

i) "3th low limit" should be less than "2th Low limit"

j) "3th high limit" should be greater than "2th High limit"

4.2.5.2 Analog input (ACAI)

4.2.5.2.1Steps for adding ACAI module

Step one: Add a virtual channel under channel list. (As 2.5.1.1 step one)

Step two: Select "102.Arith_AC AI Gathering" for this virtual channel. (As 2.5.1.1 step two)

Step three: Add an ACAI module according different usages, e.g. 1725IED_AC_2, 1725IED_AC_2, 1725IED_AC_61850 etc.

4.2.5.2.2 Setting for public parameters

As 4.2.5.1.2.

4.2.5.2.3 Setting parameters one by one

As 4.2.5.1.3.

4.2.5.3 Step position information (BCD)

4.2.5.3.1 Setting for parameters

	BCD name	Short name	Obj addr	Code type	Proc time(0.1s)	Start SPI No.	Total
1	0 BCDInput	0 BCDIn	0x00000000	Decimal	10	0	4

Fig. 2-16 Point by point parameter property of step position information

1) ***Step info name:*** Max of 59 characters (not including ending character of character string) is allowed to input.

2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.

3) ***Obj addr:*** Setting is necessary according to communication protocol. Input range [an integer which consists of 4 bytes]

4) ***Code type:*** You may select decimal code, binary code, BCD code, one to one code, remainder three codes, gray code, new gray code, or ones place and tens place code separately. Default setting is decimal code.

5) ***Proc time (0.1s):*** Time for stabilization of BCD input. Input range [0～255]. Default: 10.

6) ***Start SPI No.:*** The start number of the DI channel (point) used as the first bit. [Range of input of this module].

7) ***Total:*** The number of SPI in this step position information, Input range [Range of input of this module]. Default: 4.

4.2.5.4 Digital input (DI)

4.2.5.4.1Steps for adding DI module

Step one: Add a virtual channel under channel list. (As 2.5.1.1step one)

Step two: Select “101.Arith_DI/Counter Gathering” for this virtual channel. (As 2.5.1.1step two)

Step three: Add a DI module for this virtual channel, e.g. 1725IED_DI_32, 1725IED_DI.

4.2.5.4.2 Setting for parameters

	DI type	Debounce time(ms)
1	SPI	20

Fig. 2-17 DI point by point parameter property

- ✓ ***DI type:***

SPI: Single Point Input (Form one input)

SPCHC: Single Point Counter every Half Cycle (counter incrementing on each pulse half cycle that is count on 0 to 1 and 1 to 0)

SPCFC: Single Point Counter every Full Cycle.

SPI+SPCHC:

DPI1: The first point of Double Point Input (Form C inputs)

DPI2: The second point of Double Point Input (Form C inputs)

DPCHC1: The first point of Double Points Counter every Half Cycle.

DPCHC2: The second point of Double Points Counter every Half Cycle.

DPCFC1: The first point of Double Points Counter every Full cycle.

DPCFC2: The second point of Double Points Counter every Full cycle.

DPI+DPCHC1: The first point of Double Points Input & Double Points Counter per Half Cycle active simultaneously

DPI+DPCHC2: The second point of Double Points Input & Double Points Counter per Half Cycle active simultaneously

- ✓ ***Debounce Time:*** needed for status of digital input point, Input range [0～65535]. Unit [ms]

4.2.5.4.3 Single point input

4.2.5.4.3.1 Setting for public parameters

	SPI SOE buffer size	SPI COS buffer size
1	8	8

Fig. 2-18 Common parameter property of single point input

- ✓ **SPI SOE buffer size:** Set up the size of the buffer which used for storing SPI SOE in this module. Input range [0～65535].
- ✓ ***SPI COS buffer size:*** Set up the size of the buffer which used for storing SPI COS in this module. Input range [0～65535].

4.2.5.4.3.2 Setting for parameters point by point

	SPI name	Short name	Obj addr	DI No.	Invert	Receiver make SOE	SOE->COS	Tc(ms)	Nc	T recover
1	0 SPI	0 SPI	0x00000000	0	NO	NO	NO	0	0	0
2	1 SPI	1 SPI	0x00000001	1	NO	NO	NO	0	0	0

Fig. 2-19 Point by point parameter property of single point input

1) ***SPI name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input, which be displayed on LCD.
3) ***Obj addr:*** Setting is necessary according to communication protocol. Input range [an integer which consists of 4 bytes]
4) ***DI No.:*** Wiring position No. that corresponds to this input point. Calculate from 0. Only available to Dongfang RTU module.
5) ***Invert:*** Invert the status of input signal. Double click to select among:
    a) *I/O:* Invert by I/O (for DF1000 I/O only)
    b) *Main:* Invert by main communication processor module
    c) *Disable:* This function is disabled.
6) ***Receiver make SOE:*** If select "Yes", this processor will make SOE when it get a digital change.
7) ***SOE->COS:*** Whether to generates COS through SOE or not.
8) ***Tc (ms):*** Chatter filter period. Range: [3* debounce period~60000ms].0: Chatter filter inactive
9) ***Nc:*** Number of Changes. Range [0~127].0: Chatter filter inactive.
10) ***T recover:*** Chatter filter reset time. Range:[1~255]*Tc

4.2.5.4.4 Double points input

4.2.5.4.4.1 Setting for pubic parameters

	DPI SOE buffer size	DPI COS buffer size
1	8	8

Fig. 2-20 Common parameter property of double points input

- ✓ ***DPI SOE buffer size:*** Input range [0～65535].
- ✓ ***DPI COS buffer size:*** Input range [0～65535].

4.2.4.4.4.2 Setting for parameters point by point

	DPI name	Short name	Obj addr	DI No.	Invert1	Invert2	Receiver make SOE	SOE->COS	Tc(ms)	Nc	T Recover	Proc time(10ms)
1	0 DPI	0 DPI	0x00000000	2	NO	NO	NO	NO	0	0	0	1000

Fig. 2-21 Point by point parameter property of double points input

1) ***DPI name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.
3) ***Obj addr:*** Setting is necessary according to communication protocol. Input range [an integer which consists of 4 bytes], needn't to change in physical module.
4) ***DI No:*** Wiring position No. of the input point. Calculate from 0. Only available to Dongfang RTU module. Editing is forbidden.
5) ***Invert1:*** Invert the status of the first channel, double click to select among:

   I/O: Invert by I/O (for DF1000 I/O only);

   Main: Invert by main communication processor module

   Disable: This function is inactive.
6) ***Invert2:*** Invert the status of the second channel, double click to select among:

   I/O: Invert by I/O (for DF1000 I/O only);

   Main: Invert by main communication processor module

   Disable: This function is inactive.
7) ***Receiver make SOE:*** If select "Yes", this processor will make SOE when it get a digital change.
8) ***SOE->COS:*** Whether to generate COS according to SOE or not.
9) ***Tc (ms):*** Chatter filter period. Range: [3* debounce period~60000ms].0: Chatter filter inactive.
10) ***Nc:*** Number of Changes. Range [0~127].0: Chatter filter inactive.
11) ***T recovery:*** Chatter filter reset time. Range:[1~255]*Tc
12) ***Proc time (s):*** Double Point Processing time, range [0~255] s

4.2.5.4.5 Counter

4.2.5.4.5.1 Setting for parameters

	Counter name	Short name	Obj addr	DI No.	Counter type	DPI Proc(ms)
1	0 Kwh	0 Kwh	0x00000000	0	NULL	0
2	1 Kwh	1 Kwh	0x00000001	0	NULL	0

Fig. 2-22 Point by point parameter attributes of counter

1) ***Counter name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.

3) ***Obj addr:*** Setting is necessary according to communication protocol. Input range [an integer which consists of 4 bytes], Not used for physical module.
4) ***DI No.:*** Wiring position No. of the input point. Calculate from 0. Only available to Dongfang RTU module. Editing is forbidden.
5) ***Counter type:*** Counter type, no input required, including the following types:
    a) *Integral:* Kwh value by integration calculation
    b) *SPCHC:* Single Point Counter Half Cycle
    c) *SPCFC:* Single Point Counter Full Cycle
    d) *DPCHC:* Double Points Counter Half Cycle
    e) *DPCFC:* Double Points Counter Full Cycle
6) ***DPI Proc (ms):*** Not used for counter. Max: 60000ms.

4.2.5.5 Control output (DO)

4.2.5.5.1 Steps for adding Control output module

Step one: Add a virtual channel under channel list. (As 2.5.1.1 step one)

Step two: Select "104.Arith_DO/CO" or "284.Arith_CO16" for this virtual channel according to the CO module you want to use. (As 2.5.1.1 step two)

Step three: Add a CO module for this virtual channel. E.g. 1725IED_CO, 1725IED_CO_16 (As 2.5.1.1 step three)

4.2.5.5.2 Setting for parameters

	DO name	Short name	Obj addr	DO attri	Command format	Output mode	Virtual CO	Reset tin
1	204F4 FECHA	0 DO	0x00000001	Close	-	Pulse ON	Normal	
2	204F4 ABRE	1 DO	0x00000001	Trip	-	Pulse ON	Normal	

Reset time(s)	On time(long:ms)	On time(short:ms)	DO enable Device ID	Enable SPI No.	Enable value
30	2000	0	-	-	Open
30	2000	0	-	-	Open

Fig. 2 -23 Point by point parameter property of control output

1) ***DO name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.
3) ***Obj addr:*** DO address. Working together with DO attribute, and the combination of these two shall be given an absolute address to suit different protocols. No input required for DF1000 I/O module. Range [0~65535].
4) ***DO attri:*** Working together with DO address, and the combination of these two shall be given an absolute address to suit different protocols. No input required for DF1000 I/O module. Double click to select among 'trip', 'close' and '-'(no attribute).
5) ***Command format***: Command format, including "Signal", "Double" and "Rise/Fall".

6) ***Output mode:*** Output mode, including the following types: "-", "Pulse on", "Pulse off", "Latching on", "Latching off". Default: "Pulse on".
7) ***Virtual CO:*** Not used for physical module and it is used by internal logic of program.
8) ***Reset time:*** If the execution command is not received within the set period, this DO channel pre-select command will be canceled. Range [1~255]s. Default:60s.
9) ***On time (long: ms):*** Contact closure time. Range [100~60000ms]. Default: 2000ms.
10) ***On time (short: ms):*** Contact closure time. Range [100~60000ms]. Default: 1000ms.
11) ***DO enable Device ID:*** Appoint the device of the processor, let the device control whether allow the operation of DO.
12) ***Enable SPI No.:*** Appoint the SPI No. enabled DO.
13) ***Enable value:*** Appoint the value of enable SPI

4.2.5.6 Synchronous control

4.2.5.6.1Setting for parameters

	SYN name	Short name	Obj addr	DO No.	None voltage SYNC	SYN enable device ID	Enable DI No.	Enable DI
1	0 Sync	0 Sync	0x00000000	65535	NO	NULL	0	SPI
2	1 Sync	1 Sync	0x00000001	65535	NO	NULL	0	SPI

Enable DI type	Enable DI value	Side A voltage No.	Side A voltage coe a	Side A voltage coe b	Side B voltage No.	Side
SPI	0	0	0	0	0	
SPI	0	0	0	0	0	

Side B voltage coe a	Side B voltage coe b	Side A Freq No.	Side B Freq No	Angle difference No.	Voltage high limit(0.1V)
0	0	0	0	0	0
0	0	0	0	0	0

Voltage low limit(0.1V)	None voltage low limit(0.1V)	Freq high limit(0.01Hz)	Freq low limit(0.01Hz)	Voltage rating	Voltag
0	0	0	0	0	
0	0	0	0	0	

Voltage difference(%)	Freq difference	Angle difference(0.1度)	Freq change rate(0.01Hz/s)	SYN elapsed time	Lead time(ms)
0	0	0	0	0	0
0	0	0	0	0	0

Fig. 2-24 Point by point parameter property of synchronous control

1) ***Syn name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.
3) ***Obj addr:*** Setting is necessary according to communication protocol. Input range [an integer which consists of 4 bytes], Not used for physical module.
4) ***DO No.***: As DO No. of synchronous control, range of DO No. of the module.

5) ***SYN enable device ID:*** Appoint the device of the processor, let the device control whether allow the operation of synchronous control.
6) ***Enable DI No.***: Appoint the DI No. enabled synchronous control.
7) ***Enable DI type:*** Select SPI, DPI, and Device Status. Default: SPI.
8) ***Enable DI value:*** Appoint the value of enable DI.
9) ***Side A voltage No.***: Appoint the device's AC library point No., input range [AC library of the device].
10) ***Side A voltage coe a:*** The coefficient of side A voltage is a.
11) ***Side A voltage coe b:*** The coefficient of side A voltage is b.
12) ***Side B voltage No.***: Appoint the device's AC library point No., input range [AC library of the device].
13) ***Side B voltage coe a:*** The coefficient of side B voltage is a.
14) ***Side A voltage coe b:*** The coefficient of side B voltage is b.
15) ***Side A Freq No.***: Appoint the device's AC library point No., input range[AC library of the device].
16) ***Side B Freq No.***: Appoint the device's AC library point No., input range[AC library of the device].
17) ***Angle difference No.***: Appoint the device's AC library point No., input range [AC library of the device].
18) ***Voltage high limit (0.1V):*** Range [0~65535], Unit [0.1V].
19) ***Voltage low limit (0.1V):*** Range [0~65535], Unit [0.1V].
20) ***None voltage low limit (0.1V):*** When voltage lower than the setting voltage, it can be regard as none voltage. Range [0~65535], Unit [0.1V].
21) ***Freq high limit (0.1Hz):*** Range [0~65535], Unit [0.1Hz].
22) ***Freq low limit (0.1Hz):*** Range [0~65535], Unit [0.1Hz].
23) ***Voltage rating:*** Range [0~65535], Unit [0.1V].
24) ***Voltage difference:*** The difference between the voltage of side A and side B for synchronous control, relative value. Range [1~100], Unit [%].
25) ***Freq difference:*** The difference between the frequency of side A and side B for synchronous control, Range [0~65535], Unit [0.01Hz].
26) ***Angle difference:*** The difference between the angle of side A and side B for synchronous control, Range [0~65535], Unit [0.1 degree].
27) ***Freq change difference:*** Range [0~65535], Unit [0.01Hz/s].
28) ***Syn elapsed time:*** Input range [4 byte], Unit [ms].
29) ***Lead time:*** Input range [4 byte], Unit [ms].

4.2.5.7 BCO output

4.2.5.7.1 Setting for parameters

	BCO name	Short name	Obj addr	Start No.	Output Num	Output mode	Output count	ON time(ms)	OFF ti
1	0 BCO	0 BCO	0x00000000	0	4	Pulse ON	1	0	
2	1 BCO	1 BCO	0x00000001	0	4	Pulse ON	1	0	

OFF time(ms)	Reset time(s)	BCO enable device ID	Enable data type	Enable data No.	Enable value1	Enable value2	E
0	30	NULL	AI	0	0	0	N
0	30	NULL	AI	0	0	0	N

BCO feedback device ID	Feed back data type	Feed back data No.
NULL	AI	0
NULL	AI	0

Fig. 2-25 Point by point parameter property of BCO output

1) ***BCO name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.
3) ***Obj addr:*** Setting is necessary according to communication protocol. Input range [an integer which consists of 4 bytes], Not used for physical module.
4) ***Start No.:*** Input range [0～65535].
5) ***Output Num:*** Input range [1～32].
6) ***Output mode:*** You may select Ignore, pulse ON, pulse OFF, bolt lock ON, or bolt lock OFF. Default setting is Ignore.
7) ***Output count:*** Input range [1～255].
8) ***ON time (ms):*** Contact closure time. Input range [0~65535]. Unit [ms]
9) ***OFF time (ms):*** Contact open time. Input range [0~65535]. Unit [ms]
10) ***Reset time (ms): I***f the execution command is not received within the set period, this BCO channel pre-select command will be canceled. Range [1~255] s. Default: 60s.
11) ***BCO enable device ID:*** Appoint the device of the processor, let the device control whether allow the operation of BCO.
12) ***Enable data type:*** Select Invalid type, AI, BCD, SPI, DPI:1, DPI:2, Device Status, Counter, DO, SYN, BCO, AO, Event report, Protect AI, Setting and Soft BO. Default: AI.
13) ***Enable data No.***: Appoint the data No. enabled BCO.
14) ***Enable value1:*** Appoint the first value of enable data.
15) ***Enable value2:*** Appoint the second value of enable data.
16) ***BCO feedback device ID:*** Appoint the device of the processor; let the device feedback the result of BCO.
17) ***Feedback data type:*** Select Invalid type, AI, BCD, SPI, DPI:1, DPI:2, Device Status, Counter, DO, SYN, BCO, AO, Event report, Protect AI,

Setting and Soft BO. Default: AI.

18) ***Feedback data No.***: Appoint the data No. feedback BCO.

4.2.5.8 Analog output (AO)

4.2.5.8.1 Setting for parameters

	AO name	Short name	Obj addr	Type	SBO	FULL Scale	Reset time(s)	AO enable device ID	Enable SPI No.	Enable value
1	0 AO	0 AO	0x00000000	0 ~ 5 V	SBO	0x3FFF	30	-	-	Open
2	1 AO	1 AO	0x00000001	0 ~ 5 V	SBO	0x3FFF	30	-	-	Open

Fig. 2-26 Point by point parameter property of analog output

1) ***AO name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.
3) ***Obj addr:*** Setting is necessary according to communication protocol. Input range [an integer which consists of 4 bytes], Not used for physical module.
4) ***Type:*** You may select 0~5V，0~10V，-5~+5V，0~1mA，0~5mA，0~10mA，0~20mA，or 4~20mA. Default setting is 0~5V.
5) ***SBO:*** You may select direct mode or select before operation mode. Default setting is direct mode.
6) ***FULL Scale:*** Value corresponding to full scale, in HEX, range [FFH~7FFFH], default: 0FFFH.
7) ***Reset time:*** f the execution command is not received within the set period; this AO channel pre-select command will be canceled. Range: [1~255]s, default: 30s.
8) ***AO enable device ID:*** Appoint the device of the processor, let the device control whether allow the operation of AO.
9) ***Enable SPI No:*** Appoint the SPI No. enabled AO
10) ***Enable value:*** Appoint the value of enable SPI.

4.2.5.9 Event report (SER)

4.2.5.9.1 Setting for public parameters

	Control byte of event report	Event report buffer size
1	0	64

Fig. 2-27 Common parameter property of event report

- ✓ **Control byte of event report:** Set up the byte that used for control the event report in this module. Editing is forbidden.
- ✓ ***Event report buffer size:*** Set up the size of the buffer which used for storing event report in this module. Default: 8.

4.2.5.9.2 Setting for parameters

	Event report name	Short name	Fun Code	INF ID	Judge state when convert to SOE	Auto reset	Enable to be reset
1	0 SER	0 SER	0	0	NO	NO	NO
2	1 SER	1 SER	0	0	NO	NO	NO

Enable to be reset by command	Reset time(s)	DI type after conversion	DI No. after conversion	Type
NO	0	SPI	0	NULL
NO	0	SPI	1	NULL

Fig. 2-28 Point by point parameter property of event report

1) ***Event report name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.
3) ***Fun Code:*** It's used to distinguish different event report.
4) ***INF ID:*** Relay action information ID.
5) ***Judge state when convert to SOE***: Whether need to judge When convert event report to SOE. Default: NO.
6) ***Auto reset:*** Whether need to auto reset. Default: NO.
7) ***Enable to be reset by command:*** Allow reset through DO. Default: NO.
8) ***Reset time (s):*** The time of auto reset.
9) ***DI type after conversion:*** Convert event report to DI, choose SPI or DPI.
10) ***DI No.*** after conversion: The virtual DI No. of event report converted.
11) Type: Virtual DI type, including "Protect action info", "Device self check info", "Fault measure event", "Warning event".

4.2.5.10 Element enable (Soft BO)

4.2.5.10.1 Setting for public parameters

	Element enable control by
1	0

Fig. 2-29 Common parameter properties of element enable

✓ **Element enable control by:** Set up the byte that used for control the element enable.

4.2.5.10.2 Setting for parameters

	Element enable name	Short name	Obj addr	Enable
1	0 SoftBO	0 SoftBO	0x00000000	NO
2	1 SoftBO	1 SoftBO	0x00000001	NO

Fig. 2-30 Point by point parameter properties of element enable

1) ***Event report name:*** Max of 59 characters (not including ending character of character string) is allowed to input.

2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input
3) ***Obj addr:*** Setting is necessary according to communication protocol. Input range [an integer which consists of 4 bytes], Not used for physical module.
4) ***Enable:*** Default: NO

4.2.5.11 Protection

4.2.5.11.1 AI for relay (Protect AI)

4.2.5.11.1.1 Setting for public parameters

	Protect AI control byte	AI Event buffer size
1	0	8

Fig. 2-31 Common parameter properties of protect AI

- ✓ **Protect AI control byte:** Set up the byte that used for control the protect AI in this module. Editing is forbidden.
- ✓ ***AI event buffer size:*** Set up the size of the buffer which used for storing the protect AI in this module. Default: 8.

4.2.5.11.1.2 Setting for parameters

	AI name	Short name	Obj addr	IO force to 0	Main force to 0	Index	AI num under this index
1	0 BhAI	0 BhAI	0x00000000	NO	NO	0	0
2	1 BhAI	1 BhAI	0x00000001	NO	NO	0	0

Fig. 2-32 Point by point parameter properties of protect AI

1) ***AI name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.
3) ***Obi adder:*** Setting is necessary according to protocol. Input range [an integer which consists of 4 bytes], Not used for physical module.
4) ***IO force to 0:*** Whether the value defined in “Force to 0” used by IO or not.
5) ***Main force to 0:*** Whether the value defined in “Force to 0” used by Main processor or not.
6) ***Index:*** Index.
7) ***AI num under this index:*** AI num under this index.

4.2.5.11.2 Protection setting (Setting)

4.2.5.11.2.1 Setting for public parameters

	Setting control byte
1	0

Fig. 2-33 Common parameter property of setting

- ✓ **Setting control byte:** Set up the byte that used for control the setting in this module. Editing is forbidden.

4.2.5.11.2.2 Setting for parameters

	Setting name	Short name	Obj addr	Enable	Type	x	Length	High limit	Low limit	Unit	Step	M
1	0 BHDZ	0 BHDZ	0x00000000	NO	General	0	4	0.000	0.000		0.000	
2	1 BHDZ	1 BHDZ	0x00000001	NO	General	0	4	0.000	0.000		0.000	

Max digit of precision integral part	Max digit of precision decimal part
0	0
0	0

Fig. 2-34 Point by point parameter properties of setting

1) ***Setting name:*** Max of 59 characters (not including ending character of character string) is allowed to input.
2) ***Short name:*** Max of 14 characters (not including ending character of character string) is allowed to input.
3) ***Obi adder:*** Setting is necessary according to protocol, input range (an integer which consists of 4 bytes), not used for physical module.
4) ***Enable:*** Default: NO.
5) ***Type:*** The type of setting, including “General”, “Config Word”, “Control BYTE”.
6) ***X:*** Combo number, Range [UCHAR].
7) ***Length:*** Range [FLOAT].
8) ***High limit:*** High Operational Limits, Range [FLOAT].
9) ***Low limit:*** Low Operational Limits, Range [FLOAT].
10) ***Unit:*** String, 8 bytes.
11) ***Step:*** Range [FLOAT].
12) ***Max digit of precision integral part:*** Range [UCHAR].
13) ***Max digit of precision decimal part:*** Range [UCHAR].

4.2.5.11.3 Assistant setting

4.2.5.11.3.1 Setting for public parameters

As 4.2.5.11.2.1

4.2.5.11.3.2 Setting for parameters

As 4.2.5.11.2.2

4.2.6 Edit database in Logic Module

While editing a logical module’s database, all real modules’ data can be mapped into this logical module’s database.

Click a data quantity listed under a logic module of a processor of a substation in the left side of the main interface. For example: Logic module, Analog input. Setting interface of this data shows in the right side of the main interface, As Fig.3-35.

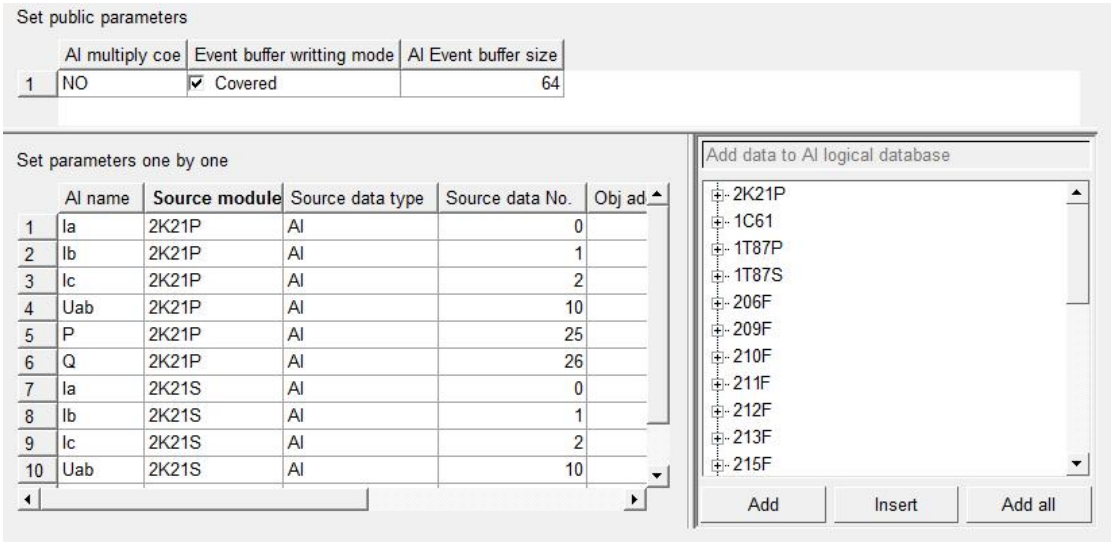


Fig. 2-35 Add AI to logical database from real module

All real modules of this processor that include quotable data types are in the right side of the point by point parameter setting.

Add: Select a module or data quantity of a module in the list, and then click Add. All data quantities of this module or this data quantity will be added into the left list.

Add all: Click Add all to add all data quantities of all listed modules into the left list.

Insert: Select a position in the left list and select a data quantity in the right list, then click Insert. The left selected position will be inserted by the right selected data quantity.

4.2.6.1 Analog input

Analog input logical database includes the following data types. Analog quantity input (analog input), step position information, and counter.

4.2.6.1.1 Setting for public parameters

	AI multiply coe	Event buffer writting mode	AI Event buffer size
1	NO	☑ Covered	64

Fig. 2-36 Public parameter property of logic analog input

- ✓ ***AI multiply coe:*** You may select Yes or No to decide whether use a, b and c or not.
- ✓ ***Event buffer writing mode:*** "Covered" means old events will be covered when buffer full.
- ✓ ***Buffer size:*** Set up the size of the buffer which used for storing analog input events in this module, input range [0~65535].

4.2.6.1.2 Setting for parameters point by point

	AI name	Source module	Source data type	Source data No.	Obj addr	a	b	c	Dead band	Force to 0	Smooth F(%)
1	Ia	2K21P	AI	0	0	1	1	0	0	0	100
2	Ib	2K21P	AI	1	1	1	1	0	0	0	100

Fig. 2-37 Point by point parameter property of logic analog input

1) ***Obj addr:*** Setting is necessary according to specific protocol. Range [an integer which consists of 4 bytes]
2) ***a:*** Say the value of a AI in RTU is X, And Y is the value should be sent to master station, Then Y=a*X/b + c.
3) ***b:*** Refer to a column.
4) ***c:*** Refer to a column.
5) ***Dead band:*** It's used for transmission process of protocol. Measured value will be transmitted only when the difference value between this value and last transmitted value greater than or equal to "dead band". This is only available to Dongfang RTU module. Input range [0～65535].
6) ***Force to 0:*** Measured value less than the set value will be reported as 0. Range [0~32767].***Smooth F:*** Smoothing factor. It shall be applied as,

$$Value_{SmoothNew} = F * Vaule_{New} + (1-F) * Value_{SmoothOld}$$

7) ***Other Items:*** Editing is forbidden.

When the properties of data points in the physical module are changed, the properties of analog input in logical library are changed accordingly

4.2.6.2 Step position information

Logic step position information database includes the following data types. Analog quantity input (analog input) and step position information.

4.2.6.2.1 Setting for parameters

	BCD name	Source module name	Source data type	Source data No.	Obj addr
1	0 BCDInput	IEDDI003	BCD	0	0

Fig. 2-38 Point by point parameter property of logic step position information

1) ***BCD name:*** Name of the actual point which corresponds to this logic point.
2) ***Obj addr:*** Setting is necessary according to specific protocol. Input range [an integer which consists of 4 bytes]
3) ***Other Items:*** Editing is forbidden.

4.2.6.3 Digital input (DI)

4.2.6.3.1 Single point input

Single digital input logical database includes the following data types. Single point input and double points input, of which, each double points input is deemed as 2 single point inputs.

4.2.6.3.1.1 Setting for public parameters

	Event buffer writting mode	SOE buffer size	COS buffer size
1	☑ Covered	128	128

Fig. 2-38 Public parameter property of logic single point input

- ✓ ***Event buffer writing mode:*** You may select Yes, No or New event cover older event when the events area is full.
- ✓ ***SOE buffer size:*** Set up the size of the buffer which used for storing SPI SOE in this module. Input range [0～65535].
- ✓ ***COS buffer size:*** Set up the size of the buffer which used for storing SPI COS in this module. Input range [0～65535].

4.2.6.3.1.2 Setting for parameters point by point

	SPI name	Invert	Transmit SOE	Transmit COS	Source module name	Source data type	Source data No.	Obj addr
1	0AI High	NO	YES	YES	IEDDC004	SPI	0	0
2	0AI Low	NO	YES	YES	IEDDC004	SPI	1	1

Fig. 2-39 Point by point parameter attributes of logic single point input

1) ***SPI name:*** Name of the actual point which corresponds to this logic point.
2) ***Invert:*** Whether change the SPI status. Default: No.
3) ***Transmit SOE:*** Protocol will send SOE of this point if select “Yes”.
4) ***Transmit COS:*** Protocol will send SOE of this point if select “Yes”.
5) ***Obj addr:*** Setting is necessary according to specific protocol. Input range [an integer which consists of 4 bytes].
6) ***Other Items:*** Editing is forbidden.

When the properties of data point in the physical module are changed, the properties of digital input in logical library are changed accordingly

4.2.6.3.2 Double points input

4.2.6.3.2.1 Setting for common parameters

	Event buffer writting mode	SOE buffer size	COS buffer size
1	☑ Covered	128	128

Fig. 2-40 Common parameter property of logic double points input

- ✓ ***Event buffer writing mode:*** “Covered" means Old events will be covered when buffer full.
- ✓ ***SOE buffer size:*** Set up the size of the buffer which used for storing DPI SOE in this module. Input range [0～65535].
- ✓ ***COS buffer size:*** Set up the size of the buffer which used for storing DPI SOE in this module .Input range [0～65535].

When the properties of data point in the physical module are changed, the properties of digital input in logical library are changed accordingly

4.2.6.3.2.2 Setting for parameters point by point

	DPI name	Invert1	Invert2	Transmit SOE	Transmit COS	Source module name	Source data type	Source data No.
1	0 DPI	NO	NO	YES	YES	IEDDI001	DPI	
2	1 DPI	NO	NO	YES	YES	IEDDI001	DPI	

Source data No.	Proc time(*10ms)	Obj addr
0	1000	0
1	1000	1

Fig. 2-41 Point by point parameter property of logic double points input

1) ***DPI name:*** Name of the actual point which corresponds to this logic point. Editing is forbidden.
2) ***Invert1:*** Whether change the DPI1 status. Default: No.
3) ***Invert2:*** Whether change the DPI2 status. Default: No.
4) ***Transmit SOE:*** Protocol will send SOE of this point if select "Yes".
5) ***Transmit COS:*** Protocol will send SOE of this point if select "Yes".
6) ***Obj addr:*** Setting is necessary according to specific protocol. Input range [an integer which consists of 4 bytes].
7) ***Other Items:*** Editing is forbidden.

When the properties of data point in the physical module are changed, the properties of digital input in logical library are changed accordingly

4.2.6.3.3 Counter

4.2.6.3.3.1 Setting for parameters

	Counter name	Freeze	Interval	User-defined Freeze Interval(min)	Source module name	Source
1	0 Kwh	Non-reset	1 hour	0	User001	Counter
2	1 Kwh	Non-reset	1 hour	0	User001	Counter

Source data type	Source data No.	Obj addr	a	b	c
Counter	0	0	1	1	0
Counter	1	1	1	1	0

Fig. 2-42 Point by point parameter property of counter in logical database

1) ***Counter name:*** Name of the actual point which corresponds to this logic point. Editing is forbidden.
2) ***Freeze:*** You may select "Reset" or "Non-Reset". Default setting is "Non-Reset".
3) ***Interval:*** Default "1 hour".
4) ***User-defined Freeze Interval (min):*** Input different freeze interval time by user. "Interval" need choose "User-defined".
5) ***Obj addr:*** Setting is necessary according to specific protocol. Input range [an

integer which consists of 4 bytes].

6) ***a:*** Coefficient for reported value.

$$Y = (X * a / b) + c$$

Y: Reported value; X: Raw value from real database;

7) ***b:*** Refer to Coe.A

8) ***c:*** Refer to Coe.A

9) ***Other Items:*** Editing is forbidden.

4.2.6.4 Control output

Logic control output database includes the following data types. Control output and synchronous control.

4.2.6.4.1 Setting for parameters

	DO name	Obj addr	DO attri	Command format	SBO	Source module name	Source data type	Source data No.
1	0 DO	0	Trip	-	SBO	User008	DO	
2	1 DO	1	Close	-	SBO	User008	DO	

Source data No.	DO enable device ID	Enable SPI No.	Enable value
0	-	-	Open
1	-	-	Open

Fig. 2-43 Point by point parameter property of logic control output

1) ***DO name:*** Name of the actual point which corresponds to this logic point. Editing is forbidden.

2) ***Obj addr:*** DO address. Working together with DO attribute, the combination of these two shall be given an absolute address to suit different protocols.

3) ***DO attri:*** DO attribute. Working together with DO address, the combination of these two shall be given an absolute address to suit different protocols..

4) ***Command format:*** Y should be set according specific protocol.

5) ***SBO***: DO type selection. Please click to select between 'Direct' and 'SBO'. 'SBO' means 'select before operation', 'Direct' means 'direct operation'. If 'SBO' is selected, only the command of 'select before operation' can be executed, the command of 'direct operation' can't be executed. If 'Direct' is selected, both the command of 'direct operation' and the command of 'select before operation' can be executed

6) ***DO enable Device ID:*** Appoint the device of the processor, let the device control whether allow the operation of DO.

7) ***Enable SPI No.:*** Appoint the SPI No. enabled DO.

8) ***Enable value:*** Appoint the value of enable SPI

9) ***Other Items***: Editing is forbidden.

4.2.6.5 BCO output

4.2.6.5.1 Setting for parameters

	BCO name	Obj addr	Source module name	Source data type	Source data No.	BCO enable device ID	Enable data
1	0 BCO	0	User001	BCO	0	NULL	AI
2	1 BCO	1	User001	BCO	1	NULL	AI

Enable data type	Enable data No.	Enable value
AI	0	0
AI	0	0

Fig. 2-44 Point by point parameter property of logic BCO output

1) ***BCO name:*** Name of the actual point which corresponds to this logic point. Editing is forbidden.
2) ***Obj addr:*** Setting is necessary according to specific protocol. Input range [an integer which consists of 4 bytes].
3) ***BCO enable device ID:*** Appoint the device of the processor, let the device control whether allow the operation of BCO.
4) ***Enable data type:*** Select Invalid type, AI, BCD, SPI, DPI:1, DPI:2, Device Status, Counter, DO, SYN, BCO, AO, Event report, Protect AI, Setting and Soft BO. Default: AI.
5) ***Enable data No.***: Appoint the data No. enabled BCO.
6) ***Enable value1:*** Appoint the value of enable data.
7) ***Other Items:*** Editing is forbidden.

4.2.6.6 Analog output

4.2.6.6.1 Setting for parameters

	AO name	Obj addr	SBO	Value format	Source module name	Source data type	Source data No.	Max valu
1	0 AO	0	Direct	Scaled	User001	AO	0	0x00003F
2	1 AO	1	Direct	Scaled	User001	AO	1	0x00003F

Max value	Min value	AO enable device ID	Enable SPI No.	Enable value
0x00003FFF	0x00000000	-	-	Open
0x00003FFF	0x00000000	-	-	Open

Fig. 2-45 Point by point parameter property of logic AO

1) ***AO name:*** Name of the actual point which corresponds to this logic point. Editing is forbidden.
2) ***Obj addr:*** Setting is necessary according to specific protocol. Input range [an integer which consists of 4 bytes]
3) ***SBO:*** You may select Direct mode or select before execute mode..
4) ***Value format:*** You may select Ignore, Scaled value, normalized value, or short floating point value. Default setting is degree scale value.

5) ***Max value*** The minimum value of the corresponding type.

6) ***Min value:*** The maximum value of the corresponding type.].

7) ***AO enable device ID:*** Appoint the device of the processor, let the device control whether allow the operation of AO.

8) ***Enable SPI No:*** Appoint the SPI No. enabled AO

9) ***Enable value:*** Appoint the value of enable SPI.

10) ***Other Items:*** Editing is forbidden.

4.2.6.7 Event report

4.2.6.7.1 Setting for public parameters

	Use obj add	Buffer writting mode	Event report buffer size
1	NO	☑ Covered	64

Fig. 2-46 Public parameter property of logic event report

- ✓ ***Use obj add:*** Whether use the object address, you may select Yes or No.
- ✓ ***Buffer writing mode:*** You may select Yes, No or New event cover older event when the events area is full.
- ✓ ***Event report buffer size:*** Set up the size of the buffer which used for storing event report in this module. Input range [0～65535].

4.2.6.7.2 Setting for parameters point by point

	Event report name	Fun Code	INF ID	Source module name	Source data type	Source data No.
1	0 SER	0	0	User001	Event report	0
2	1 SER	0	0	User001	Event report	1

Fig. 2-47 Point by point parameter property of logic event report

1) ***Event report name:*** Name of the actual point which corresponds to this logic point. Editing is forbidden.

2) ***Other Items:*** Editing is forbidden.

4.2.6.8 Protect AI

4.2.6.8.1 Setting for public parameters

	Obj addr	AI buffer size for relay
1	NO	64

Fig. 2-48 Public parameter property of logic protect AI

- ✓ ***Obj add:*** Whether use the object address, you may select Yes or No.
- ✓ ***AI buffer size for relay:*** Set up the size of the buffer which used for storing protect AI in this module. Input range [0～65535].

4.2.6.8.2 Setting for parameters point by point

	AI name for relay	Obj addr	Source module	Source data name	Source data No.	a	b	c
1	Uab	0x00000000	IEDAC003	Protect AI	0	1	1	0
2	Ucb	0x00000001	IEDAC003	Protect AI	1	1	1	0

Fig. 2-49 Point by point parameter property of logic protect AI

1) ***AI name for relay:*** Name of the actual point which corresponds to this logic point. Editing is forbidden.
2) ***Obj addr:*** Setting is necessary according to specific protocol. Input range [an integer which consists of 4 bytes].
3) ***a:*** Say the value of a protect AI in RTU is X, And Y is the value should be sent to master station, Then Y=a*X/b + c.
4) ***b:*** Refer to a column.
5) ***c:*** Refer to a column.
6) ***Other Items:*** Editing is forbidden.

4.3 Interactive between software and device

4.3.1 Communication Setting

Right click the icon of the Engineer Tool in the Windows system, and then the menu of communication setting will pop out. Setting related property.



Fig.3-1Communication setting menu

- ***Settings****:* Open the link setting dialog.
- ***Offline:*** Disconnect the communication between Engineer Tool and main processor.
- ***Online:*** Start the communication between Engineer Tool and main processor.
- ***ViewBuf:*** Open the receiving and sending buffer interface of the port used for communication. Check the data in receiving and sending buffer..
- ***Quit:*** Exit the communication setting and return to the main interface of Engineer Tool.

4.3.1.1 Serial Port Communication

The COM1 port or other useable serial port of the computer shall be connected to the processor or I/O module by RS232 maintenance cable before starting the maintenance software. (If the computer only has USB port without RS232 serial port, USB-RS232 converter and the driver of converter shall be installed to use RS232 serial port maintenance processor or I/O module after conversion)

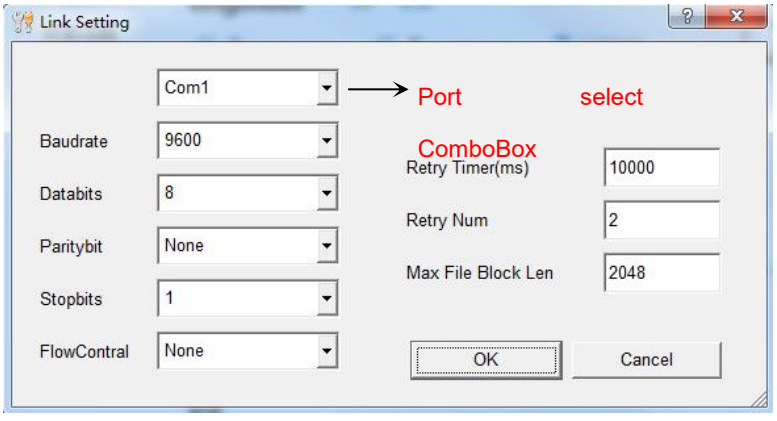


Fig.3-2 Communication setting

- ***Port***: The serial ports Com1～Com8 configured by the computer can be selected.
- ***Baud rate***: Within the range of 110～115200. Default: 9600.
- ***Data bits***: 5/6/7/8. Default: 8.

- ***Parity bit***: None/Odd/Even/Mark/Space. Default: None.
- ***Stop bits***: 1/1.5/2. Default: 1.
- ***Flow Control***: None/Hardware/XonXoff. Default: None.
- ***Retry Timer***: Can be set as required. Default: 1000ms.
- ***Retry Number***: Can be set as required. Default: 2.
- ***Max File Block Len***: 64～2048. Default: 2048 bytes.

Operate steps:

Step one: Click "Settings" or click to pop the "Link Setting" dialog.

Step two: Select serial port from combo box, as Fig. 3-2.

Step three: Set the parameters of serial port. (You can use the default parameters)

Step four: Click "OK". (Build the communication)

Notice: When the two interfaces of the computer and processor are on, insertion or pull-out of RS232 cable may damage the serial port of upper machine. RS232 optoelectronic isolator is suggested to adopt for this occasion.

4.3.1.1.1 Modify baud rate

Click the baud rate icon (also , , or , depending on the baud rate currently used) of tool bar, the "Modify baud rate" interface pops up and the communication baud rate of the Parameter Manager Tool can be set.

You may make modification when selecting required baud rate. If the modification is successful, "Modify OK" displays in the interface. The baud rate icon of the main menu changes into current baud rate icon.

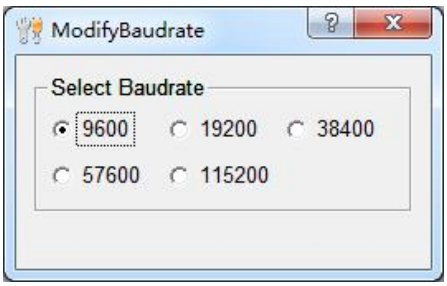


Fig. 3-3 modify baud rate

4.3.1.2 Ethernet Communication

4.3.1.2.1 Point to point Setting

The Ethernet port of the computer or processors shall be connected to the processor by Ethernet cable before starting the engineer tool.

Operate steps:

Step one: Click "Settings" or click to pop the "Link Setting" dialog.

Step two: Select "TCP/IP (Win Socket)" form combo box, as Fig. 3-4.

Step three: Set the parameters of Ethernet port.

Step four: Click “OK”. (Build the communication)

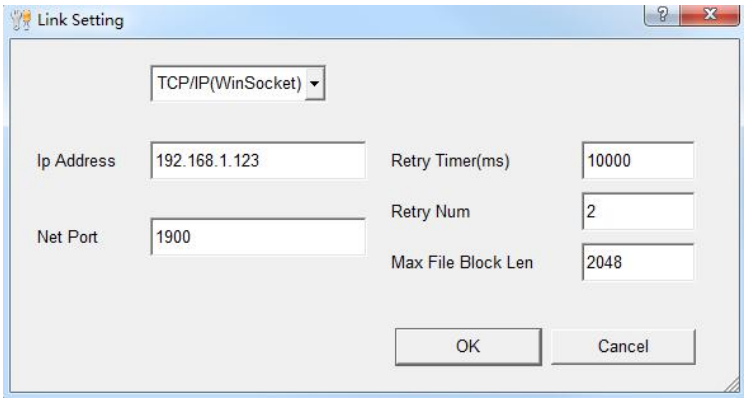


Fig.3-4 Ethernet Communication setting

- ***IP Address:*** IP address of device.
- ***Net Port:*** Net port. Default: 1900.
- ***Retry Timer:*** Can be set as required. Default: 1000ms.
- ***Retry Number:*** Retry number. Default: 2.
- ***Max File Block Len:*** 64～2048. Default: 2048 bytes.

Notice:

The “IP Address” refers to device IP address and it must be the same network segment with the computer or other device. The “Net Port” only can be set 1900 for the maintenance protocol.

The maintenance protocol uses the question-and-answer message and the “Retry Timer” refers to the time interval between the computer to send and receive messages.

4.3.1.2.2 Device auto search Setting

When you want to click the button , you can get all the IP addresses of devices in the LAN and point out its current active network card.

Device auto search function can let engineers quickly find the devices in the LAN through the network and positioning and maintain the device. You can click the button to start the function.

Search&Connect：

Step one: Click “Search” button, get all the IP addresss of decices. As Fig.3-5.

Step two: Select an ip from the list as fallow.

Step three: Click the button “Connect”, then you will connect the selected device and the dialog will hide. (You should make sure the selected IP address is the same segment with you computer IP address, otherwise the software will hint “Connected failed!”.)

Notice：If the IP address you selected is not the same segment with you computer’s, you can add a appropriate IP address to computer or modify the device’s IP.

Modify Device:

Step one: Selected an IP from the list as fallow. As Fig.3-6.

Step two: Input the new IP address you want to modify.

Step three: Click “Modify” button.

Notice：

The modify IP address must be the standard IPV4 address format, otherwise the soft will pop out warning message as Fig.3-7.

The modify IP address just add a templet IP to the net card and it will be disabled when the device reboot.

Positioning&Stop:

If there are a lot of equipments in the LAN, you need to positioning before maintain them.

Step one: Selected an IP from the list.

Step two: Click “Positioning” button and now the correct device’s front leds will flashing. (The button will change to “Stop” button as Fig.3-8)

Step three: Click “Stop” button to let the device’s leds stop flashing.

Notice: Devices positioning don’t need computer to build connection with the device, and also needn’t in the same segment.

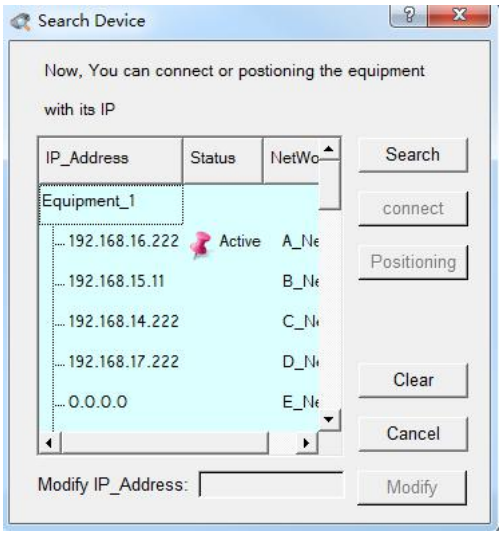


Fig. 3-5 Search device

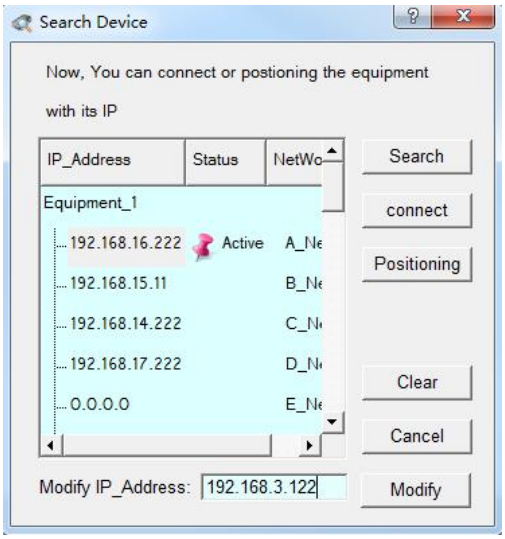


Fig.3.6 Modify IP Address

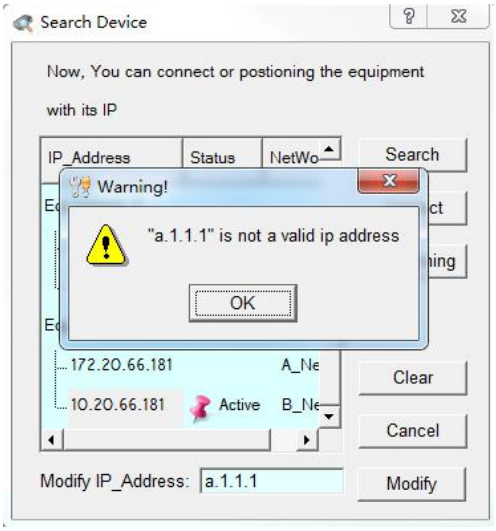


Fig.3-7 IP address warning

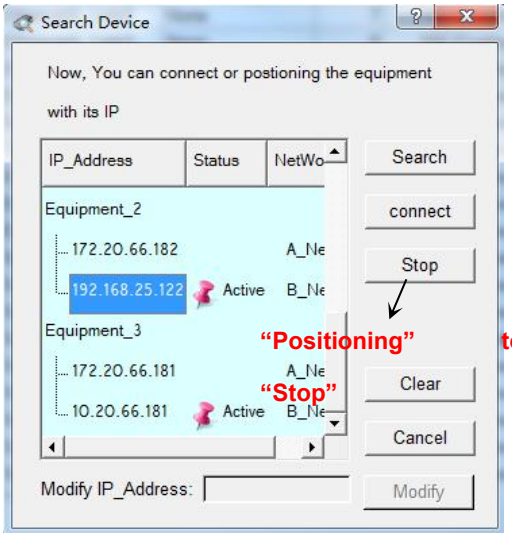


Fig.3-8 Positioning the device

- ***Search:*** Search the MCU's IP addresses in the LAN.
- ***Connect:*** Select an IP item and build connection
- ***Positioning:*** Positioning the device. Select an IP item and click the Positioning button, the LEDs of device will flash.
- ***Clear:*** Clear the list.
- ***Cancel:*** Cancel and exit.
- ***Modify IP:*** Add a temporary IP address to the MCU in order to connect the MCU with the computer.

4.3.1.3 Communication test

Sometimes the computer and the device already build physical connection, but due to the baud rate or head shot isn't consistent, they can't exchange the data. Now we should

test the serial and Etherent link communication. Click the icon pop out the communication test dialog.

Operate steps:

Step one: Select “COM” or “TCP/IP(winSocket)” from combo box. As Fig.3-9.

Step two: Click “Link” button.

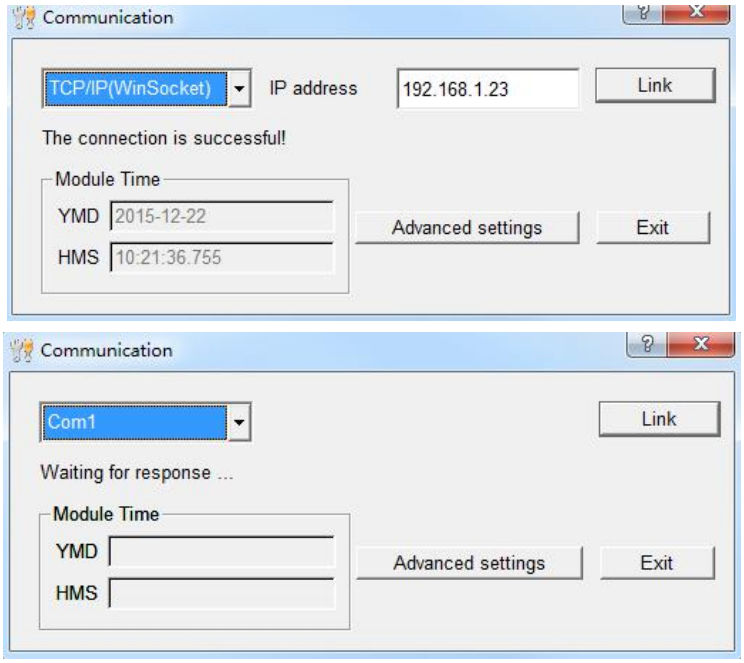


Fig.3.9 Communication test

- ***Module Time:*** Link successful, and show current time of the MCU.
- ***BeginTest:*** Begin test, and it can test etherent port and serial port com1~com8.
- ***Advanced settings:*** Set some special communication parameters.
- ***Exit:*** Exit the dialog.

When you click “Advanced settings”, it will pop out the dialog as Fig. 3-10.

- ***Wait time(ms):*** Time of waiting the response of device. Default: 1000.
- ***Retry times:*** Retry times. Default: 2.
- ***Max file block:*** Max file block. Default: 2048.
- ***Frame header:*** tradition frame header 0x01; new frame header 0xA5A5

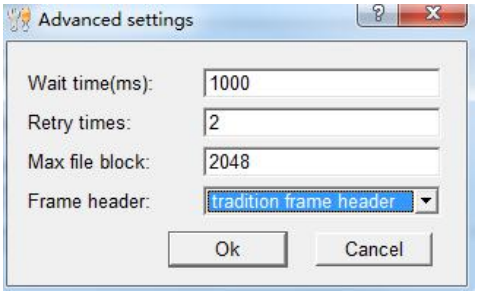


Fig.3.10 Communication advanced setting

4.3.2 File Management

The engineer tool soft ware configures the equipment through parameters files; it involves the upload/download files, program burning and permissions management.

4.3.2.1 Authority Management

When launch the software, it will pop-up the login interface firstly.



Fig. 3-11 The login interface

Input correct user name and password, then click ”Login” button and you will enter the main interface of the engineer tool soft ware.

There is three-level permission: Administrator, Maintainer and Inspector.

Inspector: The least permission, it only can do data inquiring and other basic operation.

Maintainer: The second permission, it can download/upload parameters files and the permission of Inspector.

Administrator: The highest permission, it can do add/delete user, change permission level, download program download and etc.

Operating steps:

Step one: Select “Monitor” from menu bar and click “Account manage”, as Fig.3-12. The software pop out “Login” interface, as Fig.3-13.

Step two: Use administrator account, click “Login” button. Then the software will pop out “Administrator Account Interface”, As Fig.3-14.

Step three: If you want to modify administrator’s password, please input current password and new password. And if you want add/delete/modify user account, please click “User account” button. The software will pop out “User Account Interface” as Fig.3-15.

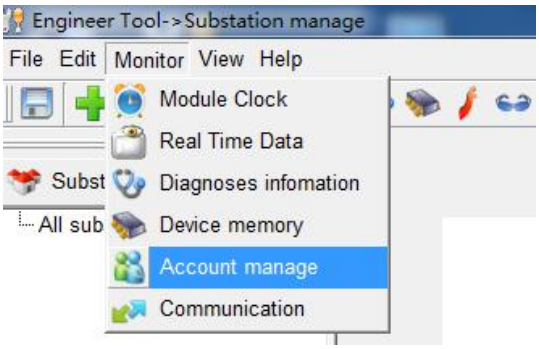


Fig. 3-12 Account manage



Fig. 3-14 Administrator Account Interface



Fig. 3-13 Administrator Login

Step four: Add/delete/modify the user account and click “OK” button

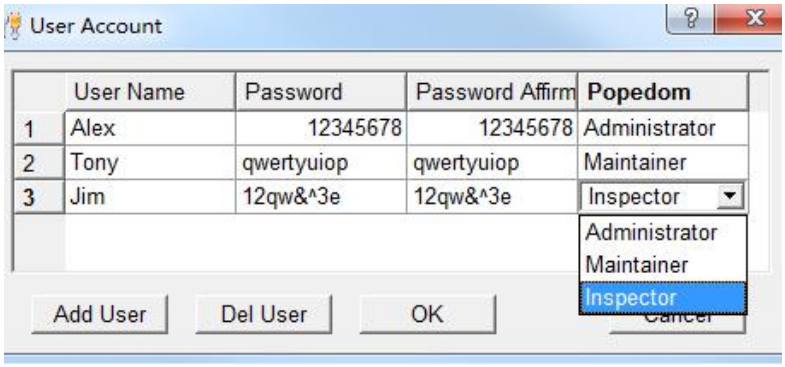


Fig.3-15 User Account Interface

Notice: Set the appropriate operation permission and password for each account. If the password is not consistent the software will pop out warning message.

4.3.2.2 Download/Upload Files

Maintainer configures parameters for RTU through engineer tool software, and then downloaded the parameter files to the device. At the same time you can upload the parameter files to the PC from the device through the serial port or net port. The engineer tools can use FTP or special method to download/upload the parameter files.

4.3.2.2.1 Special Method

Special method refers to using the user-defined maintenance protocol to transfer the parameter files, part the file according to specify size and do CRC check between the data transmission. This method can be used under serial or Ethernet communication.

Operating steps

Step one: Right click the “processor” item from the list of IDE tree of left side; it will pop out the menu as Fig.3-16.

Step two: Click one item, the software will pop out the “Down/Up Load Files Dialog” as Fig.3-17.

Step three： Selected files you want to transfer and click upload/download button. Then the progress bar will show the current progress of transmission.

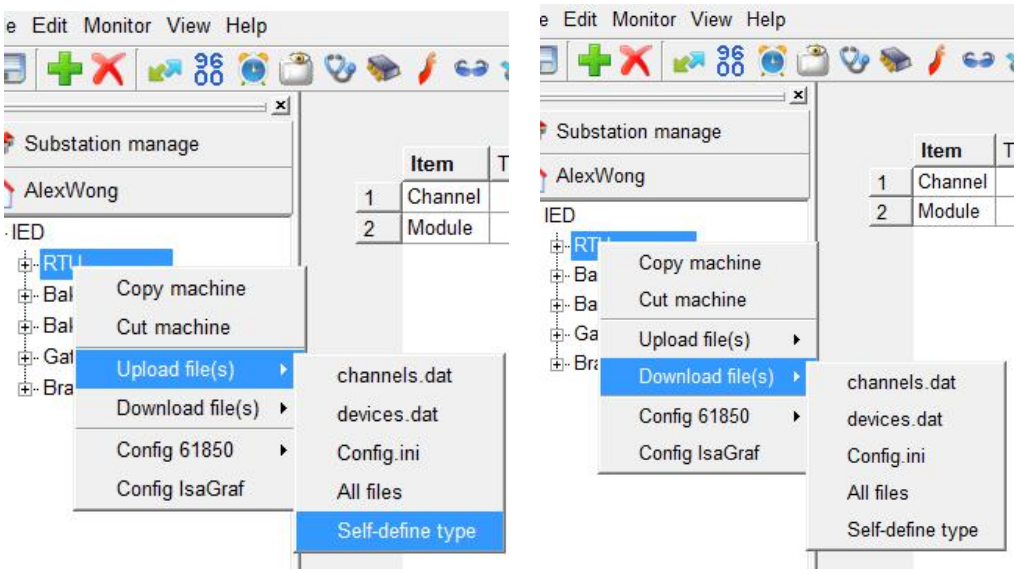


Fig.3-16 Download/Upload launch

- ***Channels.dat***: Only download/upload the channels.dat file.
- ***Device.dat***: Only download/upload the device.dat file.
- ***Config.ini***: Only download/upload the config.dat file.
- ***All file(s)***: Download/Upload all files of the current processor you selected.
- ***Self-define type***: Only pop out "Down/Up Load Files Dialog".

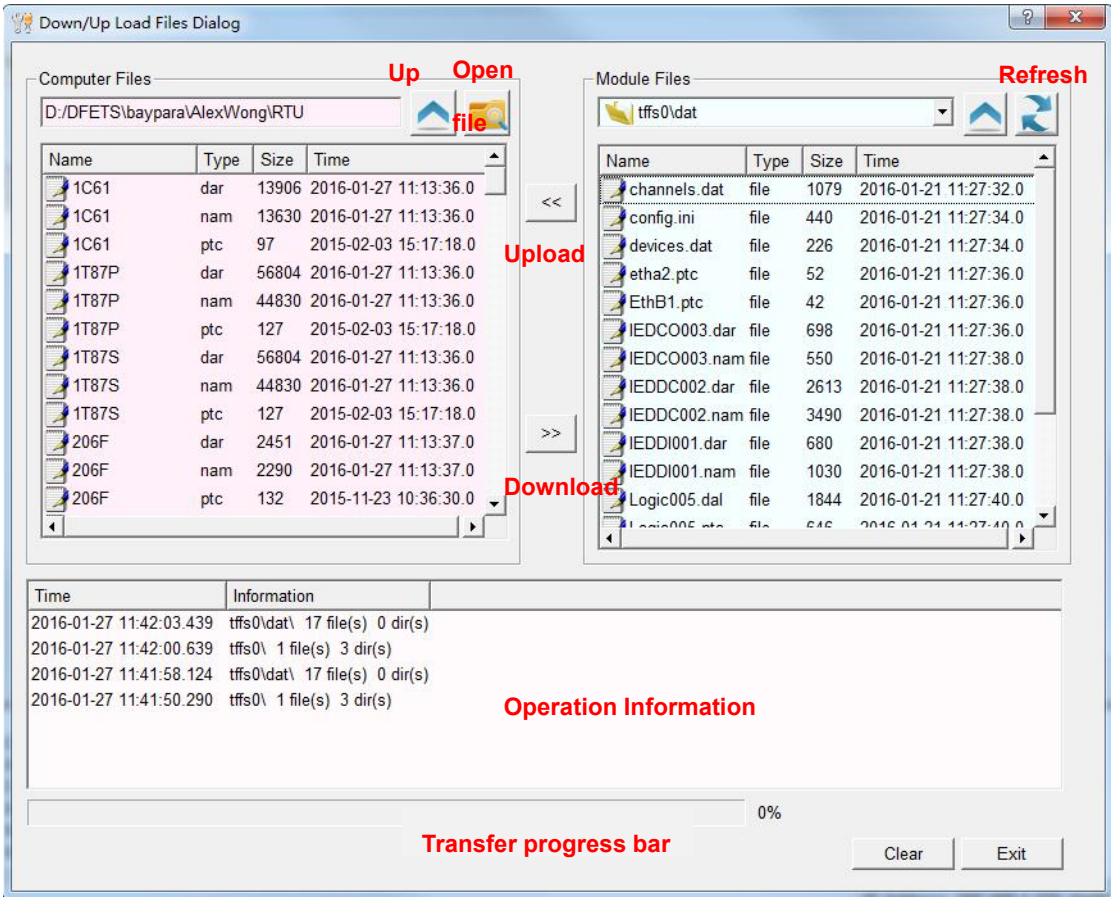


Fig.3-17 Down/Up Load Files Dialog

4.3.2.2.2 FTP Method

The engineer tool software add ftp method for quickly transferring files, but this method only can be used under Ethernet.

Operating steps

Step one: Click button at tool bar, the software will pop out the dialog as Fig3-18.

Step two: The software will auto add the “Ftp Server” address according to the current link information and you should input the “User Name”, “Password” and “Port”. Then click “Connect” button.

Step three: Selected files you want to transfer and click upload/download button. Then the progress bar will show the current progress of transmission.

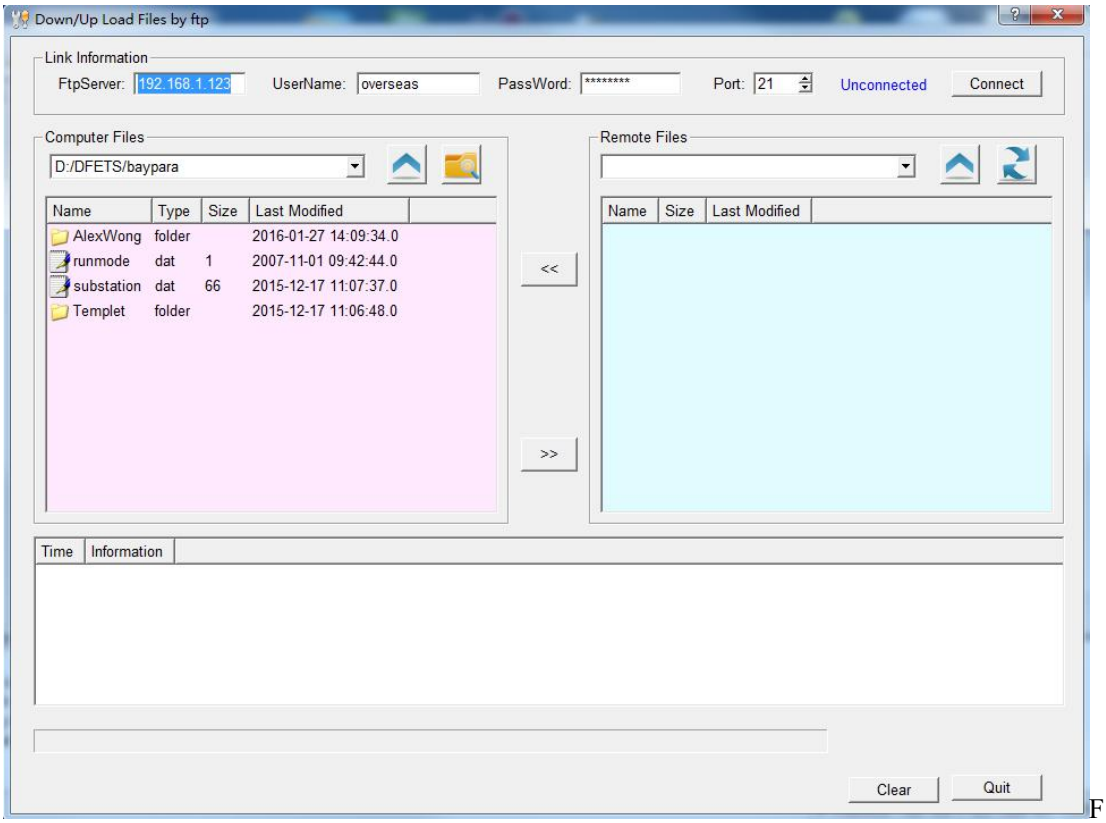


Fig.3-18 Down/Up Load Files by FTP

Notice：There is some short cut as Fig.3-18, you can use right click to pop it.

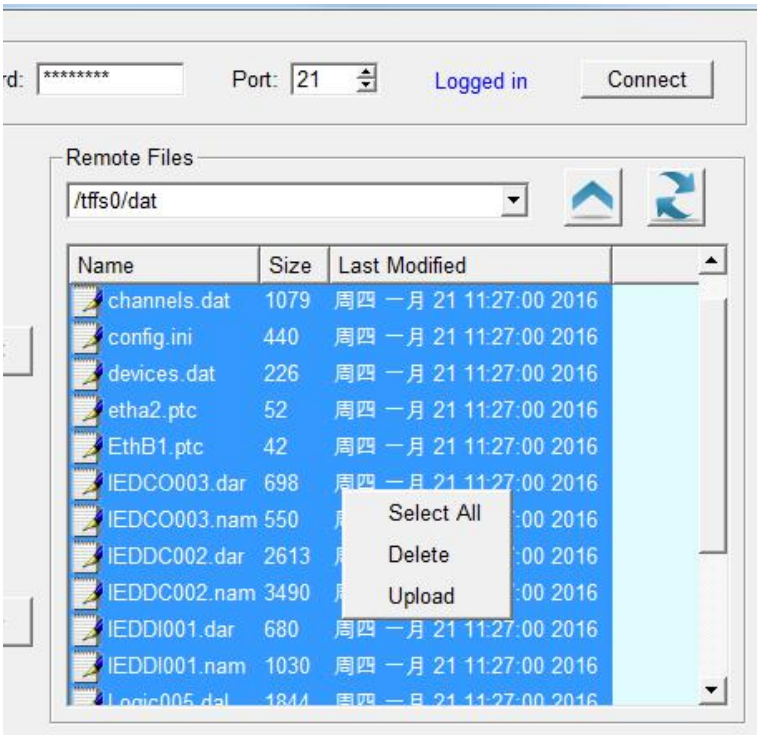


Fig.3-19 short cut of files transfer

4.3.2.3 Download Program

There are three methods that used to download the software of DF1725IED through maintenance com or net port. (vxWorks_romCompress.bin)

Download program by the serial terminal

Step one: Pressing “a” on the keyboard and restart the IED then input “s” behind Command>: and ENTER, as Fig. 3-20.

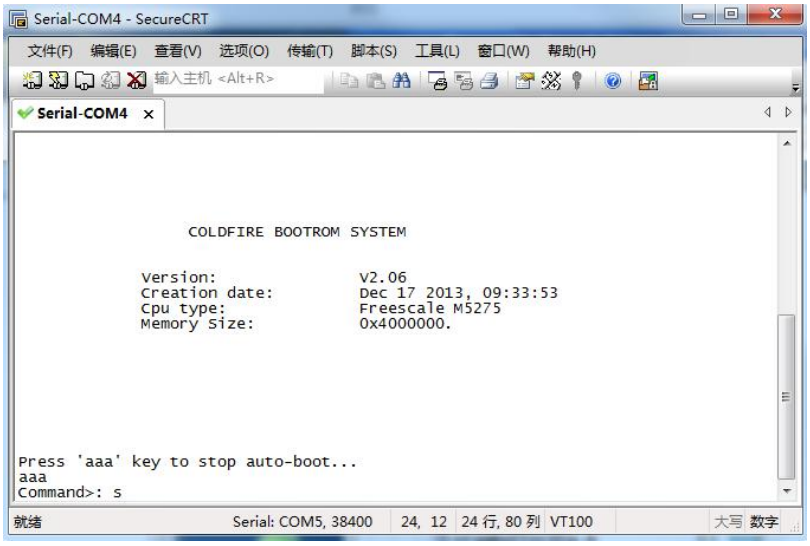


Fig. 3-20 Interface of SecureCRT

Exit from the Super Terminal or SecureCRT, and open the maintenance software, and select self-define type to download the program, as Fig. 3-20.

Step two: Download the vxWorks_romCompress.bin by self-define type.

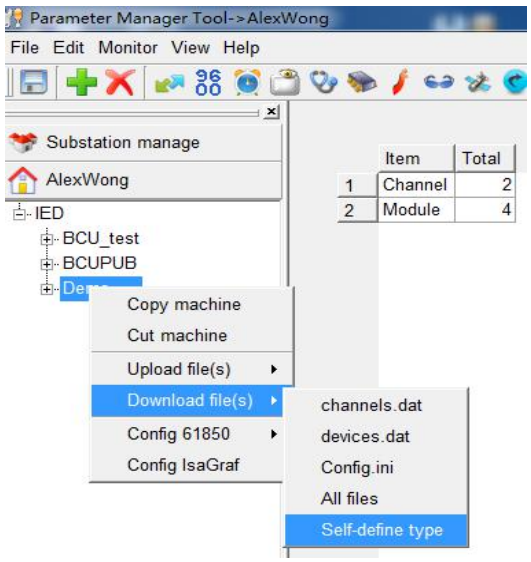


Fig. 3-21 Select self-define type to download files

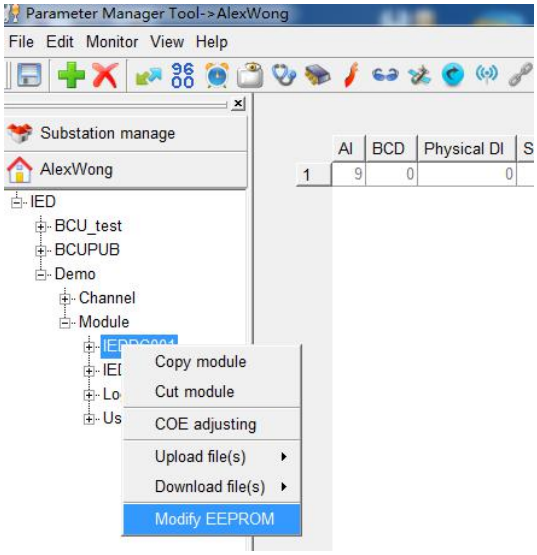


Fig. 3-23 Modify EEPROM

Step three: Double click the vxWorks_romCompress.bin file or chick >>, and then select OK after finishing. Restart IED and work in new main program.

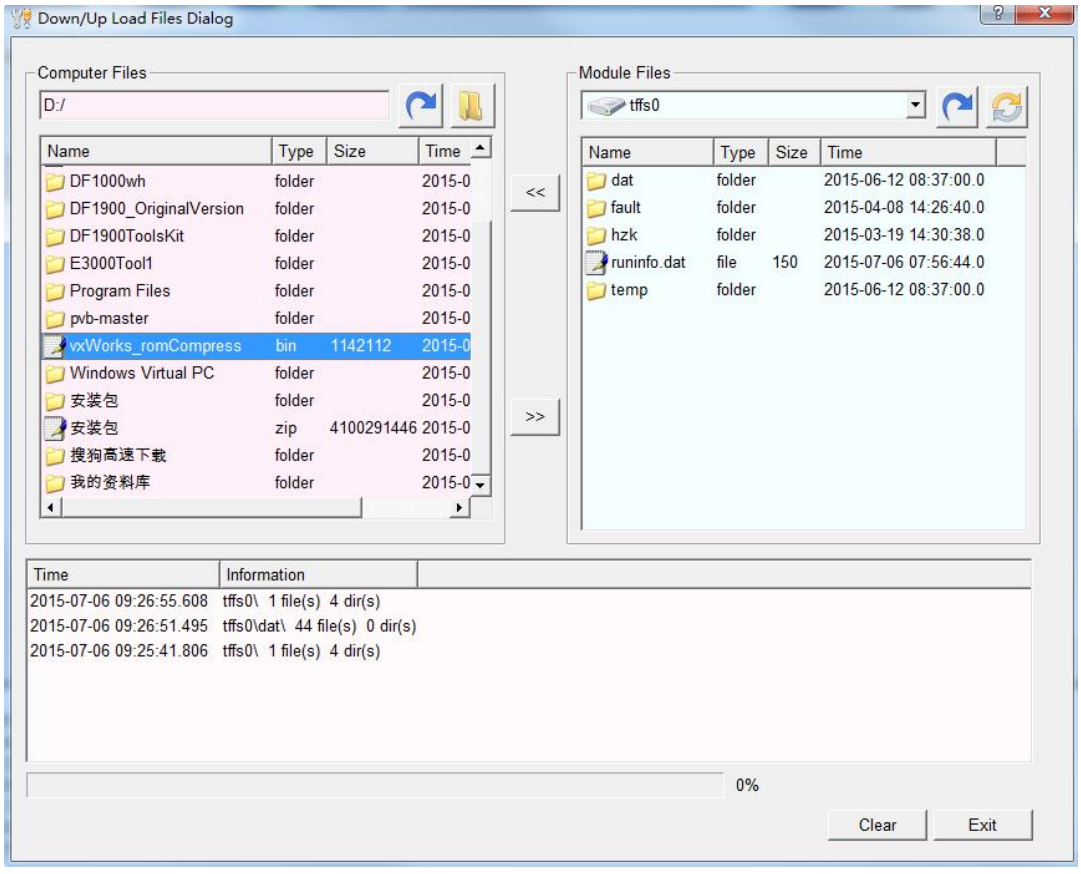


Fig. 3-22 Download the main program file

Download program by modify EEPROM.

Step one: Open the maintenance software, and then right click any module under module level as Fig.3-23.

Step two: The soft will pop out “SetEeprom” dialog, select 1725IED_SYS and click “Read Data” button, as Fig.3-23.

Step three: Changing RUNFLAG (0)1234 to 8888, then click “Write Data” button and restart the MCU.

Step four: Download the vxWorks_romCompress.bin by self-define type.

Step five: Changing RUNFLAG (0) from 8888 to 1234, then click ‘Write Data’ and restart IED and work in new main program now.

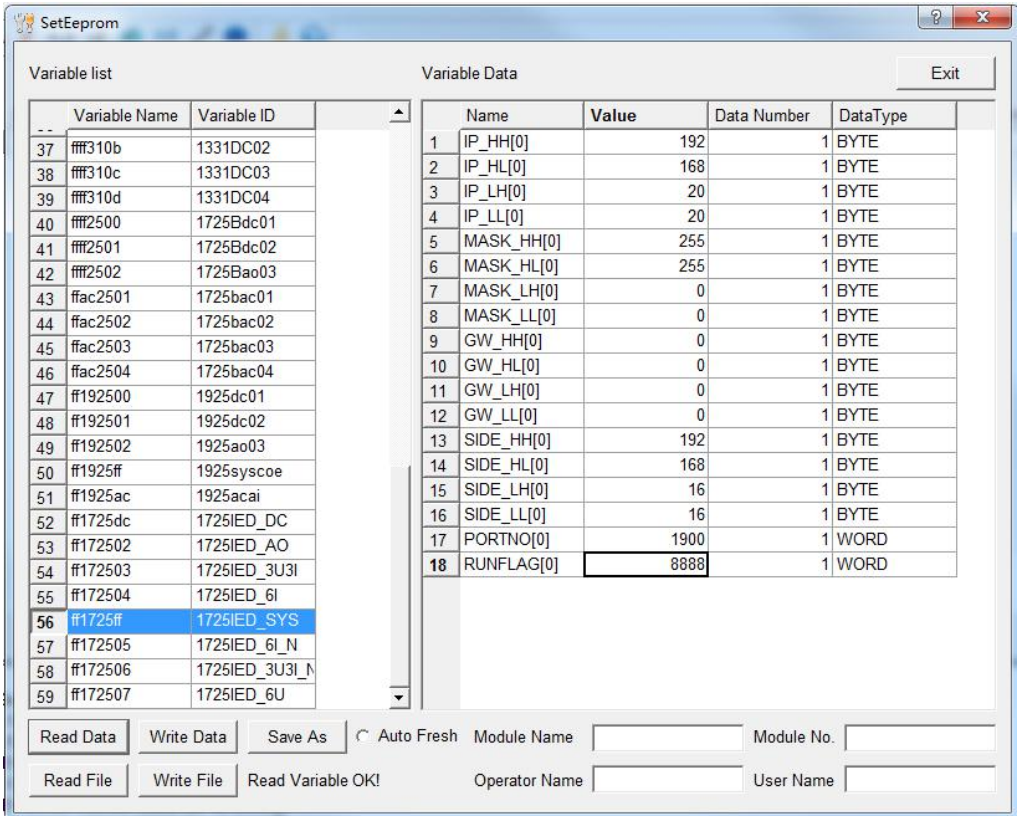


Fig. 3-24 Interface of modify EEPROM

Rapid burning by ftp

Only the program above ver.4.02.001 of MCU support rapid burning by ftp. Otherwise please use the method above.

Step one: Clicking the button in the toolbar of engineer tool software.

Step two: As Fig. 3-25, select the file (vxWorks_romCompress.bin) from the computer.

Step three: The software will send the file to the MCU, then the MCU will erasing the original program and flashing the new one automaticly, as Fig. 3-26.

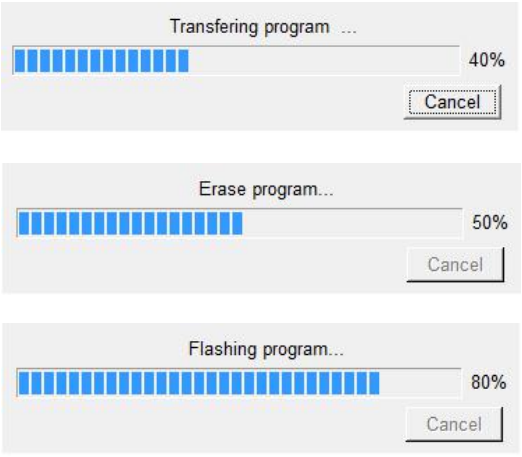


Fig. 3-26 Flash program by ftp automatically

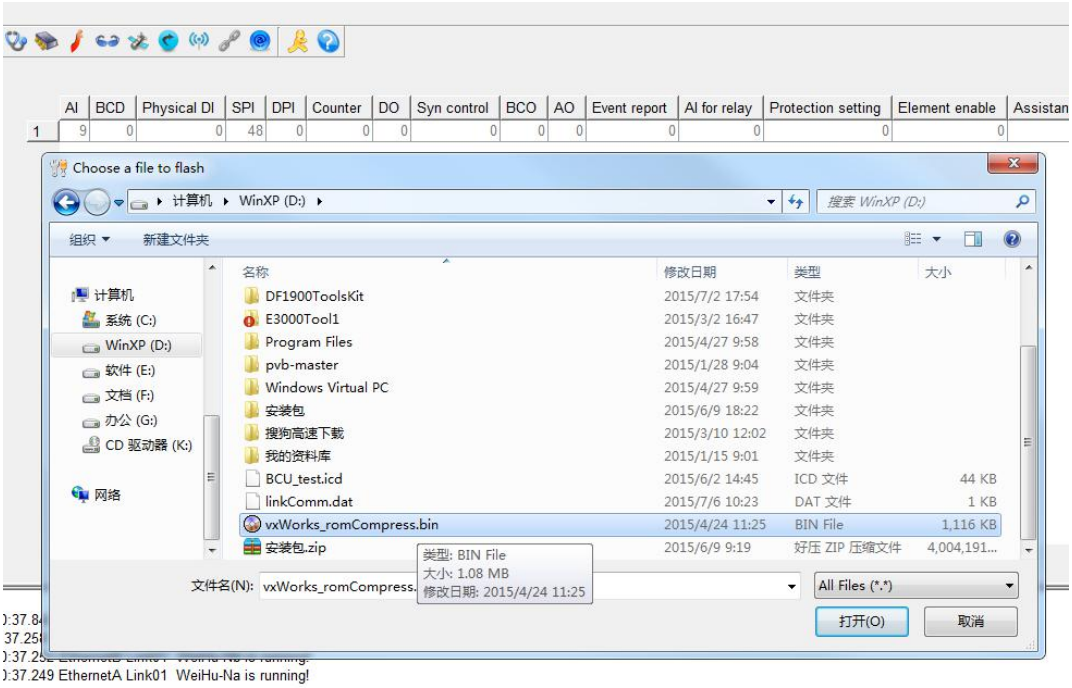


Fig. 3-25 Select the program file



Fig. 3-27 Progress burning successful

Waitting, until the software hint you “Progrom flash successful!”, restart the device.

Notice: Keep in communication when you flash the program file

4.3.3 Data View

When the device connects with PC, the engineer tool software will exchange data with the device through the maintenance protocol.

4.3.3.1 View Data Buffer

Right click the icon of the Engineer Tool in the windows system, and then the menu of communication setting will pop out and click the “ViewBuffer”.



Fig.3-28 Raw Buffer

- ***Fresh***: Start current connection.
- ***Pause***: Pause current connection.
- ***Clear***: Clear the data of list.
- ***Exit***: Close but keep connection.
- ***Txd***: Message to the device.
- ***Rxd***: Message from the device.

- ***Identification***: 434 is the number of message(0~1000) and 19302 is the current time(format of "sszzz")

Notice: The engineer tool software used the maintenance protocol to exchange the data by ask-answer, usually the send and receive messages come in pairs.

4.3.3.2 View Rxd/Txd Buffer

The engineer tool software can view messages of different channels. When you want to view it, please make sure the connection and the project setting is right.

Operate step：

Step one： The current level can't be "Station" or "IED", you must select a project and then click from the tool bar, it will pop out the interface as Fig.3-29.

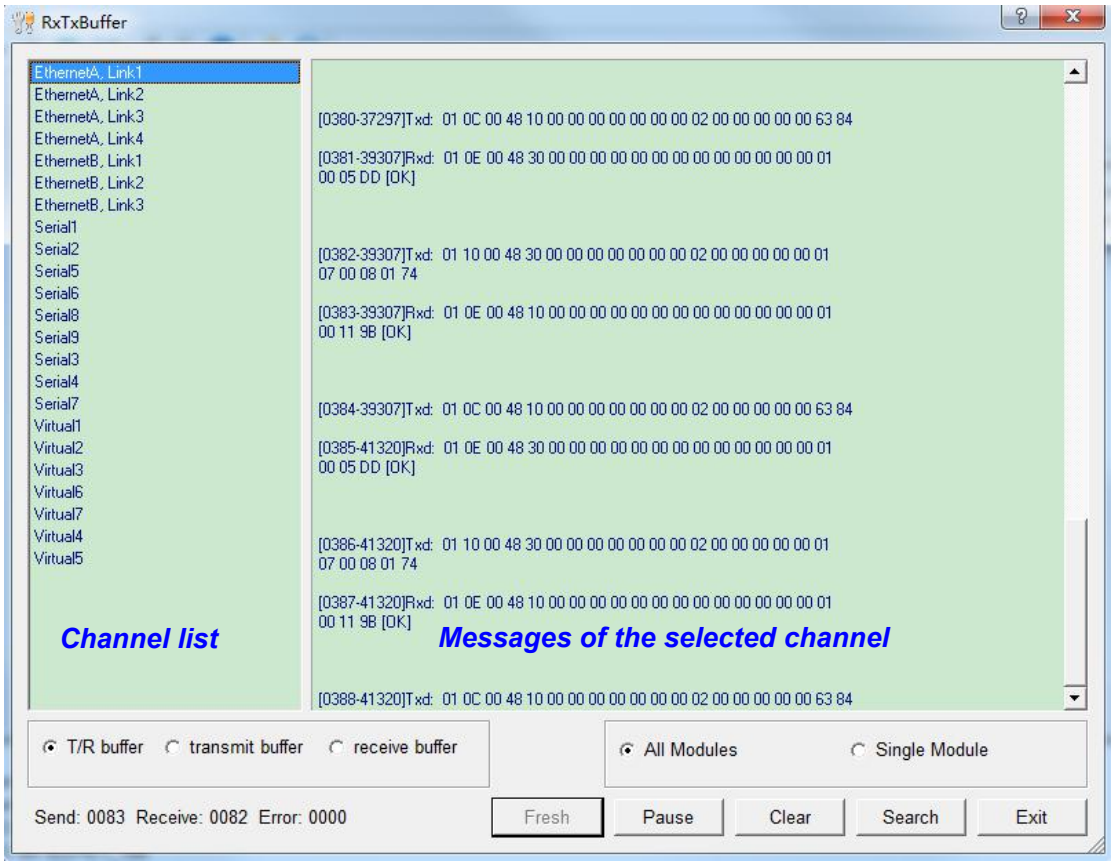


Fig.3-29 Rx/Tx Buffer Interface

- ***Fresh***: Start the current connection.
- ***Pause***: Pause the current connection.
- ***Clear***: Clear the data of list.
- ***Search***: Search the code from the list and mark it with different color, as

Fig.3-30.

- ***T/R buffer***: Display the transmit and receive buffer.
- ***Transmit buffer***: Just display the transmit buffer
- ***Receive buffer***: Just display the receive buffer.
- ***All Modules***: Watch all module data.
- ***Single Module***: Watch single module data.

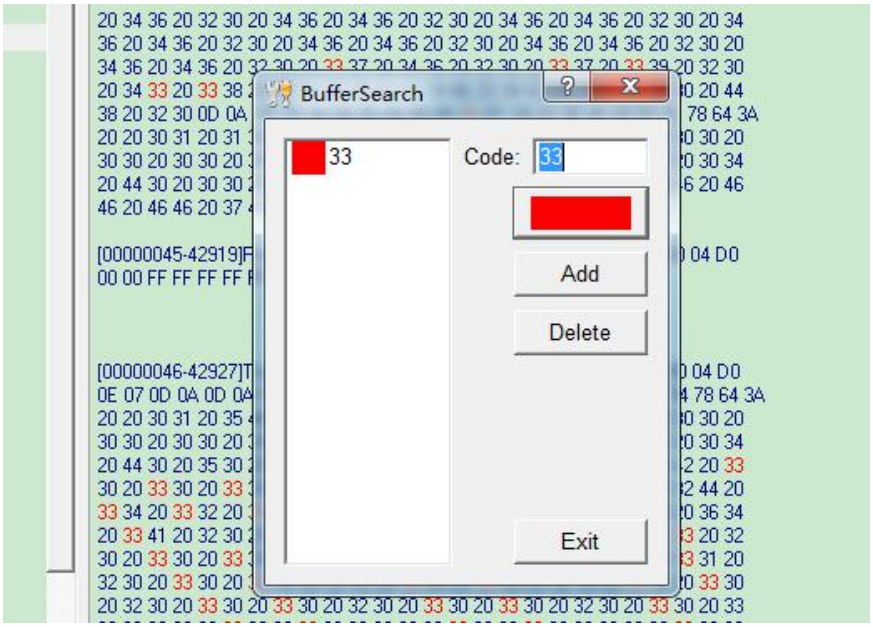


Fig.3-30 the buffer search interface

4.3.3.3 Event Query

The engineer tool software can view the running information, operation information and self-check information, the running information and the self-check information will display in the bottom dock window and it will check new information on time.

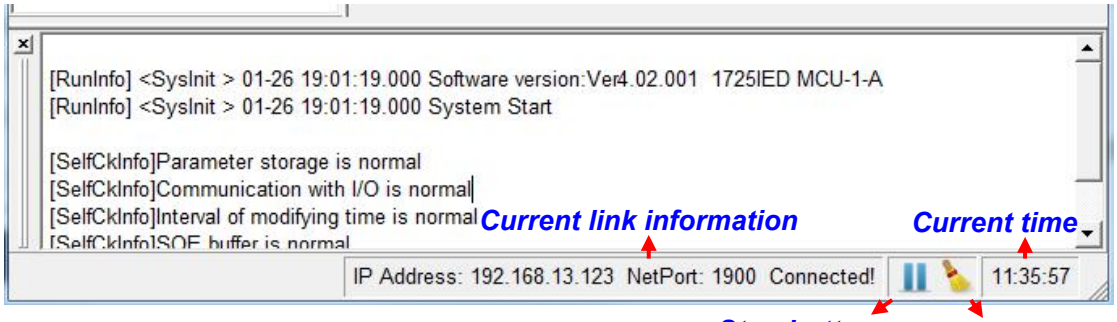


Fig3-31 the bottom dock window

Operate step:

Step one: Click from the tool bar, it will pop out the interface as Fig.3-31.

Step two: Select the information type you want to get.

Step three: Click “Fresh” button.

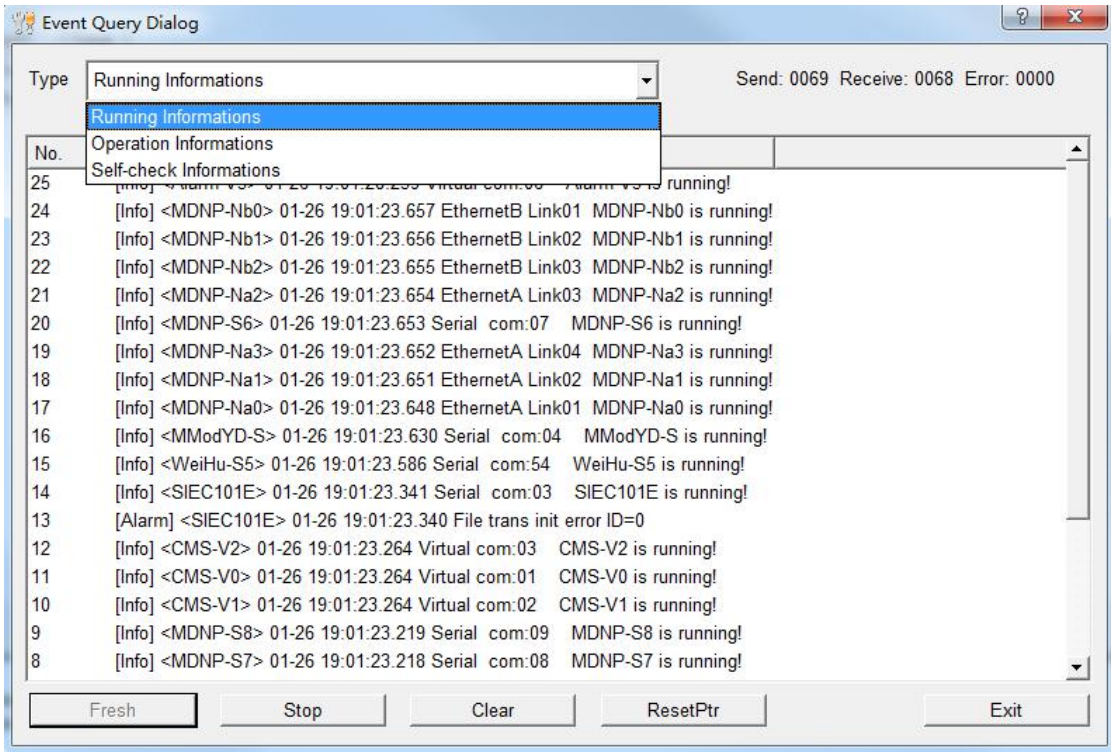


Fig3-32 Event query interface

- ***Fresh***: Get current event information of the device.
- ***Stop***: Stop questioning current event information.
- ***Clear***: Clear the data above.
- ***ResetPtr***: Reset the pointer and question current event information renew.
- ***Exit***: Close the dialog.

Notice: The engineer tool will always query event information, upper left corner of the interface are given the current number of inquiry, response, and error messages number.

4.3.3.4 View and modify memory

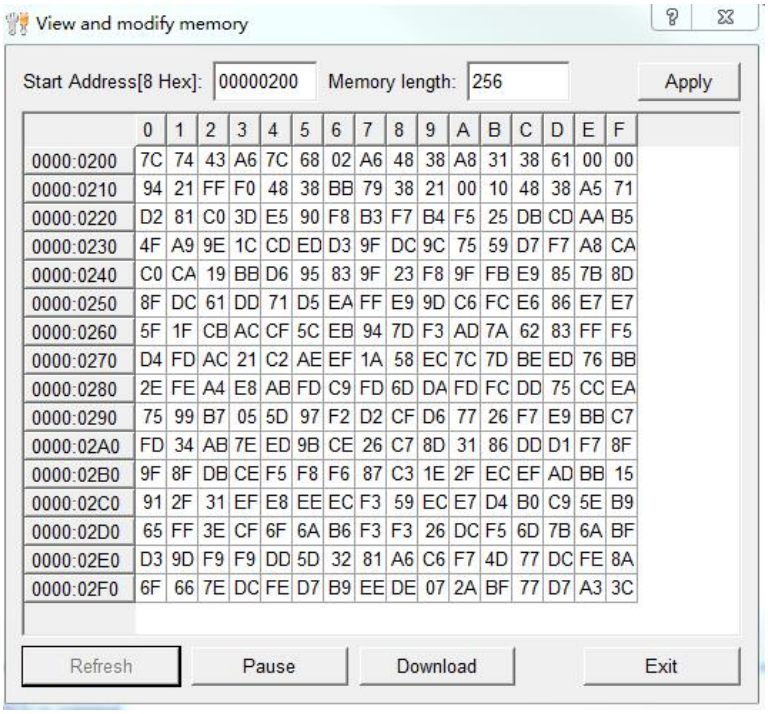
The equipment memory cannot be modified at will and it is only used for module ex-work debugging by technicians.

Operate step:

Step one: Click the icon  to open the interface of View and modify memory (View and modify memory).

Step two: Input the start address, memory length and click “Apply” button.

Step three: Click “Refresh” button.



	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000:0200	7C	74	43	A6	7C	68	02	A6	48	38	A8	31	38	61	00	00
0000:0210	94	21	FF	F0	48	38	BB	79	38	21	00	10	48	38	A5	71
0000:0220	D2	81	C0	3D	E5	90	F8	B3	F7	B4	F5	25	DB	CD	AA	B5
0000:0230	4F	A9	9E	1C	CD	ED	D3	9F	DC	9C	75	59	D7	F7	A8	CA
0000:0240	C0	CA	19	BB	D6	95	83	9F	23	F8	9F	FB	E9	85	7B	8D
0000:0250	8F	DC	61	DD	71	D5	EA	FF	E9	9D	C6	FC	E6	86	E7	E7
0000:0260	5F	1F	CB	AC	CF	5C	EB	94	7D	F3	AD	7A	62	83	FF	F5
0000:0270	D4	FD	AC	21	C2	AE	EF	1A	58	EC	7C	7D	BE	ED	76	BB
0000:0280	2E	FE	A4	E8	AB	FD	C9	FD	6D	DA	FD	FC	DD	75	CC	EA
0000:0290	75	99	B7	05	5D	97	F2	D2	CF	D6	77	26	F7	E9	BB	C7
0000:02A0	FD	34	AB	7E	ED	9B	CE	26	C7	8D	31	86	DD	D1	F7	8F
0000:02B0	9F	8F	DB	CE	F5	F8	F6	87	C3	1E	2F	EC	EF	AD	BB	15
0000:02C0	91	2F	31	EF	E8	EE	EC	F3	59	EC	E7	D4	B0	C9	5E	B9
0000:02D0	65	FF	3E	CF	6F	6A	B6	F3	F3	26	DC	F5	6D	7B	6A	BF
0000:02E0	D3	9D	F9	F9	DD	5D	32	81	A6	C6	F7	4D	77	DC	FE	8A
0000:02F0	6F	66	7E	DC	FE	D7	B9	EE	DE	07	2A	BF	77	D7	A3	3C

Fig. 3-33 Dialog box of memory viewing

- ***Start Address (8 Hex):*** Set the start address of memory viewing and modifying.
- ***Memory length:*** To check the memory length, press “Apply” to notify the module.
- ***Refresh:*** Refresh the displayed content.
- ***Pause:*** Pause refreshing.
- ***Download:*** Download the modified memory into the module.
- ***Exit:*** Exit.

4.3.3.5 View real time data

Click the icon  (Real Time Data) of the main menu to see the real time data. The data value line of the wanted data list will display the real time data.

If there is analog input data, the real time data will display the analog input events which may occur.

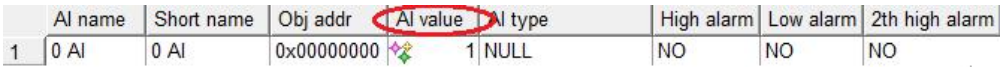
If there is input, the real time data will display the digital input events and SOE which may occur.

To stop refreshing, click the button of “Stop refresh” and then click the button of “Start refresh” to refresh again.

4.3.3.5.1 View real time data of the physical module

Analog input

The “Analog input value” line will be added in the analog input list. When communication is normal, real time analog input value displays.



	AI name	Short name	Obj addr	AI value	AI type	High alarm	Low alarm	2th high alarm
1	0 AI	0 AI	0x00000000	1	NULL	NO	NO	NO

Fig. 3-34 View real time data of analog input

Analog input events

When viewing the real time data of analog input, analog input events which may exist also displays as following:

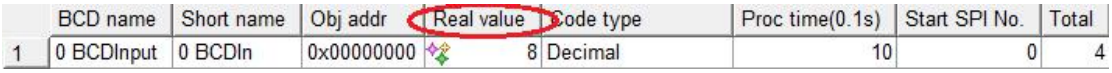


AI No.	Event type	AI Value over limit	Time	Obj addr	AI Name

Fig. 3-35 View analog input events

Step position information

The “Real time value” line will be added in the step position information list. When communication is normal, the real time step position information value displays.

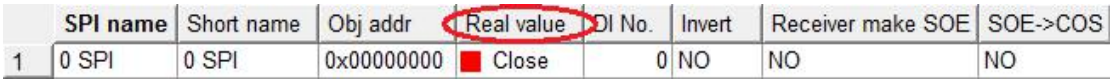


	BCD name	Short name	Obj addr	Real value	Code type	Proc time(0.1s)	Start SPI No.	Total
1	0 BCDInput	0 BCDIn	0x00000000	8	Decimal	10	0	4

Fig. 3-36 View real time data of step position

Single point input

The “Real time value” line will be added in the single point input list. When communication is normal, the real time single point input value displays.

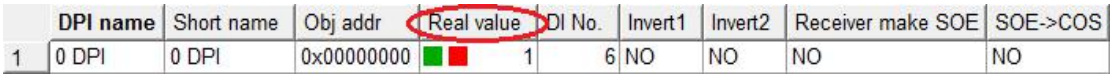


	SPI name	Short name	Obj addr	Real value	DI No.	Invert	Receiver make SOE	SOE->COS
1	0 SPI	0 SPI	0x00000000	Close	0	NO	NO	NO

Fig. 3-37 View real time data of single point input

Double points input

The “Real time value” line will be added in the double points input list. When communication is normal, the real time double points input value displays.



	DPI name	Short name	Obj addr	Real value	DI No.	Invert1	Invert2	Receiver make SOE	SOE->COS
1	0 DPI	0 DPI	0x00000000	1	6	NO	NO	NO	NO

Fig. 3-38 View real time data of double points input

Digital input events

When viewing the real time data of single point or double points input, digital input events which may exist also displays in following form:

	DI No.	Time tag	DI Value	Time	Obj addr	DI Name
1	5	✓	Close	2015-04-06 15:16:33.678	0x00000005	5 SPI
2	5	✓	Open	2015-04-06 15:16:34.069	0x00000005	5 SPI

Fig. 3-39 View digital input events

SOE

When viewing the real time data of single point or double points input, the SOE which may exist also displays in following form:

	DI No.	Time tag	DI Value	Time	Obj addr	DI Name
1	5	✓	Close	2015-04-06 15:16:33.678	0x00000005	5 SPI
2	5	✓	Open	2015-04-06 15:16:34.069	0x00000005	5 SPI

Fig. 3-40 View SOE

Counter

The “Real time value” line will be added in the counter list. When communication is normal, the real time counter value displays.

	Counter name	Short name	Obj addr	Real value	DI No.	Counter type	DPI Proc(s)
1	0 Counter	0 Counter	0x00000000	2	10	Pulse:SPCHC	0

Fig. 3-41 View real time data of the counter

Protect AI

The “Analog input value” line will be added in the protect analog input list. When communication is normal, real time analog input value displays.

	AI name	Short name	Obj addr	AI value	IO force to 0	Main force to 0	Index	AI num under this index
1	0 BhAI	0 BhAI	0x00000000		NO	NO	0	0
2	1 BhAI	1 BhAI	0x00000001		NO	NO	0	0

Fig. 3-42 View real time data of the Protect AI

Soft BO

The “Element enable status” line will be added in the element enable input list. When communication is normal, real time element enable status value displays.

	Element enable name	Short name	Obj addr	Element enable status	Enable
1	0 SoftBO	0 SoftBO	0x00000000		NO
2	1 SoftBO	1 SoftBO	0x00000001		NO

Fig. 3-43 View real time data of the Soft BO

Action report & Alarm report

When viewing the real time data of event report, report events which may exist also displays in following form:

Device para ID	Device name	Device Type	Event Type	INF	Event Name	Status	Start time	Action time	Fault I

Fig. 3-44 View real time data of the event report

4.3.3.5.2 View real time data of the logic module

Analog input

The “Original value” and “Transmission value” lines will be added in the analog input list. When communication is normal, the original value and transmission value of analog input display.

	AI name	Raw valu	Transmitted value	Source module	Source data type	Source data No.
1	0 AI	1	1	IEDDC001	AI	0

Fig. 3-45 View real time data of the logic analog input

Analog input events

When viewing the real time data of analog input, analog input events which may exist also displays as following:

AI No.	Event type	AI Value over limit	Time	Obj addr	AI Name

Fig. 3-41 View analog input events

Step position information

The “Real time value” line will be added in the step position information list. When communication is normal, the real time step position information value displays.

	BCD name	Real value	Source module name	Source data type	Source data No.
1	0 BCDInput	2	IEDDI002	BCD	0

Fig. 3-46 View real time data of step position information

Single point input

The “Real time value” line will be added in the single point input list. When communication is normal, the real time single point input value displays.

	SPI name	Short name	Obj addr	Real value	DI No.	Invert
1	1 SPI	1 SPI	0x00000001	Open	1	NO

Fig. 3-47 View real time data of single point input

Double points input

The “Real time value” line will be added in the double points input list. When communication is normal, the real time double points input value displays.

	DPI name	Short name	Obj addr	Real value	DI No.	Invert1	Invert2
1	0 DPI	0 DPI	0x00000000	1	10	NO	NO

Fig. 3-48 View real time data of double points input

Digital input events

When viewing the real time data of single point or double points input, digital input events which may exist also displays in following form:

	DI No.	Time tag	DI Value	Time	Obj addr	DI Name
1	8	✓	Open	2015-04-08 18:22:35.836	0x00000009	9 SPI
2	0	✓	1	2015-04-08 18:22:37.037	0x00000000	0 DPI

Fig. 3-49 View digital input events

Counter

The “Original value” and “Transmission value” lines will be added in the counter list. When communication is normal, the real time original value and transmission value of the counter display.

	Counter name	Raw value	Transmit value	Freeze	Interval
1	0 Counter	9	9	Non-reset	1 hour

Fig. 3-50 View real time data of counter

SOE

When viewing the real time data of single point or double points input, the SOE which may exist also displays in following form:

	DI No.	Time tag	DI Value	Time	Obj addr	DI Name
1	8	✓	Open	2015-04-08 18:22:35.836	0x00000009	9 SPI
2	0	✓	1	2015-04-08 18:22:37.037	0x00000000	0 DPI

Fig. 3-52 View SOE

Protect AI

The “Analog input value” line will be added in the protect analog input list. When communication is normal, real time analog input value displays.

	AI name for relay	AI value	Obj addr	Source module	Source data name	Source data No.	a	b	c
1	Uab		0x00000000	IEDAC003	Protect AI	0	1	1	0
2	Ucb		0x00000001	IEDAC003	Protect AI	1	1	1	0

Fig. 3-53 View real time data of the Protect AI

Action report & Alarm report

When viewing the real time data of event report, report events which may exist also displays in following form:

Device para ID	Device name	Device Type	Event Type	INF	Event Name	Status	Start time	Action time	Fault I

Fig. 3-54 View real time data of the event report

4.3.4 Other function setting

4.3.4.1 Authority Management

Click the icon  to open the interface of clock synchronization to view the clock of the processor. The clock can be changed into computer time.

When the communication is normal, the “Module Time” displays in the interface including year, month, day, hour, minute and second. The “Modify Time” displays the real time of computer. Click the button of “Modify” to modify the module time according to the current time of computer.

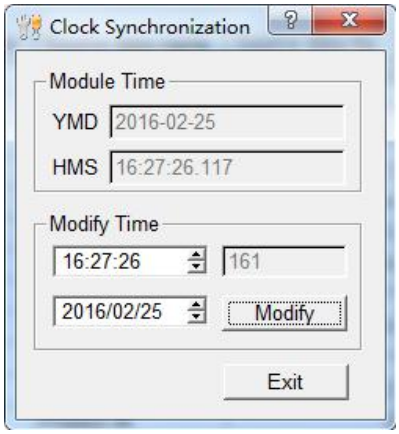


Fig. 3-55 Clock Synchronization

4.3.4.2 GPS time zone setting

Set the GPS board time zone. If there is no GPS board, skip it. Click the icon  and open the GPS time zone setting interface.

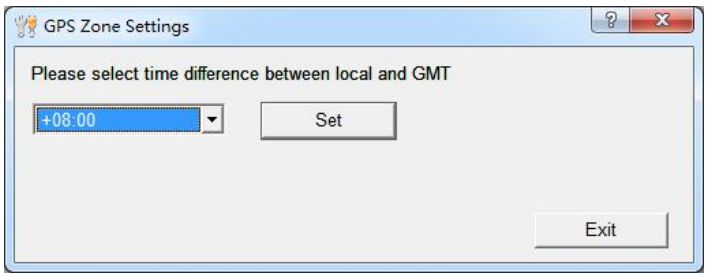


Fig. 3-56 GPS Time Zone Setting

After selecting the time difference, click “Set” to set the GPS and the interface will display whether it is successfully set or not.

4.3.4.3 Cascade

The Gateway (or the device with gateway function) connected multiple intelligent equipment, engineer tool software only need to connect one of Gateway’s communication

port, the engineer tool software can maintain all the smart devices connect to the Gateway, it supports both the Ethernet mode and serial port mode.

Operate steps：

Step one: Click the icon  and open the Cascade Operate interface, as fallow:

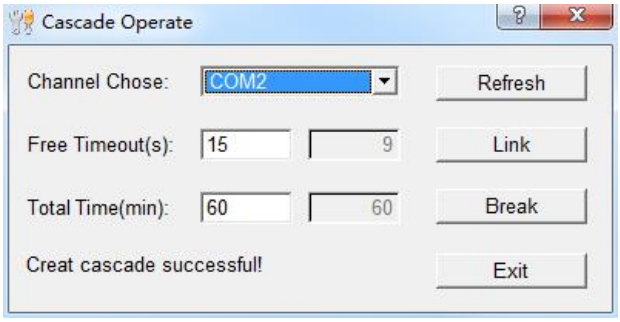


Fig. 3-57 Cascade operate

- ***Refresh:*** Get the cascade channel from the channel.dat
- ***Free Time out (s):*** The max free time of the connection without data, the cascade connection will break if time out, min time is 15s. Right edit line will show the time left.
- ***Total Time out (min):*** The total time that the cascade connection can maintain, max time is 60min. Right edit line will show the time left.
- ***Link:*** Create the cascade, the bottom left corner will show the result.
- ***Break:*** Break current cascade.
- ***Exit:*** Exit.

Step two: Click “Refresh” button and choose the channel information file (channel.dat), then the software will pick out the channel that with cascade or maintenance protocol to the combo box.

Step three: Select a channel from the “Channel Choose”.

Step four: Click “Link”, three are eight result will appear as fallow:

a) Create cascade successful!
b) Format error.
c) No objective channel, or is not a physical one, or is not active.
d) The objective channel is the source one.
e) The objective channel is not running maintenance or running bridging.
f) Auto close detecting time is less than 15s.
g) Bridging total time is less than 3 minutes.
h) Bridging total time exceeds 1 hour.

i) N/A.

Step five: Close this interface and you can maintain as the direct connect maintenance

Notice:

The communication port of Gateway connected with the engineer tool software must have protocol task. There is no limit to the specification protocol types, but the protocol must be the one that used the channel to send and receive the data.

The maintenance port of smart device must connect with the Gateway and the protocol on the corresponding Gateway's port should choose the cascade protocol (No. 285).

4.3.4.4 Reset

When the communication is normal, you can reset the equipment with the engineer tool software.

Click the icon and open the reset operation interface.

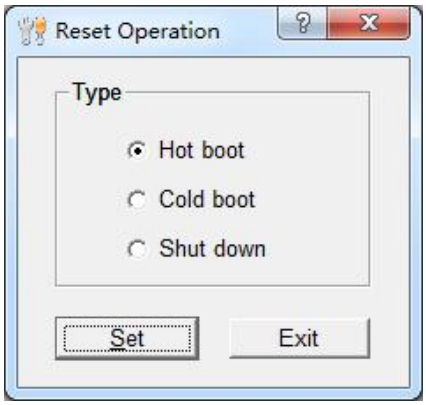


Fig. 3-58 System Reset

4.3.4.5 Help & About

Click the icon and open the about interface. From this interface you can get the software relevant information. Click the "Contents" from the "Help" menu, the user manual will pop out.



Fig. 3-59 About

4.3.4.6 Exit

Click the icon  to exit the engineer tool software.

5. Installation and Wiring

5.1 Installation & Inspection

Upon shipment arrival, please check if you have received the right materials and equipment as stated according to the shipping list. Please contact the sales department of Dongfang Group for any question. Perform installation as prescribed in the "Assembly and Wiring Drawings of the *DF1725IED"* contained in the cubicle.

5.1.1 Cubicle Installation

Move the cubicle to the right location and mount it on the two V-irons. Make sure that the base of the cubicle stands across the V-irons, keeping it horizontal to the cubicles and panels around it.

Make fixing holes in the V-irons according to the mounting-hole position of the cubicle base. After that, fix the cubicle with screws and bolts. Alternatively, you can also weld the cubicle directly on the V-irons. Good earthing of the V-irons should be ensured.

The external dimension of the cubicle is 600×800×2260 mm (unless specified otherwise) when placing order.

Inspection

Open the front door, back door and rotating door. Check if there is any damages and loosening of the cubicle and its components. They should be solidly grounded.

Unscrew all the plug-in boards and take them out by removing the connecting shielded cable behind them. Check all integrated circuits on the board and make sure they are of good contact with the sockets and well secured (taking care of PCBs). Thereafter, restore the plug-in boards, connect the shielded cable, and fasten the front securing screws.

Check the terminals at the back of the cubicle, ensuring no loose screws. Make sure there are no damages and defects on the terminal board and all plug-in components. Terminal boards and flat cables should be free of loosening & wrong connections.

All the cables and connecting wires shall be free of deformation, well organized and tightly secured.

The power supply terminals should disconnect from equipment before confirming the power supply is correct or not. In addition, variation of the power supply should be in the required limits. After finishing the above, the system should be able to be powered up.

5.2 Secondary Circuit Wiring

The connection in here refers to the connection between the field secondary circuit (or transformer) and terminals. They may be done along with work described in Section 1. All the incoming cables should be properly bounded and distributed behind or besides terminals. The wiring connections will be described below according to their functions.

Analog Input (AI)

If transducers output current signals, the common ground input connection shall be avoided. The shorting links on the remote measuring board shall be changed to double-end input configuration. The current signals shall flow from N*AI to (N+1)* AI terminals, where N is 1, 3, 5… (odd numbers) as an example. The Nth remote measuring output shall be connected to (2N-1) AI terminals, with the input to (2N) AI.

If transducers output voltage signals, the common ground input connection shall be avoided. The shorting links on the remote measuring board shall be changed to double-end input configuration.

Digital Input (DI)

In the case of sourceless contacts, one end of the signal wire from the auxiliary contacts of the field switch shall be connected to the nDI of the DI-16 board. The other end shall be connected to those of other signal wires with +24V terminal.

In the case of source contacts, the GND end shall be connected to the COM of the DI-16 board. The V+ end shall be connected to the corresponding digital input terminal. The power supply of the source contact shall be 24V or 48V unless specified otherwise.

After finishing all the connections, the corresponding LED on the DI board will be lighted when both the power supply is on and the contact is in the “On” status. Otherwise the LED will go out.

Pulses Input

Pulses from the energy meters or energy transducers are classified into two kinds: the wetting and empty types. The connection is the same as that of DI, using the same remote DI-16 board. The LED is lighted at high level and

extinguished at low levels.

The amplitude of the source pulse should be 24V or 48V. Else, it has to be specified at the time of order as this can be specially adjusted.

Binary Output and Control Output

The DF1725IED remote control is a “point-to-point” output. It has a closing relay and a cut-off relay for every circuit. Each of these relays is provided with one normally open contact.

AC Analog Input (ACAI)

The AC Analog Input is directly in series with the site CT with voltage input parallel with PT output terminals. For DF1725IED, IA,IB,IC, UA, UB and UC should be connected to measure the current, voltage, active power and reactive power, etc of a circuit. Even for a single bus bar (though many lines are from the same bus bar and the voltage are completely the same), all the same voltages have to be connected together in parallel on the exterior row of terminals. This is to enable independent measurement of current and voltage by each ACAI-3U3I circuit.

When connecting, please pay close attention to the current flowing direction and phase sequence. Otherwise, wrong power and other values will be worked out.

5.3 The size of DF1725IED

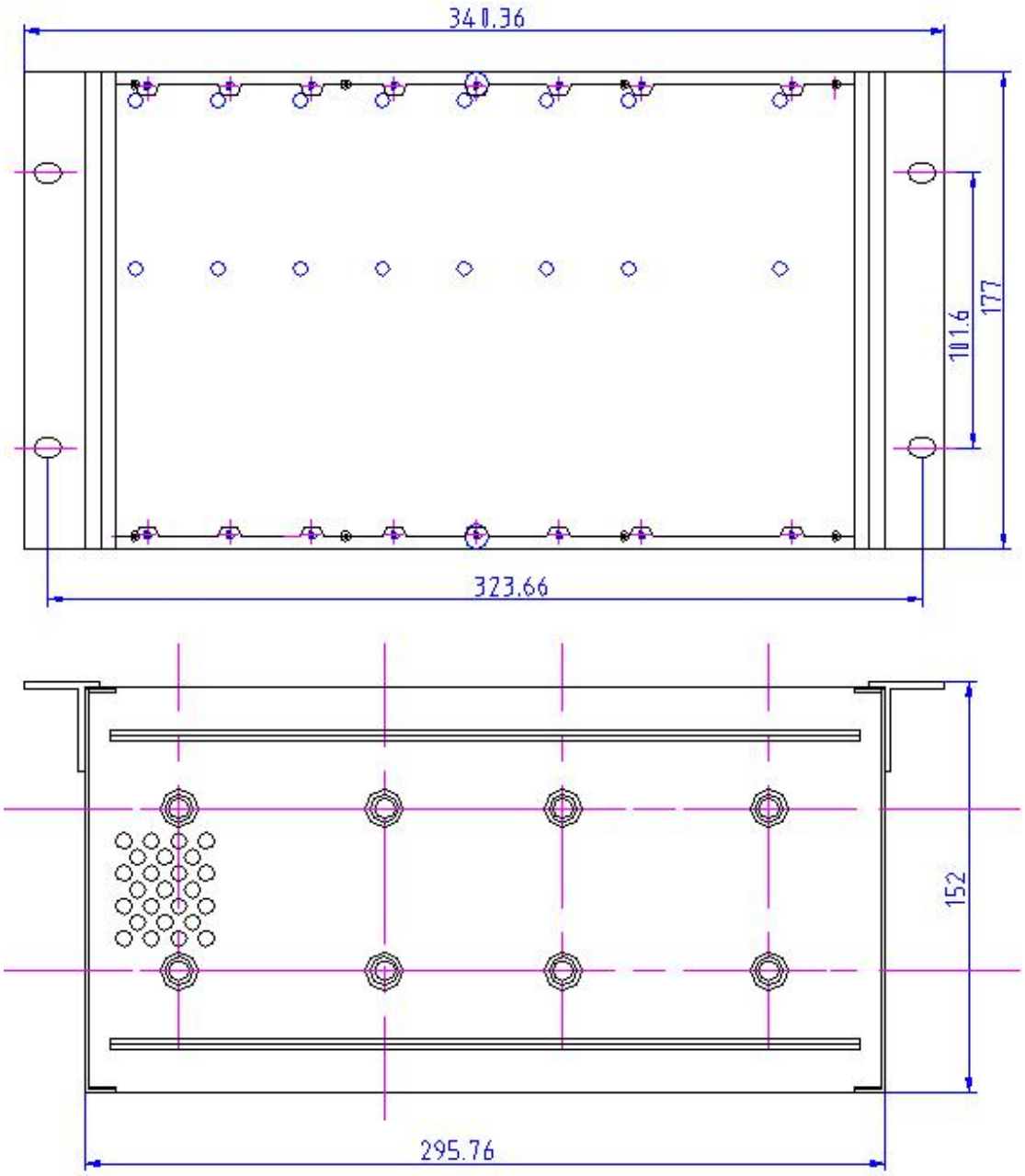


Fig9.1 Front view and top view for DF1725IED

6. Site Commissioning

This chapter caters to the powering-up commissioning after finishing installation and connection, including system configuration, parameter setting, functionality

checking and etc. It can be performed using a laptop computer.Please refer to “DF1000v User’s manual” for the detailed informations.

6.1 System Configuration

If all the “POWER” indicators are lighted and MCU “RUN” indicator flash regularly after powering up, this signifies that all plug-in components are functioning properly. Use a serial communications cable to connect the COM1 port of MCU to the servicing laptop. Upload all the files in MCU to the laptop. The types & number of components in MCU would be displayed on the computer and we can know what actually exist in the system. And we know also if new components should be added to this system.
Before configuration, the address of each functional board has to be set. This address can be set at random (between 0 ~ 7) as long as that the address is different from each other among the same type boards.

Caution: Among the same type of boards, the address should be different, otherwise, mis-opeations of remote controls, state changing of digital inputs, unstable measured values of analogue inputs, etc., may occur.

After the address of each board is set, other details like Type, Name of Information Value, Analog input Transformer Ratio, Digital input Time against chattering, Control output closed time, etc. will be configured. There are two methods for the system configuration: automatic and manual. Please note that automatic configuration means that default values are assumed and selected. For example, each DCAI-8 plug-in board is configured by default to 8 AI inputs, numbered 1 ~ 8AI, and each DI-16 plug-in board has 16 DI inputs, numbered from 1 to 16DI. The system does not automatically report the configuration to the laptop when finished. If different values/settings are required, the above setup has to be configured manually.

6.2 Function Checking

After finishing all the configuration, various functions of the system can be verified. They include correctness of AI, DI, KWh acquisition, reliability of DO, quality of communications between 1725IED and the front-end computer (or the master station) and other features required.

7. Trouble Shooting

Some typical failures and their possible causes are discussed herein for references. However, it is impossible to treat all the troubles that may happen on site. You are encouraged to analyze any failures carefully using your own discretion. You are able to find out which plug-in board is not running through remote servicing and this saves the unnecessary trouble of going down to the

substation to rectify.

7.1 General Problems

The LED “RUN” flashes abnormal

Please check the value of “RUNFLAG[0]” of “1725IED_SYS” in EEPROM,this value should be 1234,otherwise,change it to 1234,then power off 1725IED and then power on

If this problem is still exist, you should download the program of DF1725IED,the way to download please refer to “DF1000v user’s manual”

MCU can not find the board or the “status” of the board is fault.

Please check the address of it is right or not.Among the same type of boards,the address of them should be different.

Please plug out it and then plug it into the subrack again,make sure it is tight

Computer can not communicate with DF1725IED through maintenance software

Please check the baudrate of the computer and DF1725IED

Please make sure the cable between computer and DF1725IED is correct

Computer can not communicate with DF1725IED through Hyper terminal

Please check the baudrate of the computer and DF1725IED

Please make sure the cable between computer and DF1725IED is correct

Change another computer to run Hyper terminal

7.2 Analog Inputs (DCAI-8 and ACAI-3U3I/6I)

DC Sampling

Please verify if the input voltage from the terminal conforms to what is acquired by the DCAI-8 plug-in component in case of DC sampling. If they do conform, you can conclude that transducers or signal transmitting lines fail. Otherwise, if they are not consistent, please examine the input circuits of the DCAI-8 plug-in component in turn.

When there is only one DCAI input that is not correct, please check the portion from the signal input terminal board to the electronic switch. If there are 8 neighboring AI measured values that happen to be incorrect, please examine the corresponding the electronic switches. If the 8 DCAI inputs are wrong, please find out if there is any problem with the respective V/F converter and counting circuit.

The “Substitution” method may work. To decide whether the DCAI-8 board fails, replace the DCAI-8 board.

AC Sampling

Incorrect AC Sampling Connection

Judgment Method of AI Computation

If the voltage and current of a line are correct and its power is wrong, it is likely

that the connection is not right. According to various real-time measured values by the “Method of Two Meters”, we can analyze which part of the wiring connection goes wrong as illustrated below.

First, we assume that the measured values of the lines are: I1, I2, U1, U2, P1, P2, Q1, Q2, P, Q, where I1 and I2 are currents of phases A and C. Assuming the input connection is correct, let the value of current and voltage be I and U respectively. U1 and U2 are phase-to-phase voltages (UAB and UCB provided that inputs are right.). P and Q are three-phase active power and reactive power respectively. Assuming that there is no unusual load on the line, the power factor is about 0.866.

From P1, Q1, P2 and Q2, we can work out (tan-1Q/P) the angle φ1 between I1 and U1 and φ2 between I2 and U2. φ1 is usually about 60 degrees and φ2 about 0. P≈1.5IU, Q≈0.9IU.

* If φ1 is about 120 degrees and φ2 is around 0, I1 (IA) is in the reverse direction with **P≈0.5IU and Q≈0.9IU**.

* If φ1 is about 60 degrees and φ2 around 180, I2 (IC) is reverse with **P≈-0.5IU and Q≈0.9IU**.

* If φ1 is about –60 and φ2 around 120, both I1 (IC) and I2 (IA) are exchanged with **P≈0 and Q≈0.**

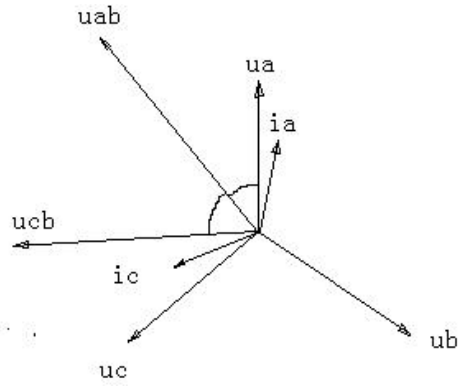
* If φ1 is about –60 and φ2 around –60, both I1 (IC) and I2 (IA) are mistaken for each other and I2 (IA) are reverse with P≈IU and Q≈-1.73 IU.

* If φ1 is about 120 degrees and φ2 around 120, both I1 (IC) and I2 (IA) are mistaken for each other and I1 (IC) are reverse with P≈-IU and Q≈1.73 IU.

* If none of the above appears, the voltage inputs may wrong or the corresponding PT is in incorrect phase rotation.

Judgment Method of the AI Wave

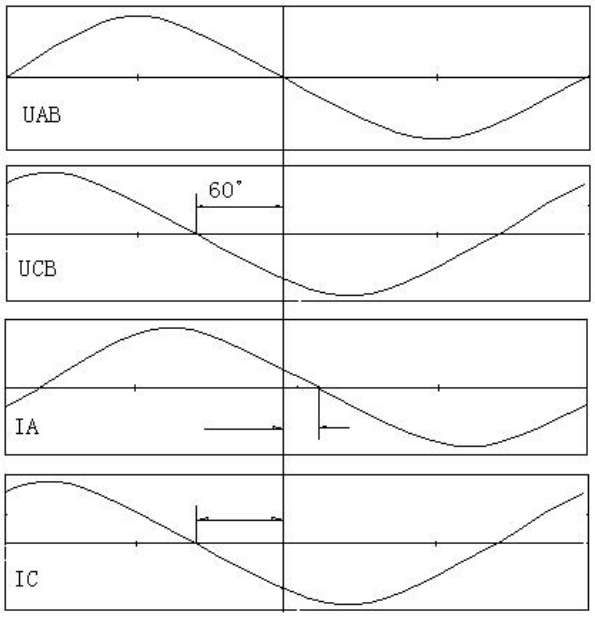
The connection mistakes can be easily detected utilizing high-speed sampling of theACAI-3U3I board. The DF1000v servicing software is used to



sample the whole suspicious line at high speed, displaying waves of UAB, UCB, IA and IC on the laptop. The waves and phase relationships between them are showed in the following figure when the inputs are correct.

The phase relationships between various values in a period (360 degrees) are given. From the graph of the voltage and current vectors, UCB should be 60 degrees ahead of UAB. IA is around 30 degrees lagging UAB (when COSΦ＝1, 30 degrees; when COSΦ＝0.866 inductive, 60 degrees; and when COSΦ＝0.866 capacitive, 0). IC shall be about 30 degrees ahead of UCB (when COSΦ＝1, 30 degrees; when COSΦ＝0.866 capacitive, 60; when COSΦ＝0.866 inductive, 0). Compare the displayed field waves with the above and the trouble will be solved at once.

Another fairly easy method is to disconnect the customer's phase B voltage input and then connect the neutral line UN of the transformer to its UB input terminal. UN is usually connected to the protective grounding network in the substation (Please attach UB to the grounding bar of the cubicle at the time of testing). Now the wave of UAB displays the UA's waveform and UCB shows the UC's waveform. UA and IA should be in phase, and UC and IC in phase (when COSΦ＝1. The phase difference should be 0 and when COSΦ＝0.866, the phase difference be±30 degrees). UC should lead Ua by 120 degrees. If they do not appear to be the above-said situation, connections are wrong. The error can be shown according to the real displayed phase relationship, e.g. if UA and IC are in phase and UC and IA in phase, UA and UC are exchanged (or IA and IC are exchanged). If the displayed current wave appears to be just 180 degrees other than what it shall be, the current is reversed in the current coil. If none of the above appears, the voltage phase sequence might be reversed, i.e.



UB and UA or UC are displaced. If IA and IC were both connected to IB in the field, it could immediately be detected by viewing the waveform.

If one or more input is/are still not "zero" when disconnected, it/they might have been configured as DC input mode when set by the laptop, which can be corrected by changing the respective setting.

Partial acquired data are incorrect and erratic

The “Substitution” method may try to decide whether the 1725ACAI or transformer board fails. If the ACAI-3U3I board breaks down, check if the 9346 settings are right. If there is no problem if that, please examine the ACAI-3U3I board. The rule is: if parts of U and I are correct, please check the MAX378 and precision resistors before examining the LF347. If the failure happens to be in the transformer board, check if the LED still remains lighted after removing the shielded cable. If the LED continues to remain unlighted (provided that there are signals from the transformers), please examine the transformers and the 39K resistor.

7.3 Digital Input (DI)

Acquired digital inputs do not conform to what they should be in the field.
Wrong connections, i.e. one auxiliary contact might have been connected to another digital input.
The field auxiliary contact is not adequate enough. Secure contacting should be ensured to keep good switching.
“Invert” settings are wrong, i.e. what shall be parameterized. “Invert” has not been set as required and vice versa.
The debounce time is too long to detect the change of status.
If 8 neighboring digital inputs do not switch, the corresponding 74HC245 or 74LS138 may break down.
If all digital inputs do not switch, the 74LS138 decoder might have been damaged.

7.4 Control Output (CO)

If relays do not operate or relays do operate but breakers do not operate, connections may wrong. The remote control lock switches on the CO-8 might have been switched to “OFF”. Else, it means that the functioning mechanism of the breakers has failed.

7.5 Communications

If communications with the master station are abnormal, please disconnect external circuits on the terminal row and short the TXD and RXD pins. With the laptop computer check if self-transmitting and self-receiving and transmit & receive buffers are normal. If the serial port is functioning properly, the “TXD” and “RXD” LEDs should light continuously. If the serial port fails, the “TXD” and “RXD” LEDs should flash in about one-second interval.

7.6 Troubles List

MCU PROBLEMS

LED Indicator	On	Off	Flashing
PWR	Power on (normal)	Hardware problem	Hardware problem
RUN	Hardware fault	Hardware problem	1 Hz - normal Other - parameter problem
ERR	Parameter problem, software problem	Normal	Parameter problem, software problem
COM1	Continuous space	Continuous mark	Data message
COM2	Continuous space	Continuous mark	Data message
COM3	Continuous space	Continuous mark	Data message
COM4	Continuous space	Continuous mark	Data message
COM5	Continuous space	Continuous mark	Data message
ETH1	Continuous space	Continuous mark	Data message
ETH2	Continuous space	Continuous mark	Data message

DI-16-220/110/48/24V PROBLEMS

LED Indicator	On	Off	Flashing
V1	Voltage input	No input	Hardware problem
V2	Voltage input	No input	Hardware problem
V3	Voltage input	No input	Hardware problem
V4	Voltage input	No input	Hardware problem
V5	Voltage input	No input	Hardware problem
V6	Voltage input	No input	Hardware problem
V7	Voltage input	No input	Hardware problem
V8	Voltage input	No input	Hardware problem
V9	Voltage input	No input	Hardware problem
V10	Voltage input	No input	Hardware problem
V11	Voltage input	No input	Hardware problem
V12	Voltage input	No input	Hardware problem
V13	Voltage input	No input	Hardware problem
V14	Voltage input	No input	Hardware problem
V15	Voltage input	No input	Hardware problem
V16	Voltage input	No input	Hardware problem

CO-8 PROBLEMS

LED Indicator	On	Off	Flashing
V1	Voltage output	No output	Hardware problem
V2	Voltage output	No output	Hardware problem
V3	Voltage output	No output	Hardware problem
V4	Voltage output	No output	Hardware problem
V5	Voltage output	No output	Hardware problem
V6	Voltage output	No output	Hardware problem
V7	Voltage output	No output	Hardware problem
V8	Voltage output	No output	Hardware problem

Appendix 1: List of Acronyms and Abbreviation

AC	Alternating current
A/D	Analog to digital
BCD	Binary Coded Decimal
BCU	Bay Control Unit
C	Capacitor
CCITT	Consultative Committee on inter-national
	Telegraphy and Telephony
CE	Chip enable
CD	Carrier detected
COM	Common communication port
COS	Change of state
CPU	Central processing unit
CS	Chip select
CT	Current transformer
CTS	Clear to send
D	Integrated circuit
DC	Direct current
DCD	Data carrier detect
DCE	Data communication equipment
DSR	Data set ready
DTR	Data terminal ready
EEPROM	Electronically erasable programmable

	Read-only memory
EMI	Electro-magnetic interference
EPLD	Erasable Programmable Logic Device
EPROM	Erasable programmable read-only memory
F	Fuse
FG	Earth ground
FSK	Frequency shift keying
GND	Ground
Hz	Hertz
I/O	Input/output
IEEE	Institute for Electrical and Electric Engineers
IED	Intelligent electronic device
INT	Interrupt
IRQ	Interrupt request
ISO	Inter-national Standards Organization
J	Connector (female)
K	Relay
LED	Light emitting diode
MCU-1	1725IED Main Communication Processing Unit,Clock stability is 1.2PPM
MCU-2	1725IED Main Communication Processing Unit, Clock stability is 0.3PPM
DI-24V	1725IED digital input board.DC2V input
DI-48V	1725IED digital input board,DC48V input
DI-110V	1725IED digital input board,DC110V input
DI-220V	1725IED digital input board,DC220V input
CO-8	1725IED digital output board
DCAI-8	DC analog input board
ACAI-3U3I	AC analog input board
ACAI-6I	AC analog input board
AO-2	Analog input output board
NVRAM	Non-volatile random-access memory
OE	Output enable
P	Plug (male connector)

PC	Personal computer
PCB	Printed circuit boards
PPS	Pulse per second
PT	Voltage current
R	Resistor, potentiometer
RAM	Random-access memory
RESET	Reset
ROM	Read-only memory
RTS	Request to send
RTU	Remote terminal unit
RV	Varistor
RXD	Receive data
RD	Receive data
S	Switch
SCADA	Supervisory Control and Data Acquisition
SOE	Sequence of events
SRAM	Static random-access memory
SWC	Surge withstand capability
T	Transformer
TTL	Transistor-transistor logic
TD	Transmit data
TXD	Transmit data
V	Diode
VDC	DC voltage

Appendix 2: Glossary of Terms

0x	A numerical prefix indicating that the number following is a

	hexadecimal number.
Application	A piece of software (a program) consisting of one or more processes and supporting functions.
Baud rate	A measurement of the speed of a communications line; the number of times a line can change state in one second; the rate at which a signal is modulated.
Binary	A number system having only two symbols (1 and o), and where values are expressed in the base two number system.
Bit	Abbreviation of binary digit. The smallest unit information in binary system. Has a value of either one (1) or zero (0).
Byte	A group of eight adjacent bits that function as a single unit. Defined as an octet by the International Electro technical Commission (IEC).
Change of state	A significant change (as defined by a particular system) in the condition of a point being monitored, for example, a change in flow rate, temperature, voltage, etc. usually is associated with dual status (that is, alarm/normal conditions). Abbreviated as COS.
COS	See change of state.
EEPROM	Electronically erasable programmable read-only memory. See electronically erasable programmable read-only memory. A type of data storage (memory chip) designed to be programmed more than once. The chips are functionally the same as EPROMs, but are erased using a particular electrical voltage. Abbreviated as EEPROM.
EPROM	Erasable programmable read-only memory. See erasable programmable read-only memory. A type of data storage (memory chip) designed to be programmed more than once. Can be read but not altered by the system, because it is a type of permanent memory. Abbreviated as EPROM.
Hexadecimal	A number system having 16 symbols, and where values are expressed in the base 16 number data. Sometimes referred to as hex. Hexadecimal numbers are often preceded by the prefix 0x.
Input/output	A type or types of hardware device(s) used to achieve human-to-machine or machine-to-machine communication.

	Abbreviated as I/O.
I/O	See input/output.
Master	The client or host station or computer, with which the RTU equipment communicates. Also referred to as a host or host computer.
Non-volatile Random Access Memory	A semi-permanent type of data storage (memory) that is backed up by batteries to maintain stored data even if system power is lost. Can be both read and changed by the system. Abbreviated as NVRAM.
NVRAM	See non-volatile random-access memory.
Phase Angle	The angle between voltage and current referenced to the voltage.
Port	An interface point connecting a communications channel and a device.
Power Factor	The cosine of the phase angle.
Ram	See random-access memory.
Random-Access Memory	A type of temporary data storage (memory) that can be read and changed while the computer is in use. Data stored in random-access memory is lost if the system loses power. Abbreviated as RAM.
Read-Only Memory	A type of permanent data storage (memory) that can be read but not altered by the system. Data stored in read-only memory is not affected by power loss to the system. Abbreviated as ROM.
Reactive power	Reactive power expressed in VARs. It is equal to volt amperes multiplied by the sine of the phase different between current factors.
ROM	See read-only memory.
Remote Terminal Unit	A piece of equipment located at a distance from a master station to monitor and control the status of outlying equipment, and to communicate the information back to the master station or host. Abbreviated as RTU.
RTU	See remote terminal unit.
SOE	See sequence of events.
Sequence of Events	A time-tagged change of state, logged as part of a chronological record of significant changes in the condition of a particular point or points being monitored. Abbreviated as SOE.

Source Code	A piece of code not yet been compiled or assembled (it is still in the language it was programmed in) and so cannot yet run on a machine.
Trip/close	A type of digital output that stops or starts an action. Trip/close outputs utilize two relay: a master relay and a select relay. Abbreviated as T/C.
Voltage sum	The magnitude of the vector sum of all the phase voltages.